

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                             feats_to_balance + feats_grouping +
                             feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

def import_data_IR():
    if target_col[0] == "i_NH3":
```

```

        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

df = pd.read_csv(file_path, sep=";")
df["datetime"] = pd.to_datetime(df["datetime"])
df.sort_values("datetime").reset_index(drop=True)

if target_col[0] == "i_NH3":
    df["campaign"] = df["datetime"].apply(assign_campaign)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

```

```

# if camp == "2025_Jun":
#     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
# elif camp == "2025_Sep":
#     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
if camp == "2025_Apr":
    #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
    #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
    mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
elif camp == "2025_Jun":
    mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
elif camp == "2025_Sep":
    mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_channel

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_channel

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

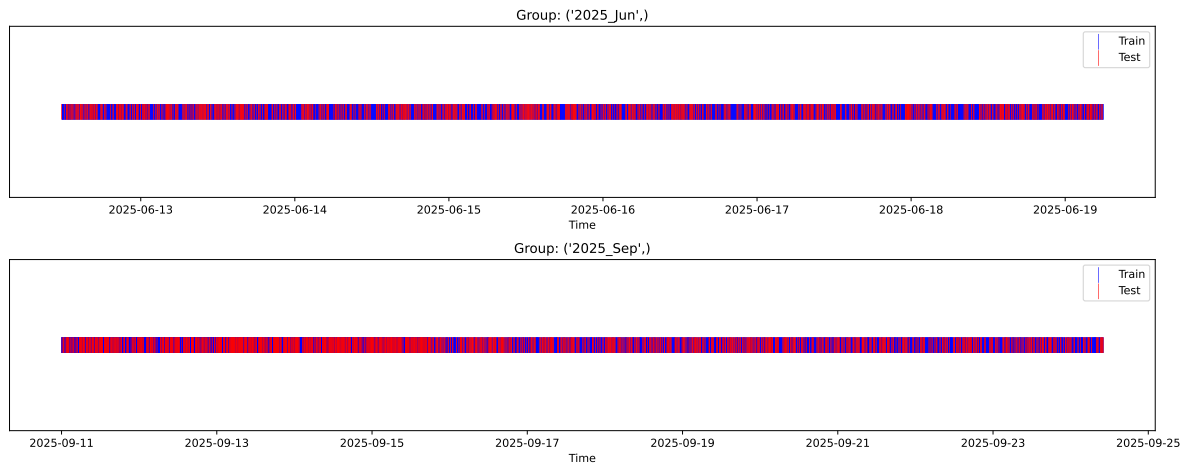
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

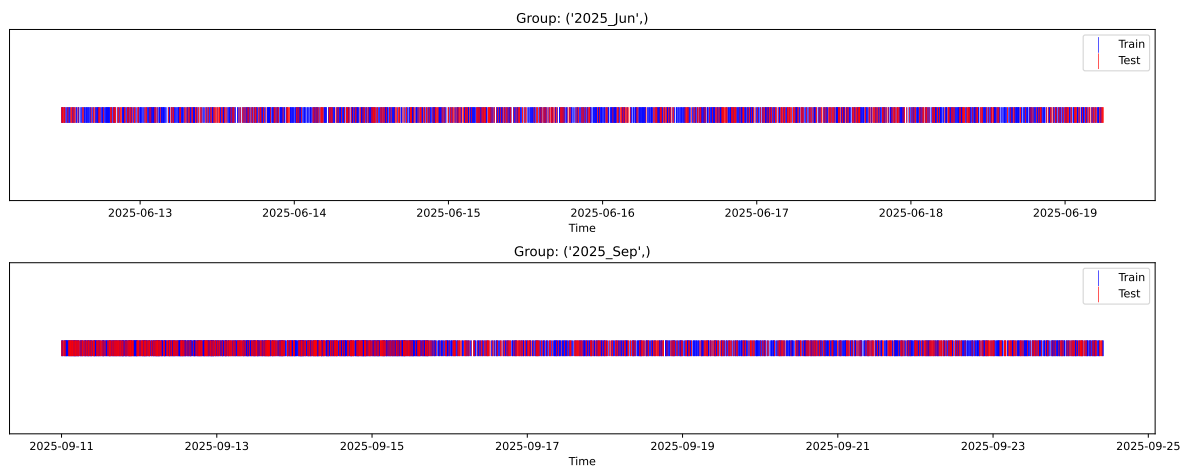
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

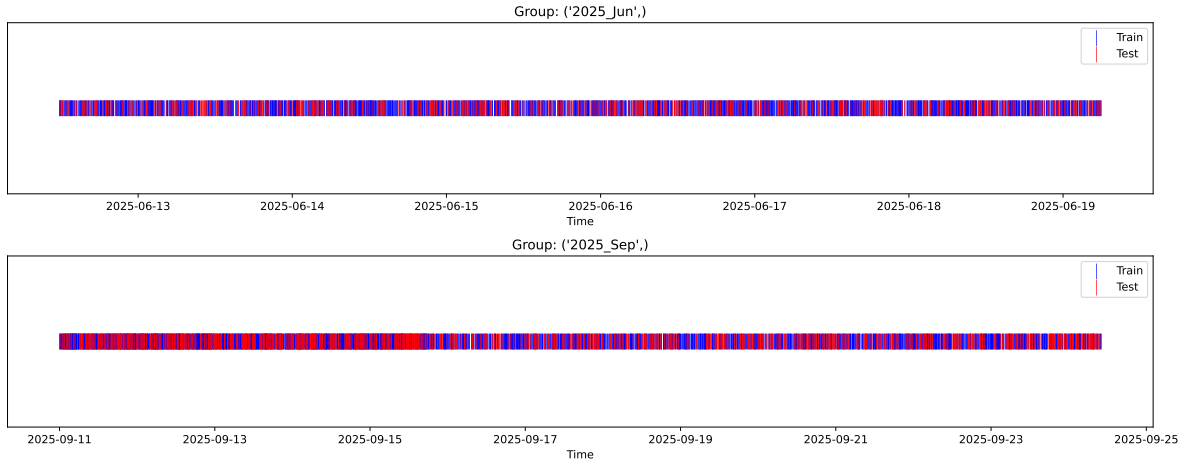
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



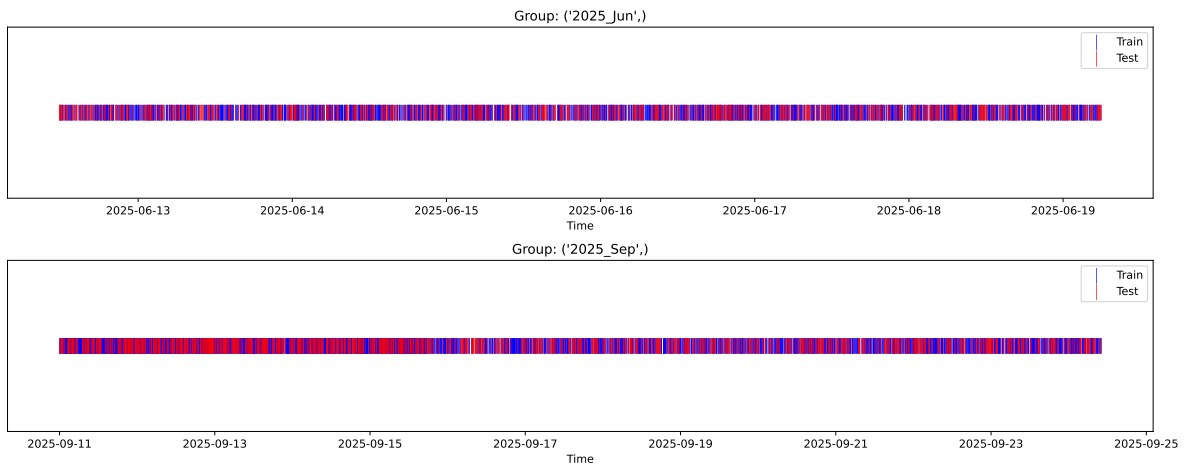
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



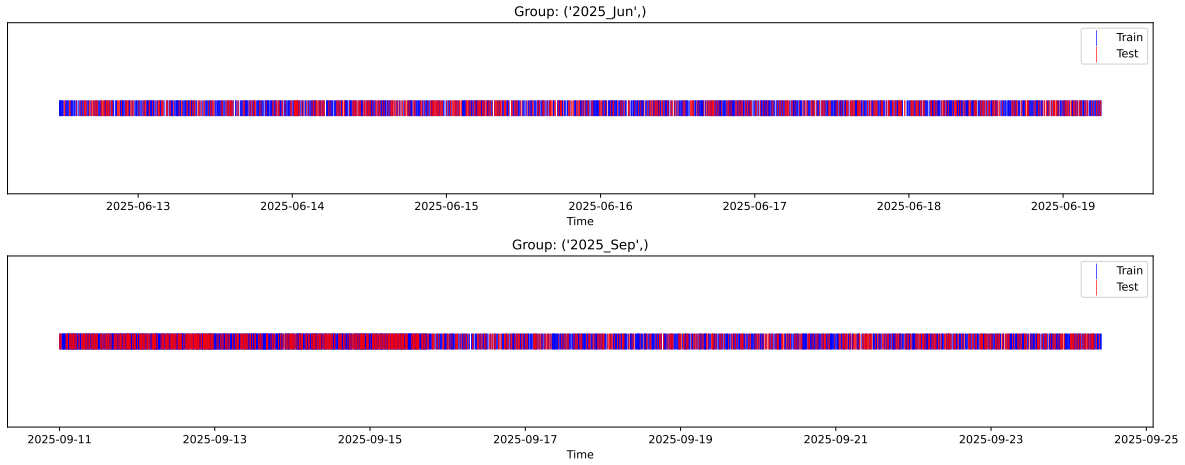
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



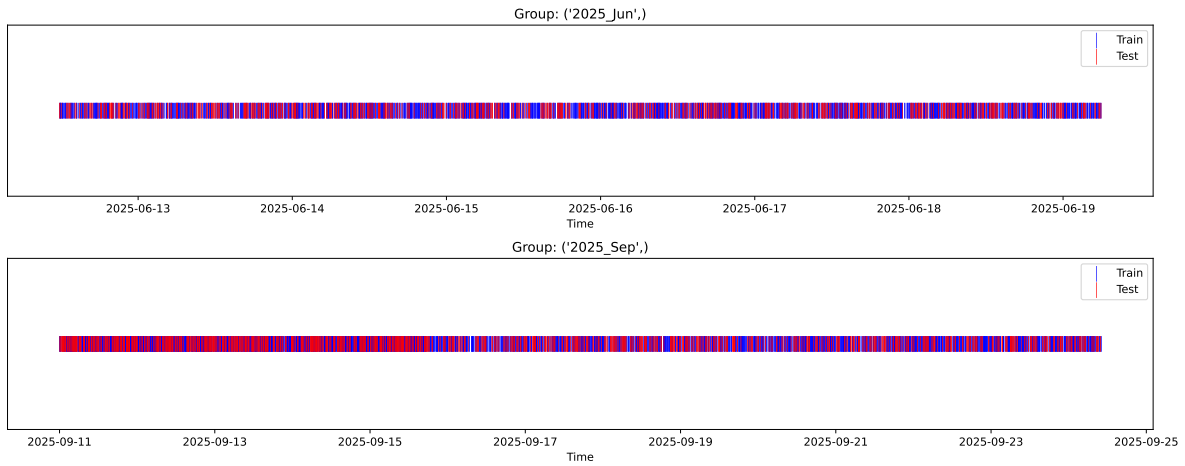
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 3

Len(group) : 1356, Len(training) : 950, Len(test) : 406

Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086
y_test: mean=2.318, std=0.740, min=1.068, max=5.447

Fold 0 — y_test: mean=2.39, std=0.76, y_train: mean=2.29, std=0.68

Fold 1 — y_test: mean=2.34, std=0.73, y_train: mean=2.31, std=0.70

Fold 2 — y_test: mean=2.33, std=0.71, y_train: mean=2.31, std=0.71

Fold 3 — y_test: mean=2.33, std=0.69, y_train: mean=2.31, std=0.71

Fold 4 — y_test: mean=2.30, std=0.73, y_train: mean=2.32, std=0.70

train scores CV: [0.62268767 0.60265136 0.58040391 0.56906438 0.60353391]

test scores CV: [0.49521601 0.53276144 0.58141377 0.60879453 0.53165288]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5957	0.55	0.0402	0.6403	0.6408	0.0908	1.1651	1.0831

Coefficients:

coef_r_CO2 = -0.0897

coef_r_NH3 = 0.0392

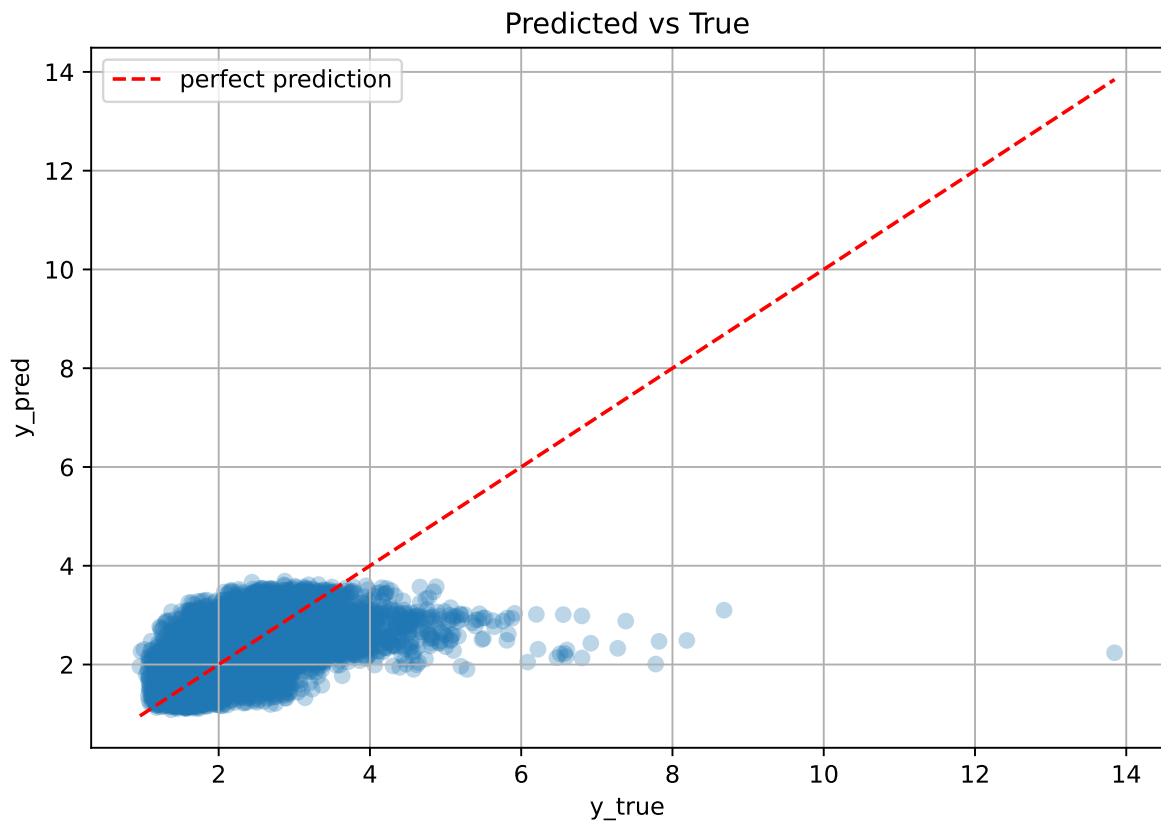
coef_r_THI = 0.4472

coef_r_temperature = -0.3735

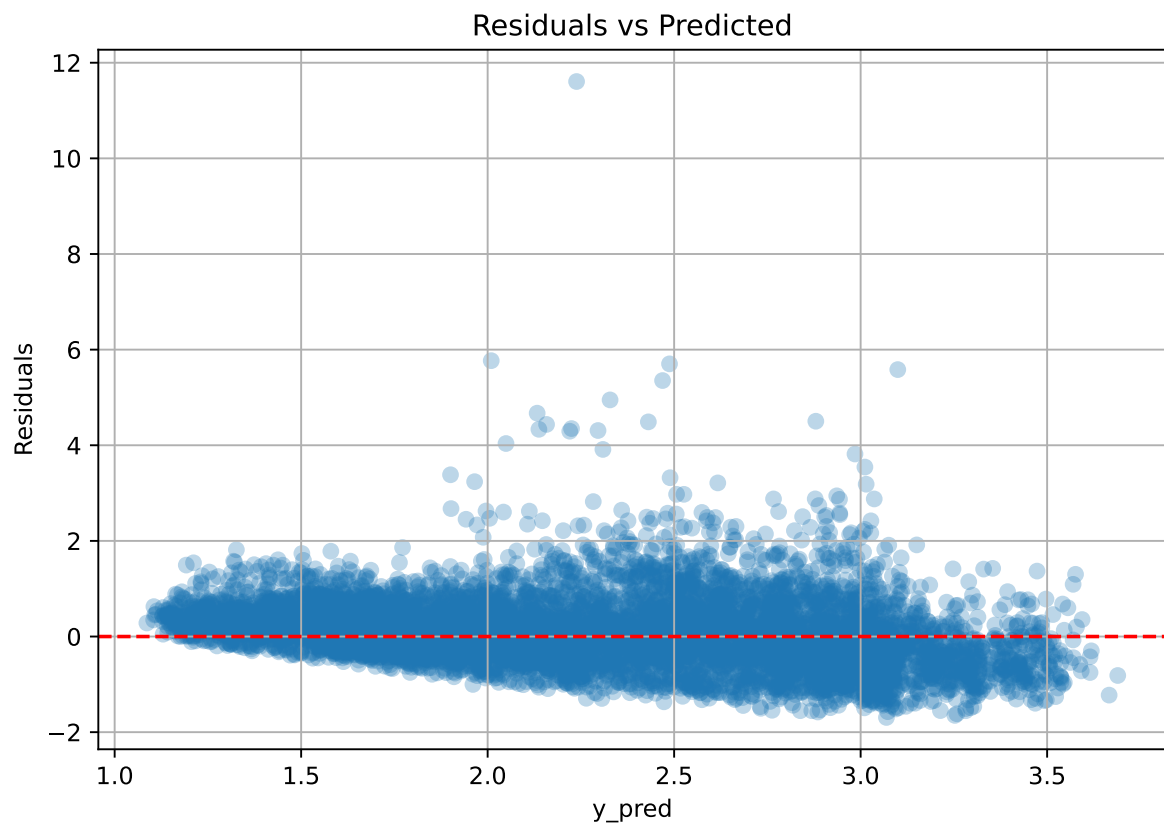
coef_r_umidity = -0.5516

intercept = 2.3165

2.3.2 Scatter



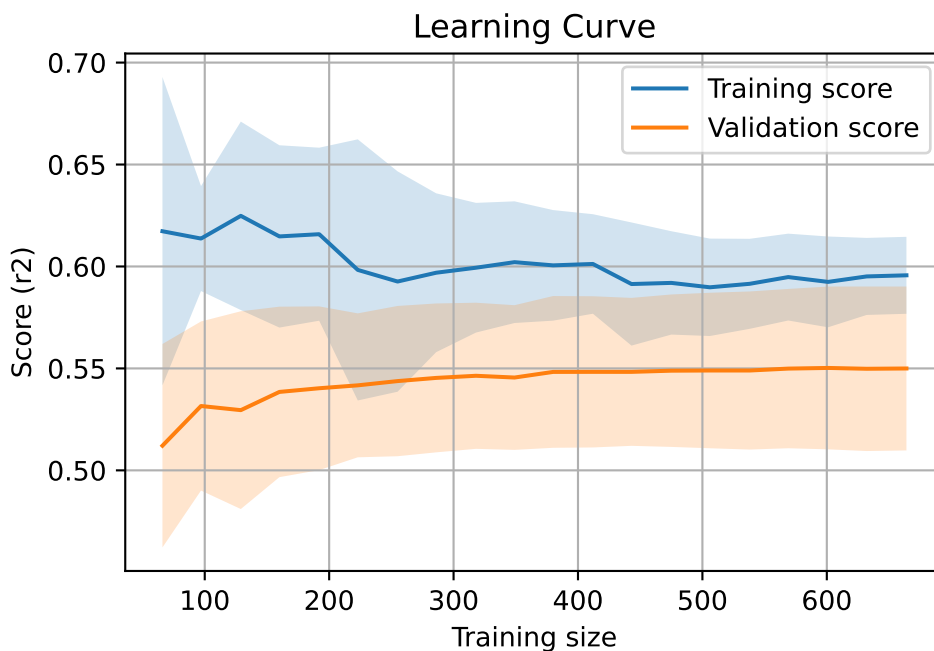
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 3

`Len(group)` : 1356, `Len(training)` : 950,

`Len(training_real)` : 665, `Len(test_of_training)` : 285, `Len(test)` : 406



2.3.5 Gridsearch — Group: 4

Len(group) : 1360, Len(training) : 952, Len(test) : 408

Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383
y_test: mean=2.202, std=0.791, min=1.062, max=7.822

Fold 0 — y_test: mean=2.22, std=0.80, y_train: mean=2.25, std=0.84

Fold 1 — y_test: mean=2.22, std=0.70, y_train: mean=2.25, std=0.88

Fold 2 — y_test: mean=2.29, std=0.86, y_train: mean=2.22, std=0.81

Fold 3 — y_test: mean=2.31, std=0.86, y_train: mean=2.21, std=0.82

Fold 4 — y_test: mean=2.22, std=0.83, y_train: mean=2.25, std=0.83

train scores CV: [0.31407089 0.27350601 0.3045029 0.31978092 0.29846023]

test scores CV: [0.23746916 0.36893865 0.26706608 0.23176705 0.28402149]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3021	0.2779	0.0494	0.2765	0.2771	-0.0008	0.9973	1.0871

Coefficients:

coef_r_CO2 = -0.1021

coef_r_NH3 = 0.1428

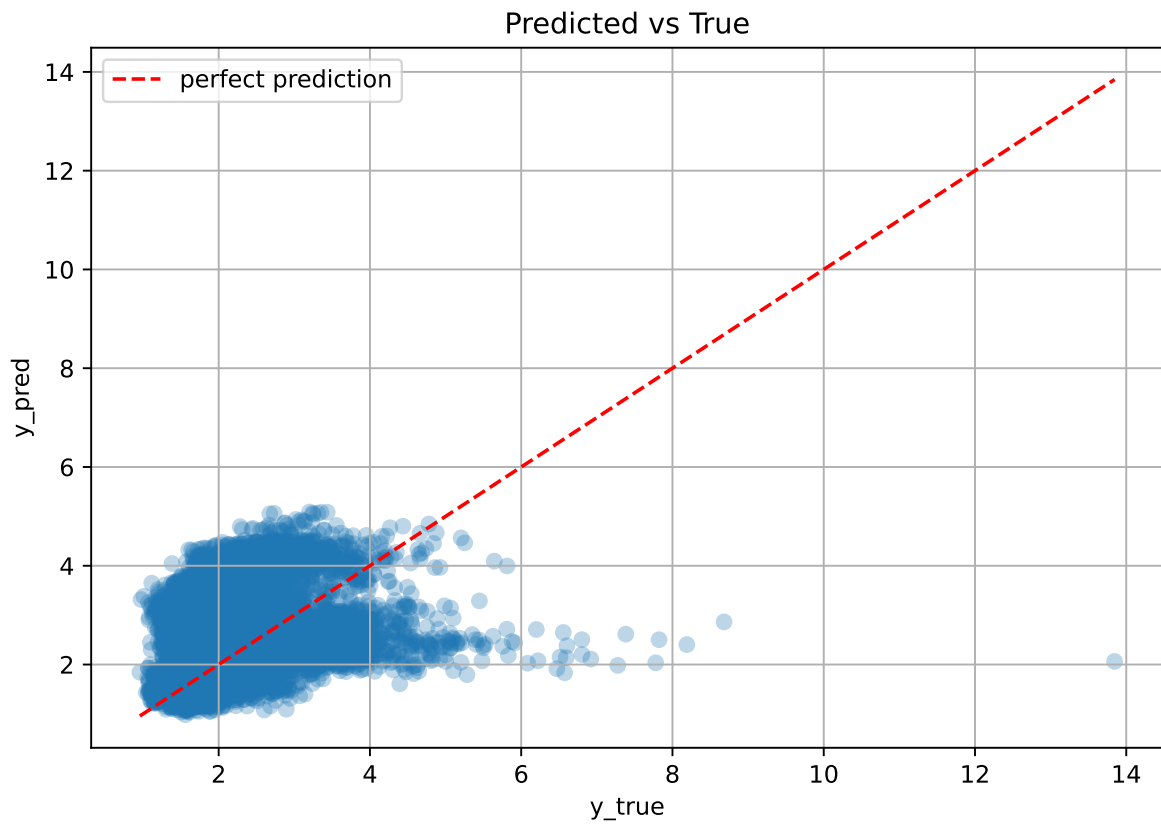
coef_r_THI = 1.0882

coef_r_temperature = -1.7734

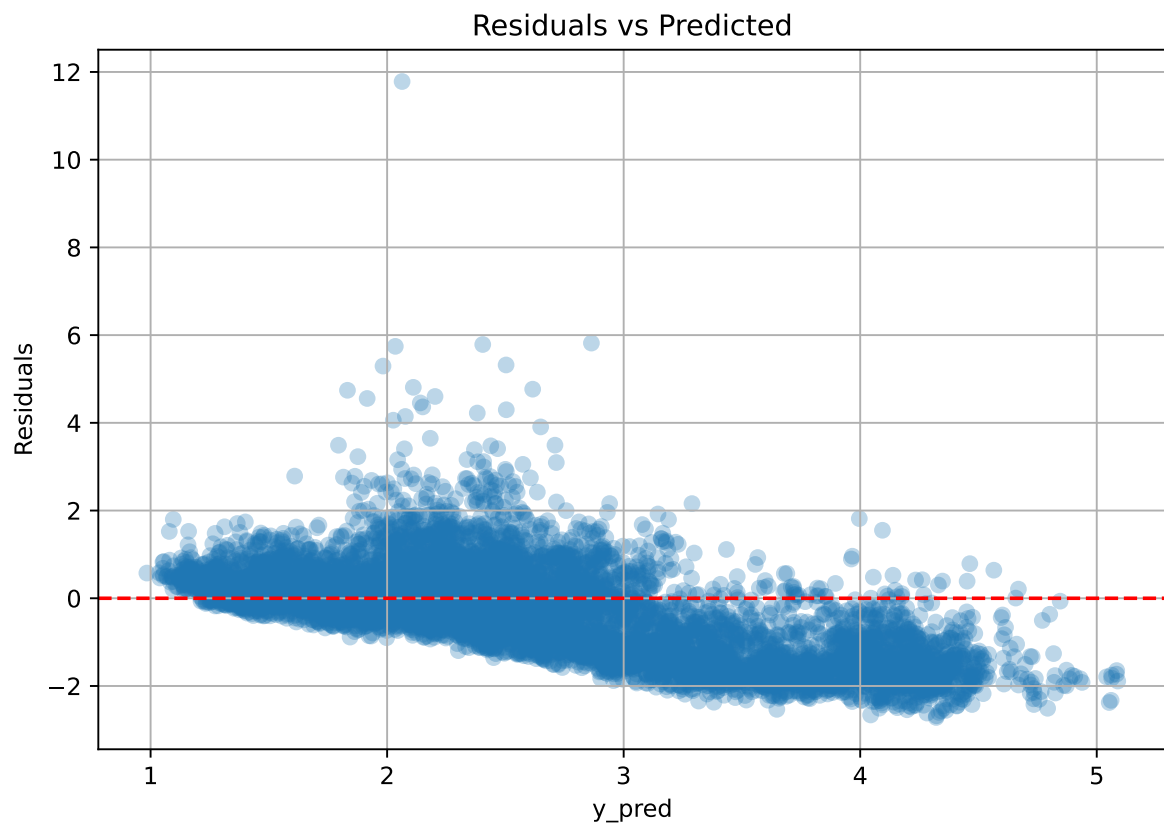
coef_r_umidity = -1.1164

intercept = 2.2387

2.3.6 Scatter



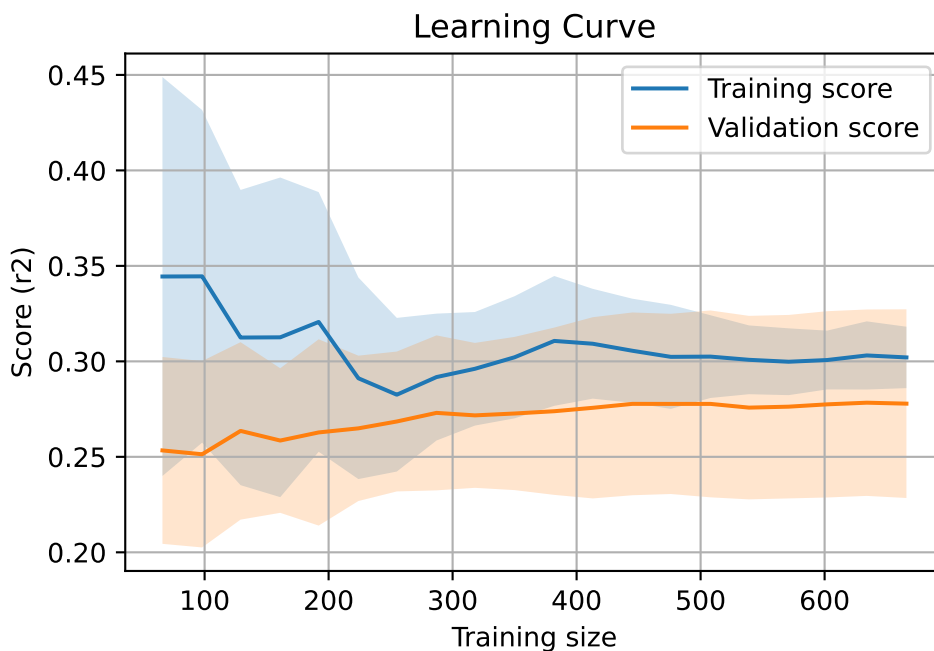
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 4

`Len(group)` : 1360, `Len(training)` : 952,

`Len(training_real)` : 667, `Len(test_of_training)` : 285, `Len(test)` : 408



2.3.9 Gridsearch — Group: 5

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

Fold 0 — y_test: mean=2.27, std=0.68, y_train: mean=2.22, std=0.68

Fold 1 — y_test: mean=2.25, std=0.71, y_train: mean=2.23, std=0.67

Fold 2 — y_test: mean=2.20, std=0.65, y_train: mean=2.25, std=0.69

Fold 3 — y_test: mean=2.21, std=0.65, y_train: mean=2.25, std=0.70

Fold 4 — y_test: mean=2.22, std=0.63, y_train: mean=2.24, std=0.70

train scores CV: [0.4277706 0.44021497 0.41172833 0.43714743 0.44005866]

test scores CV: [0.41299685 0.39069188 0.45040747 0.3878951 0.37549212]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4314	0.4035	0.0264	0.4566	0.4572	0.0537	1.1332	1.0691

Coefficients:

$\text{coef_r_CO2} = -0.0952$

$\text{coef_r_NH3} = -0.1254$

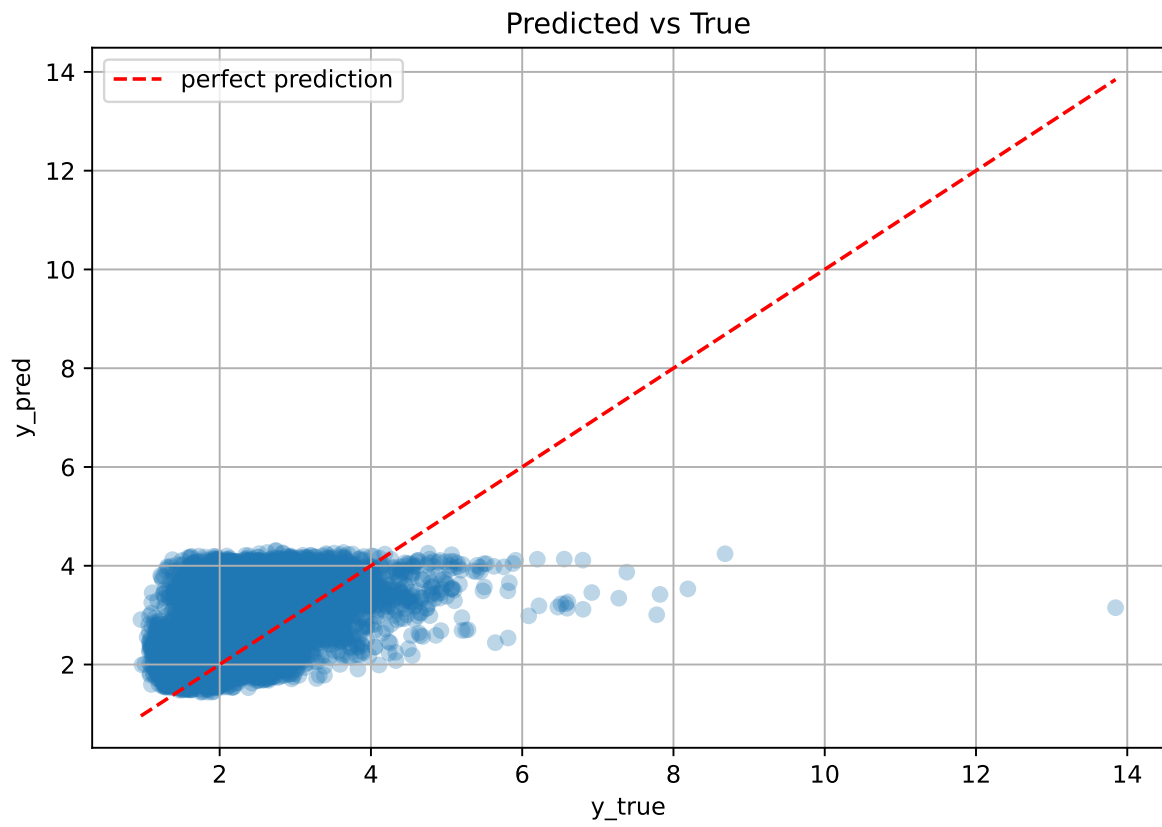
$\text{coef_r_THI} = 0.0994$

$\text{coef_r_temperature} = 0.0362$

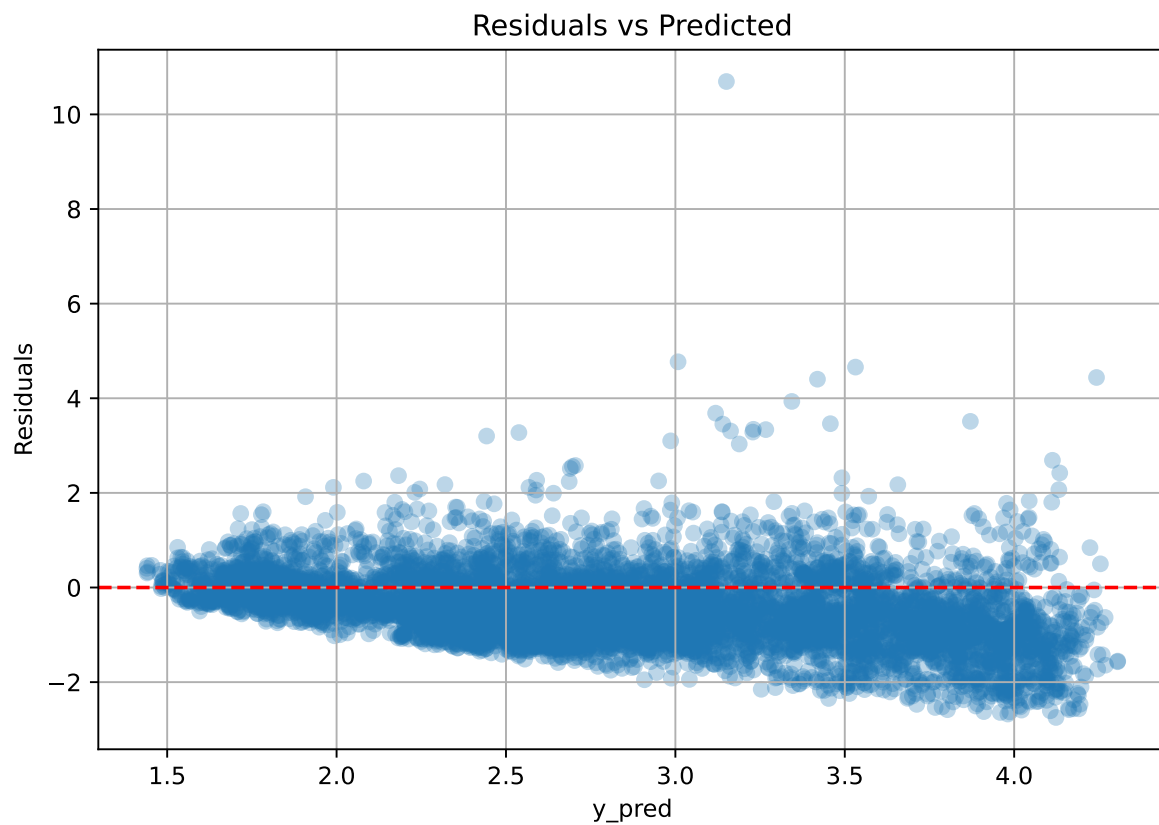
$\text{coef_r_umidity} = -0.4502$

$\text{intercept} = 2.2379$

2.3.10 Scatter



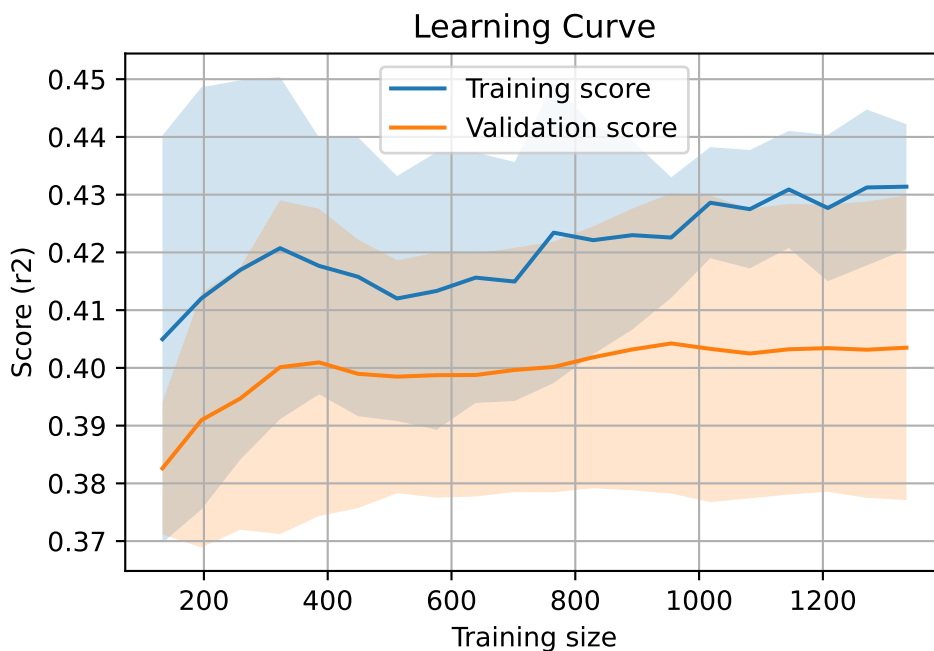
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.13 Gridsearch — Group: 11

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

Fold 0 — y_test: mean=2.58, std=0.86, y_train: mean=2.52, std=0.86

Fold 1 — y_test: mean=2.56, std=0.84, y_train: mean=2.53, std=0.87

Fold 2 — y_test: mean=2.50, std=0.84, y_train: mean=2.55, std=0.87

Fold 3 — y_test: mean=2.53, std=0.87, y_train: mean=2.55, std=0.85

Fold 4 — y_test: mean=2.53, std=0.82, y_train: mean=2.54, std=0.87

train scores CV: [0.46497694 0.46580479 0.46495948 0.46844345 0.45091902]

test scores CV: [0.45757013 0.45504757 0.45666765 0.44978339 0.49647352]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.463	0.4631	0.0169	0.3942	0.3945	-0.0686	0.8519	0.9998

Coefficients:

coef_r_CO2 = 0.0009

coef_r_NH3 = 0.0485

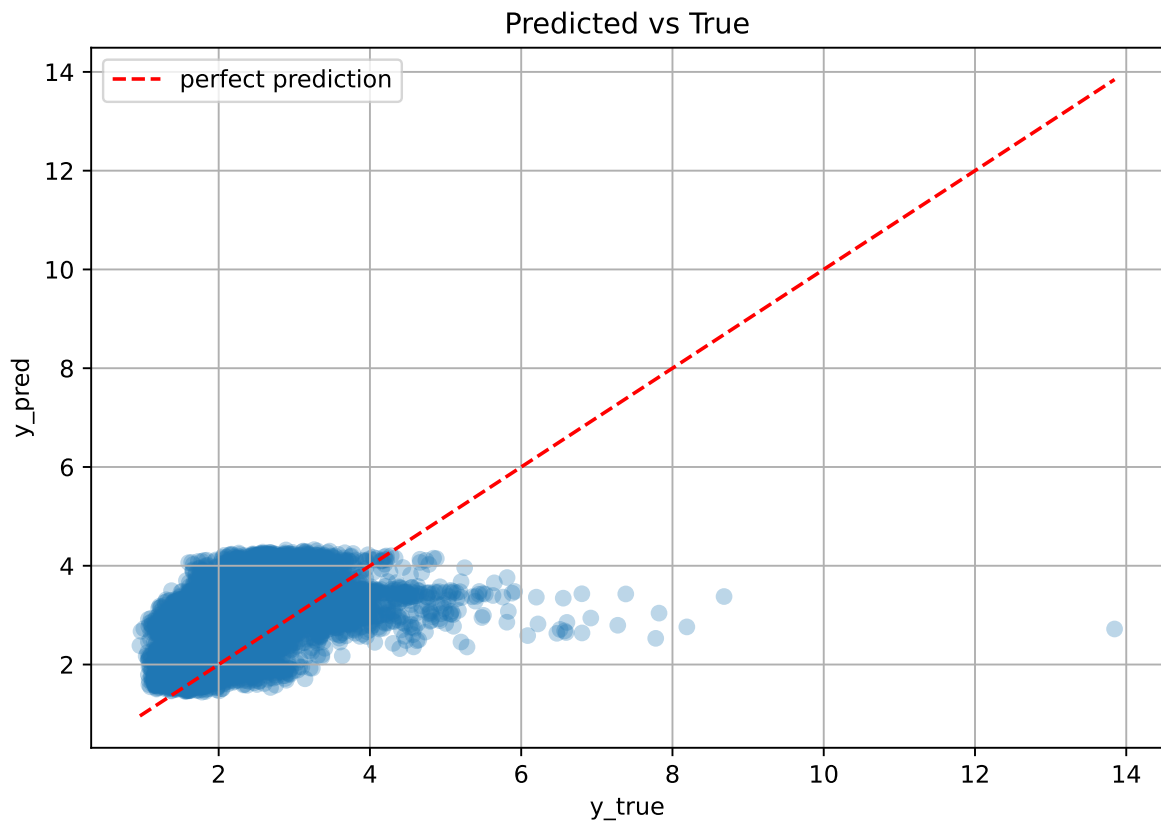
coef_r_THI = 0.8323

coef_r_temperature = -0.9731

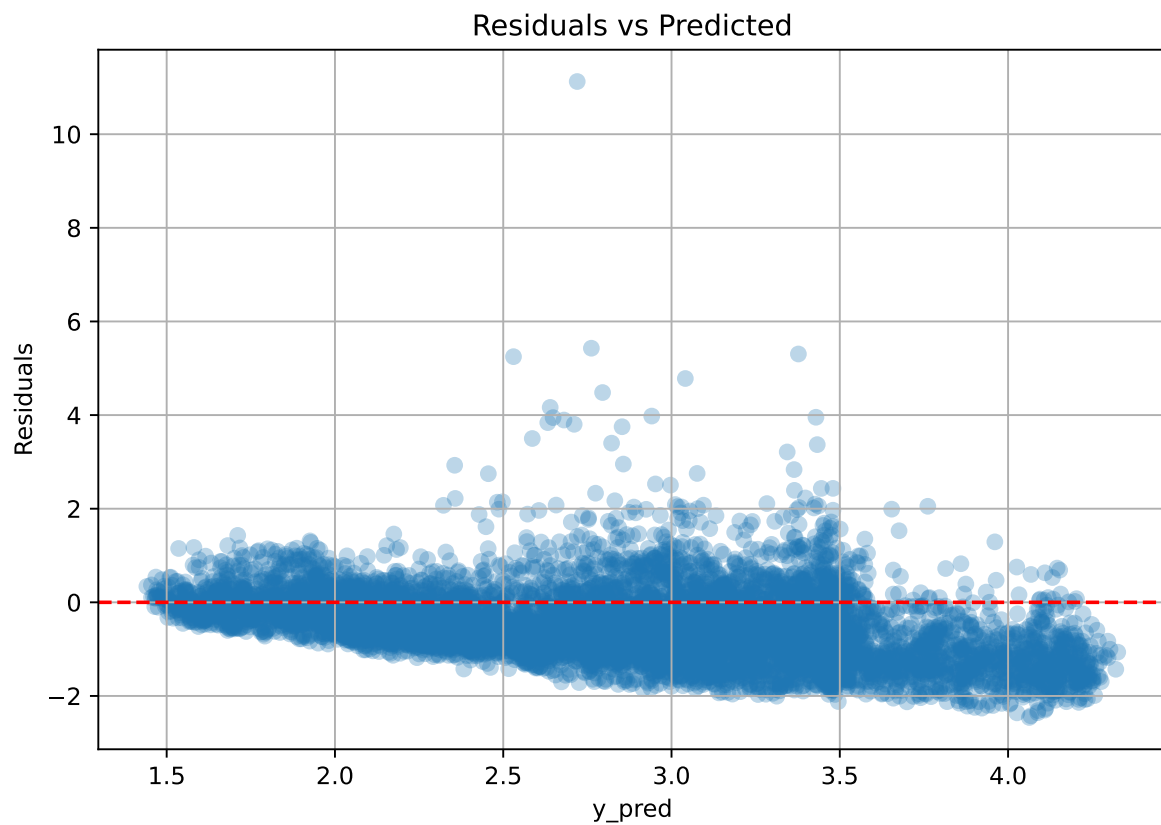
coef_r_umidity = -0.7852

intercept = 2.5398

2.3.14 Scatter



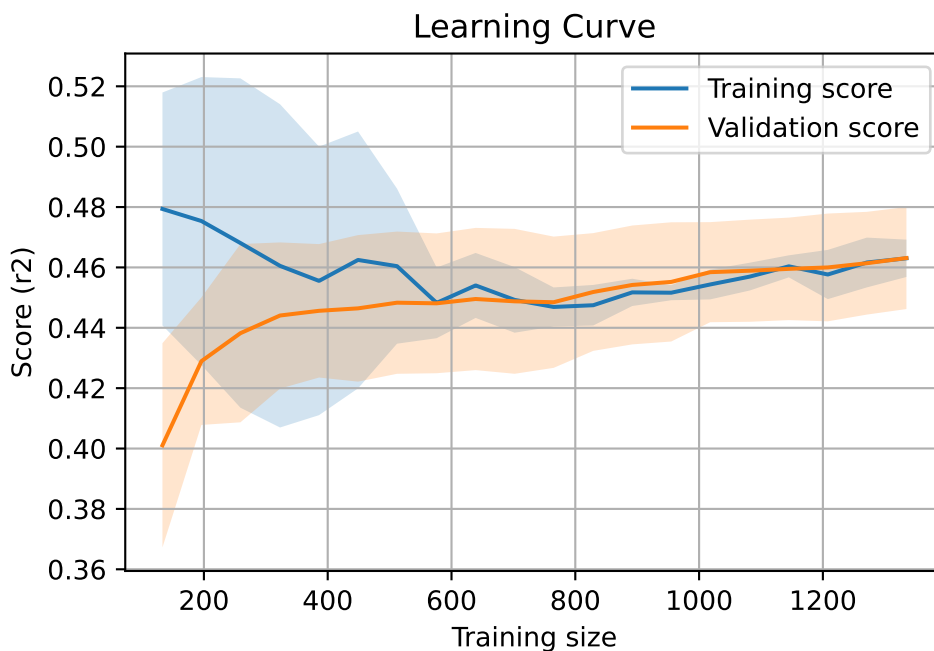
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



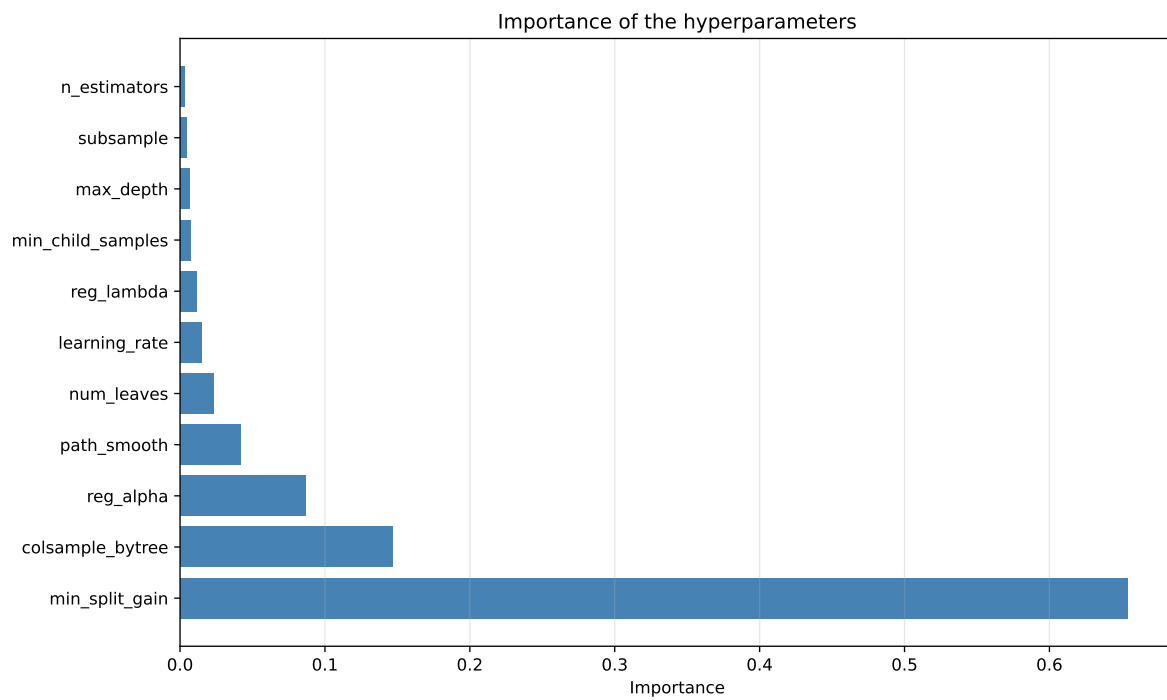
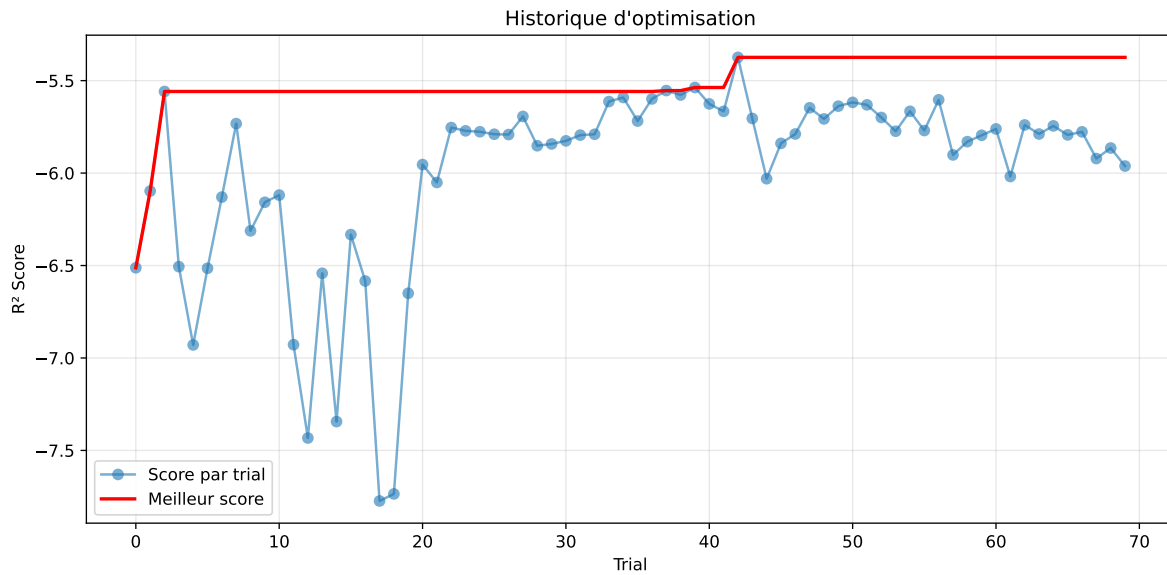
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 3

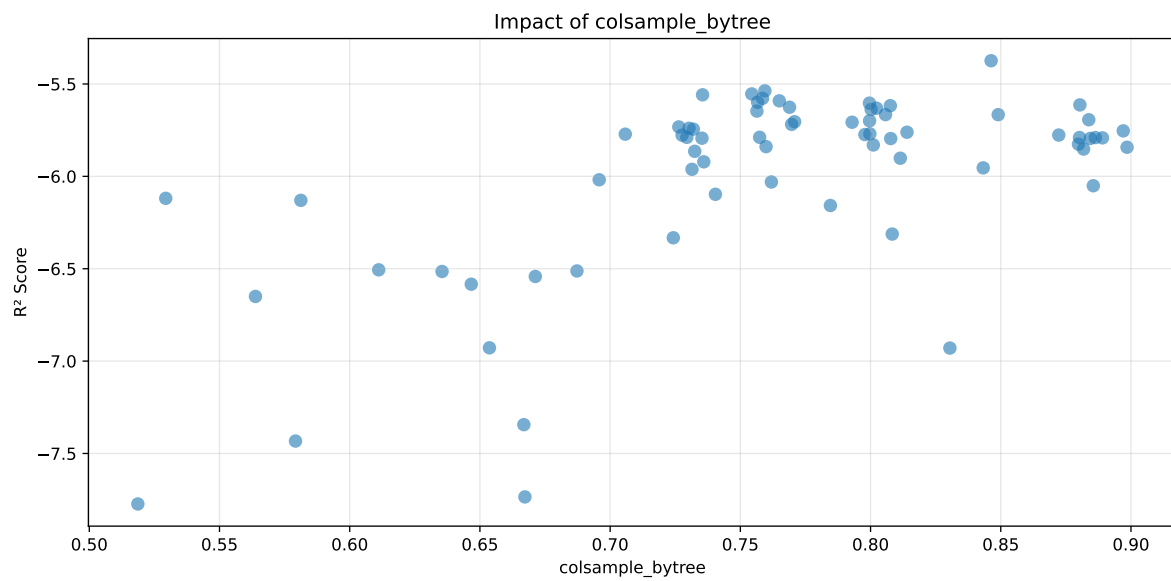
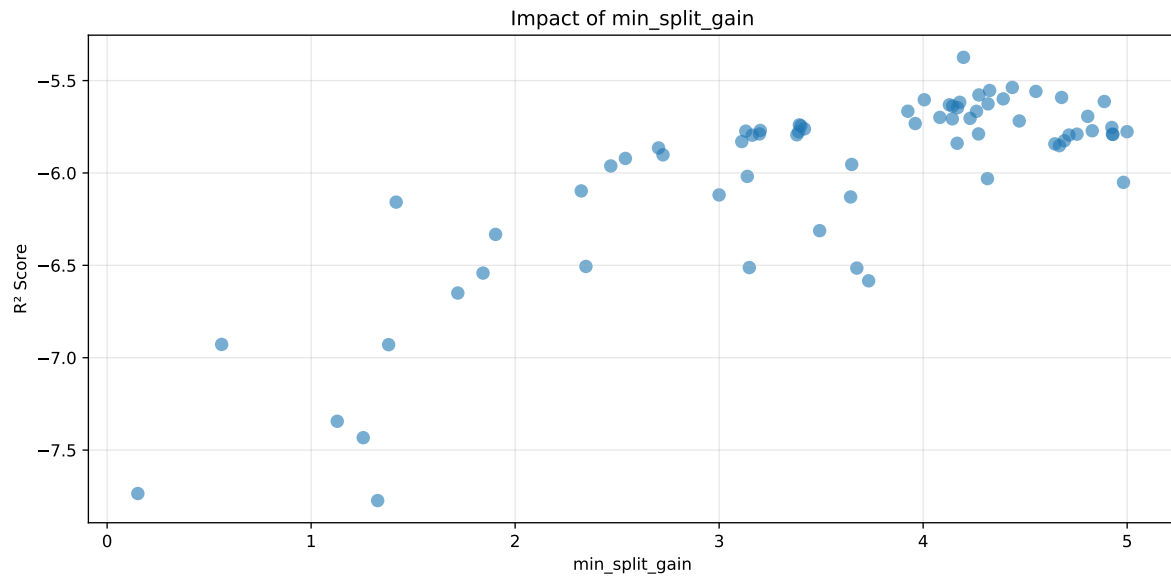
Len(group) : 1356, Len(training) : 950, Len(test) : 406

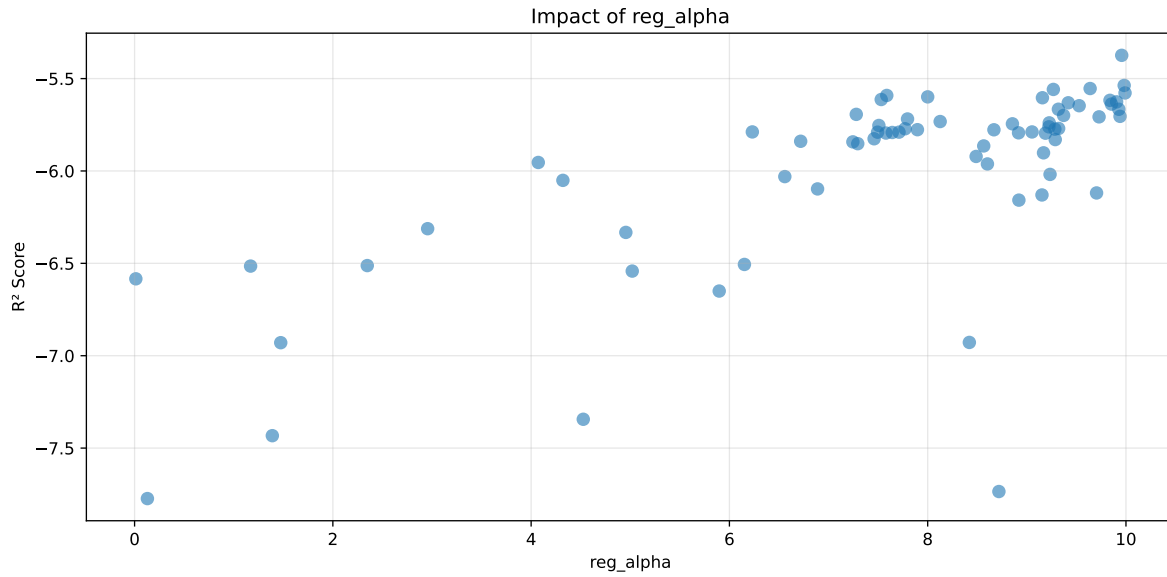
Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086
y_test: mean=2.318, std=0.740, min=1.068, max=5.447

===== TOP Importance FEATURES =====
feature importance r_umidity 230 r_NH3 83 r_temperature 51 r_THI 1 r_CO2 0



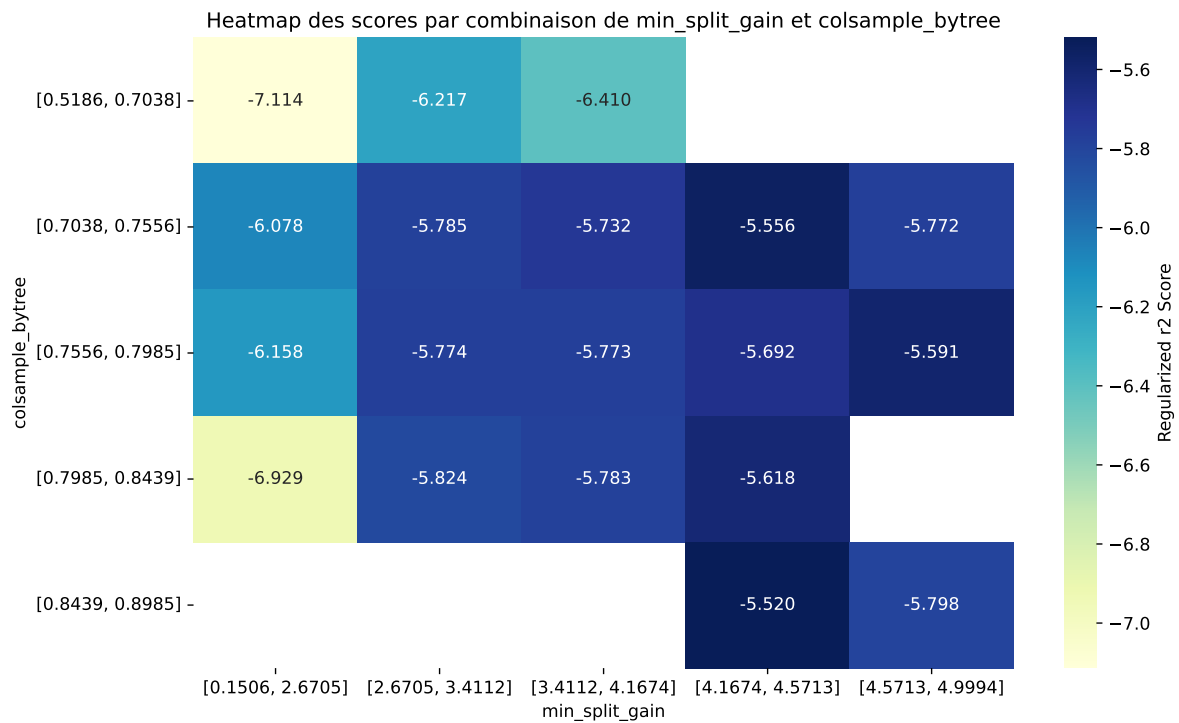
Paramètres avec >5% d'importance : ['min_split_gain', 'colsample_bytree', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5583	0.4995	0.5664	0.0397	-5.3741	0.588	0.0885	1.1772	1.1176
2	0.5699	0.5094	0.5806	0.0402	-5.537	0.6111	0.1016	1.1994	1.1187
3	0.5712	0.5106	0.5823	0.0405	-5.5537	0.6129	0.1023	1.2004	1.1188
4	0.5681	0.5074	0.5787	0.0398	-5.5587	0.6089	0.1015	1.2	1.1195
5	0.5742	0.5133	0.5854	0.041	-5.5778	0.6147	0.1013	1.1974	1.1187
6	0.5711	0.5101	0.5813	0.0406	-5.5912	0.6009	0.0908	1.1779	1.1196
7	0.5793	0.5181	0.5923	0.0401	-5.5989	0.6172	0.099	1.1912	1.1181
8	0.575	0.5139	0.5868	0.0411	-5.6037	0.6155	0.1016	1.1977	1.119
9	0.572	0.5108	0.5825	0.0406	-5.6132	0.6017	0.0909	1.1781	1.1199
10	0.5715	0.5102	0.5826	0.0407	-5.6176	0.6121	0.1019	1.1997	1.1201
---	---	---	---	---	---	---	---	---	---
70	0.6483	0.5649	0.6609	0.0377	-7.7733	0.6773	0.1123	1.1989	1.1476

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.010049247095375545, 'reg__num_leaves': 45, 'reg__subsample': 0.6149805194841786, 'reg__colsample_bytree': 0.846288869915877, 'reg__min_child_samples': 41, 'reg__reg_alpha': 9.957109681768337, 'reg__reg_lambda': 30.108173967579752, 'reg__min_split_gain': 4.197694017185606, 'reg__path_smooth': 4.25404476973982}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.020343193136041247, 'reg__num_leaves': 46, 'reg__subsample': 0.6073432092298143, 'reg__colsample_bytree': 0.7594110712206958, 'reg__min_child_samples': 43, 'reg__reg_alpha': 9.981924982662948, 'reg__reg_lambda': 29.250361916747053, 'reg__min_split_gain': 4.437053236529098, 'reg__path_smooth': 4.096544459397524}

Rank 3: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.022749333766227516, 'reg__num_leaves': 45, 'reg__subsample': 0.6304496096748545, 'reg__colsample_bytree': 0.7543894923489779, 'reg__min_child_samples': 43, 'reg__reg_alpha': 9.638522119810187, 'reg__reg_lambda': 29.612766986709776, 'reg__min_split_gain': 4.325861907162246, 'reg__path_smooth': 4.216352856402764}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.032507962176129696, 'reg__num_leaves': 25, 'reg__subsample': 0.7822589741019335, 'reg__colsample_bytree': 0.7354905211343733, 'reg__min_child_samples': 24, 'reg__reg_alpha': 9.267554221645794, 'reg__reg_lambda': 47.77026031315565, 'reg__min_split_gain': 4.552764705519141, 'reg__path_smooth': 4.611416594152529}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.021164130211578466, 'reg__num_leaves': 45, 'reg__subsample': 0.6275257519781369,

'reg__colsample_bytree': 0.758526826734625, 'reg__min_child_samples': 43, 'reg__reg_alpha': 9.990633979724164, 'reg__reg_lambda': 16.2555677398552, 'reg__min_split_gain': 4.273565447253004, 'reg__path_smooth': 4.232438440072576}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.010107858571177353, 'reg__num_leaves': 44, 'reg__subsample': 0.66590058217789, 'reg__colsample_bytree': 0.76495703798219, 'reg__min_child_samples': 44, 'reg__reg_alpha': 7.587442483607368, 'reg__reg_lambda': 9.579807195308614, 'reg__min_split_gain': 4.67883505046588, 'reg__path_smooth': 4.904830110473404}

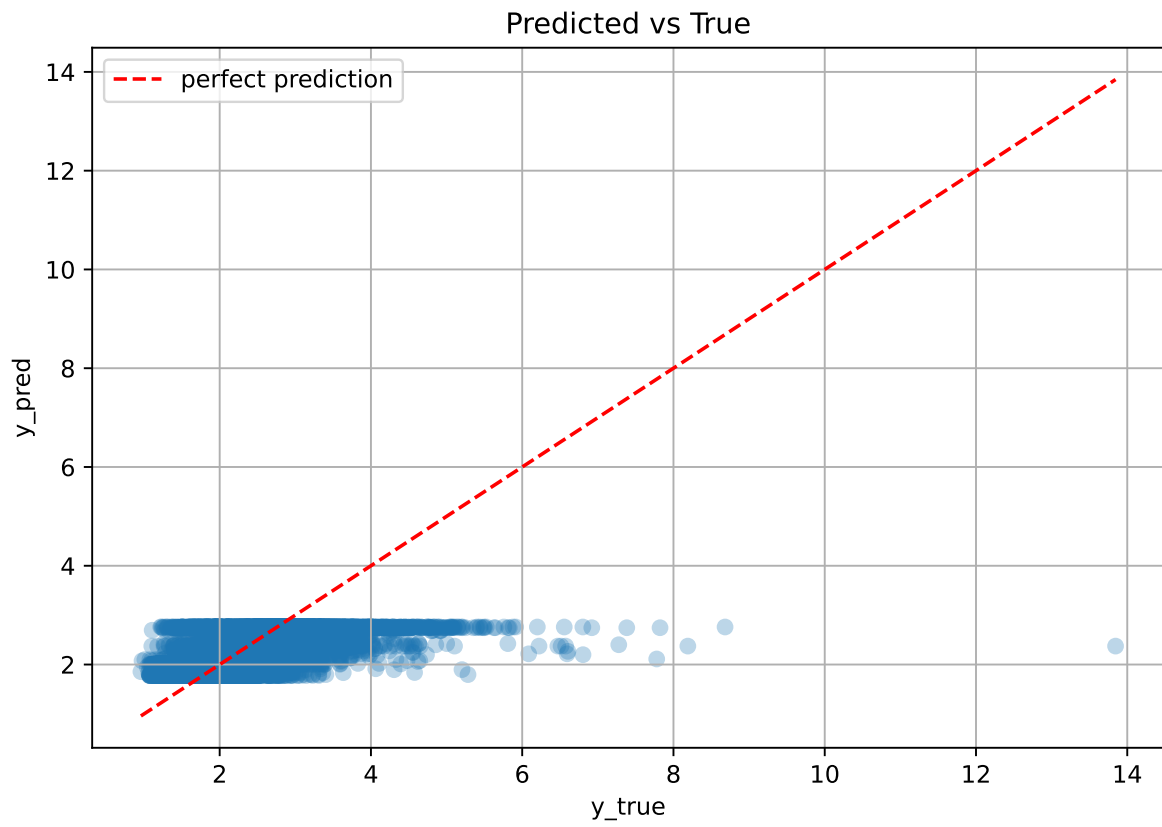
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.097245786674219, 'reg__num_leaves': 45, 'reg__subsample': 0.6339126548599773, 'reg__colsample_bytree': 0.7566334971069022, 'reg__min_child_samples': 43, 'reg__reg_alpha': 8.000071428501979, 'reg__reg_lambda': 29.959254112894353, 'reg__min_split_gain': 4.392458528202888, 'reg__path_smooth': 4.21034731809219}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.031506760418376405, 'reg__num_leaves': 49, 'reg__subsample': 0.5532427767743431, 'reg__colsample_bytree': 0.7995632231573147, 'reg__min_child_samples': 40, 'reg__reg_alpha': 9.15737679223279, 'reg__reg_lambda': 42.41034568695696, 'reg__min_split_gain': 4.005517669879697, 'reg__path_smooth': 3.844742921668057}

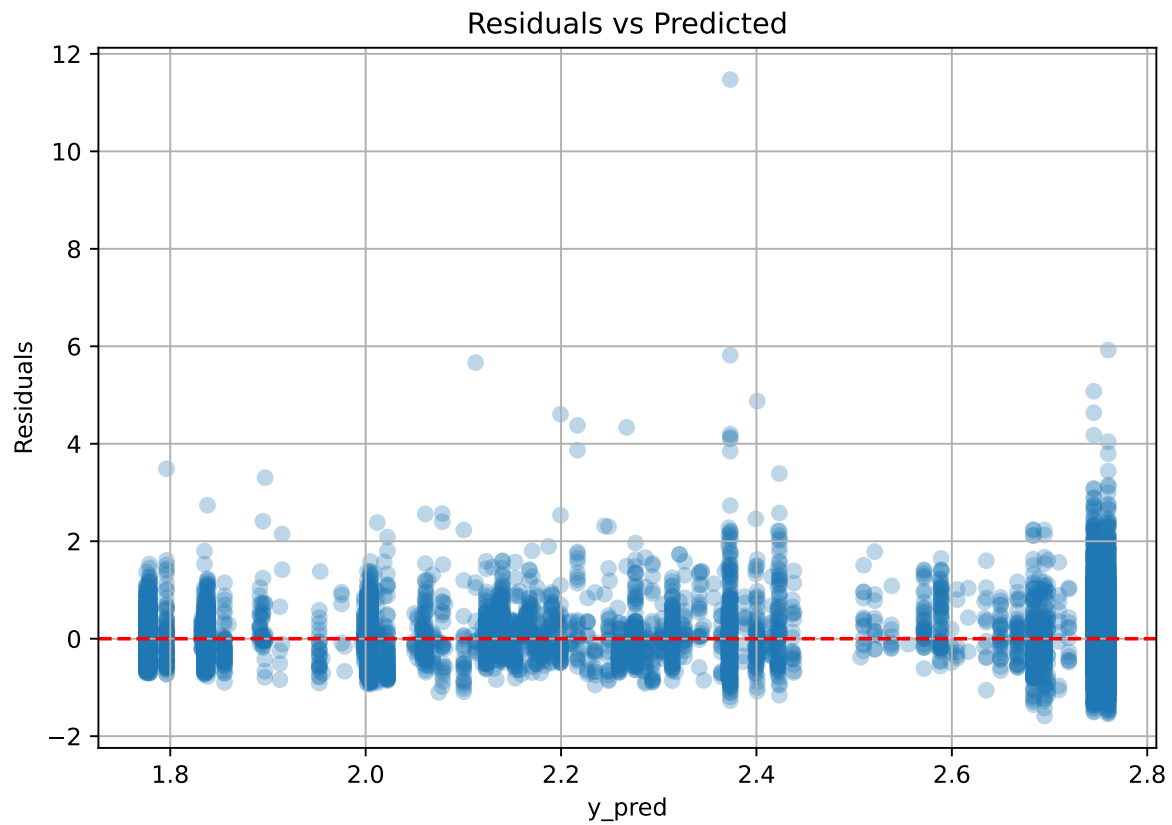
Rank 9: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.0102744387908185, 'reg__num_leaves': 43, 'reg__subsample': 0.6547950235586462, 'reg__colsample_bytree': 0.8803762526608212, 'reg__min_child_samples': 43, 'reg__reg_alpha': 7.531306320266263, 'reg__reg_lambda': 9.203937136667461, 'reg__min_split_gain': 4.887987489918337, 'reg__path_smooth': 4.895210184324342}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.03275934051027742, 'reg__num_leaves': 48, 'reg__subsample': 0.6006245050436299, 'reg__colsample_bytree': 0.8075944163568719, 'reg__min_child_samples': 40, 'reg__reg_alpha': 9.838619887081869, 'reg__reg_lambda': 40.33605823440443, 'reg__min_split_gain': 4.179332886461746, 'reg__path_smooth': 3.906697935591697}

2.4.2 Scatter



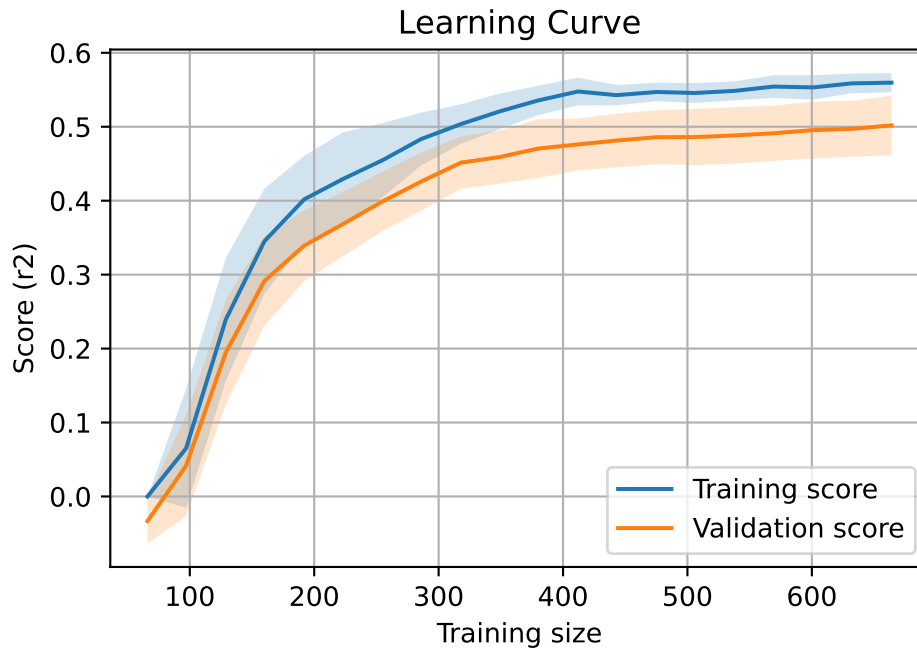
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 3

`Len(group)` : 1356, `Len(training)` : 950,

`Len(training_real)` : 665, `Len(test_of_training)` : 285, `Len(test)` : 406

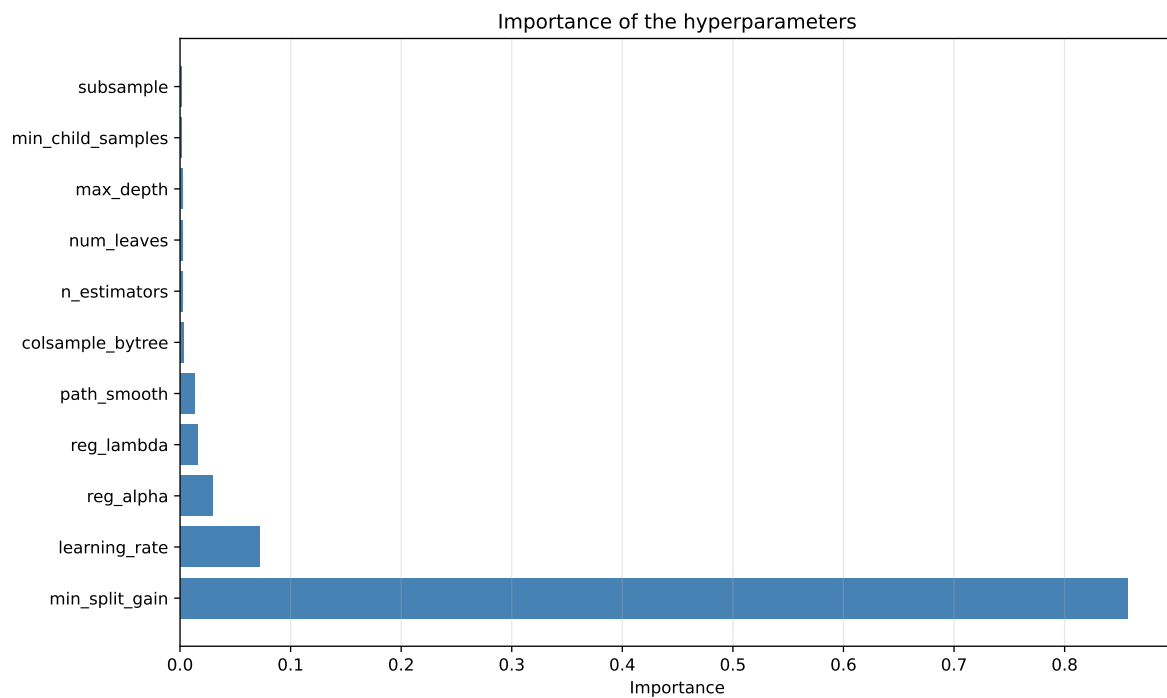
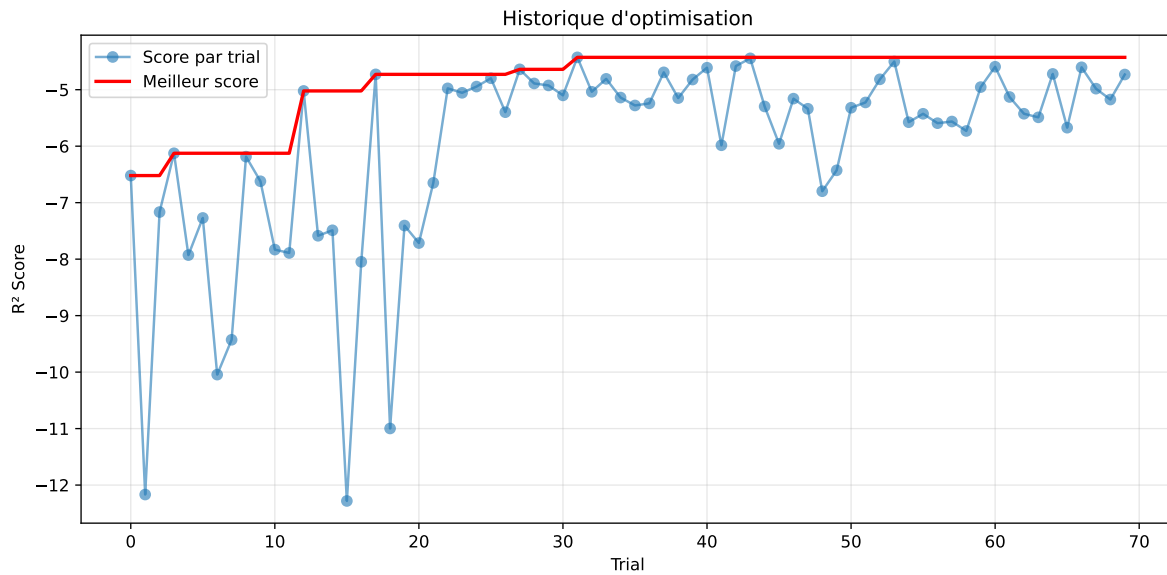


2.4.5 Gridsearch — Group: 4

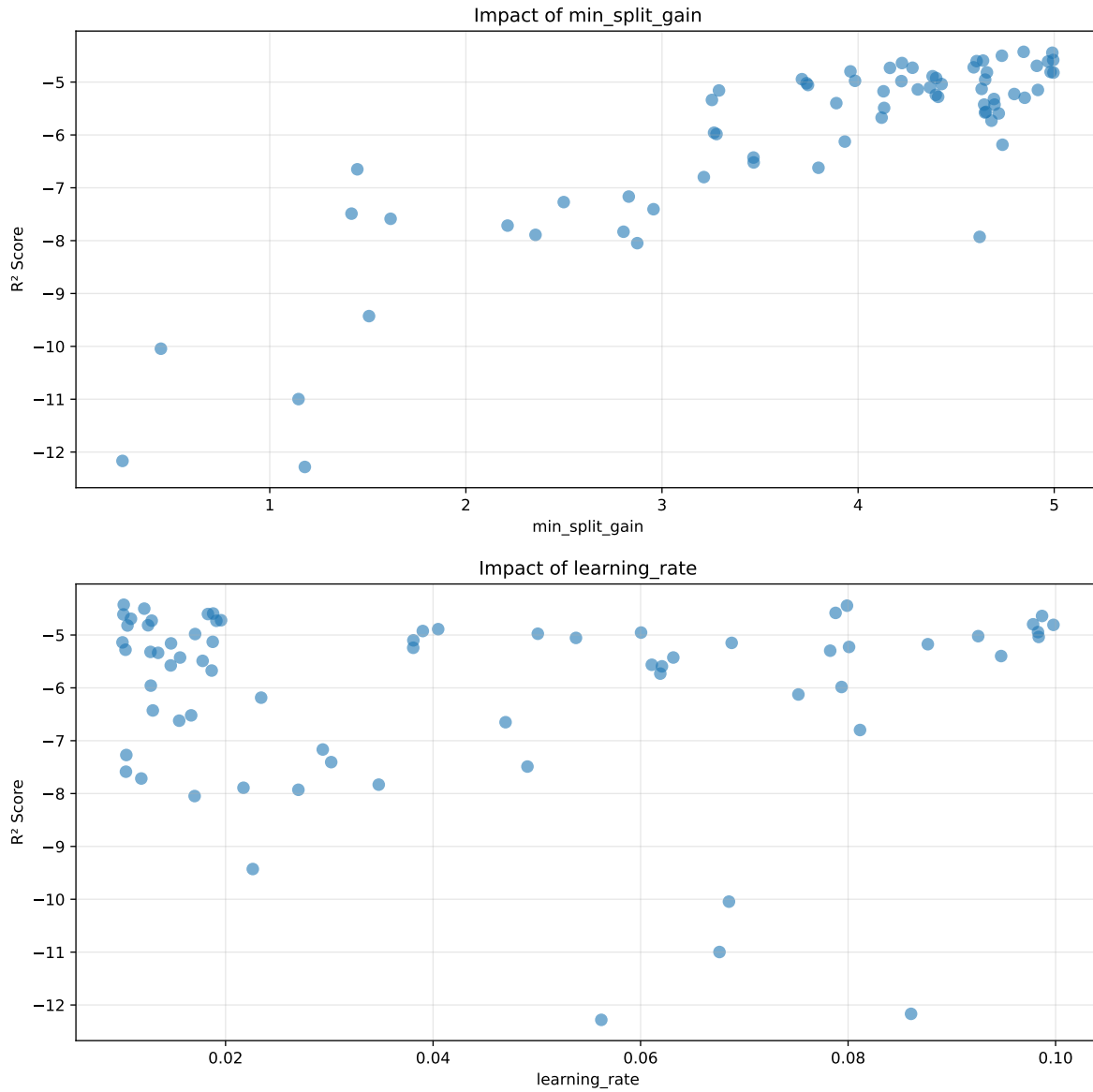
Len(group) : 1360, Len(training) : 952, Len(test) : 408

Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383
y_test: mean=2.202, std=0.791, min=1.062, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 218 r_THI 135 r_CO2 108 r_NH3 76 r_temperature 65

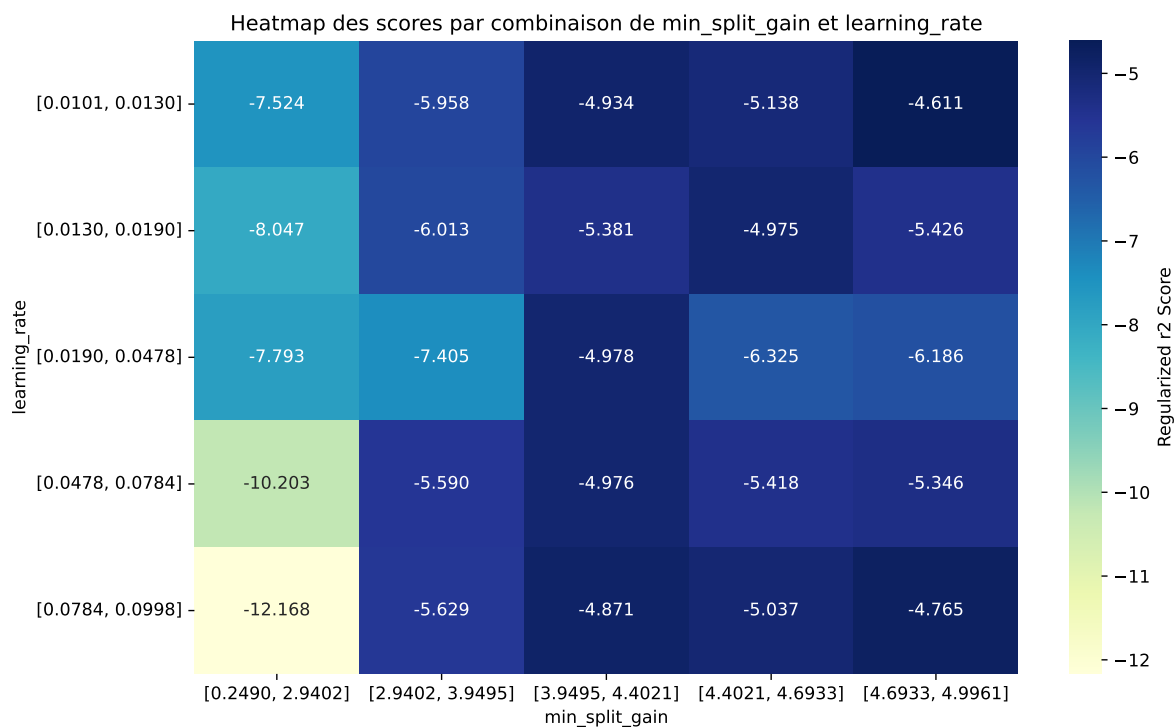


Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3354	0.2883	0.2976	0.0381	-4.4275	0.3467	0.0584	1.2027	1.1636
2	0.3267	0.2795	0.2946	0.04	-4.4451	0.3435	0.064	1.2291	1.1691
3	0.3357	0.2878	0.2985	0.0377	-4.5014	0.3587	0.0709	1.2463	1.1664
4	0.3248	0.2763	0.285	0.042	-4.5818	0.3424	0.0661	1.2394	1.1759
5	0.3398	0.2909	0.302	0.0371	-4.5936	0.3543	0.0634	1.2179	1.1679
6	0.3385	0.2896	0.2998	0.0379	-4.6033	0.3521	0.0625	1.2158	1.169
7	0.3429	0.2939	0.3041	0.0382	-4.6108	0.3589	0.065	1.221	1.1669
8	0.3362	0.287	0.302	0.0403	-4.6399	0.3474	0.0605	1.2107	1.1717
9	0.3455	0.2956	0.3065	0.0379	-4.6927	0.3622	0.0665	1.225	1.1687
10	0.3441	0.2939	0.3055	0.0376	-4.7207	0.3653	0.0714	1.2428	1.1706
---	---	---	---	---	---	---	---	---	---
70	0.5505	0.4234	0.4526	0.0452	-12.2814	0.4811	0.0576	1.136	1.3

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.010184999062020484, 'reg__num_leaves': 32, 'reg__subsample': 0.5798214639591488, 'reg__colsample_bytree': 0.7319415680725954, 'reg__min_child_samples': 35, 'reg__reg_alpha': 7.847030661757831, 'reg__reg_lambda': 21.932792518670595, 'reg__min_split_gain': 4.844070871678697, 'reg__path_smooth': 4.956386866380026}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.07989450387165228, 'reg__num_leaves': 31, 'reg__subsample': 0.5465790371192383, 'reg__colsample_bytree': 0.5723788242600937, 'reg__min_child_samples': 33, 'reg__reg_alpha': 8.96070794608733, 'reg__reg_lambda': 19.299777242576646, 'reg__min_split_gain': 4.990298387362311, 'reg__path_smooth': 4.723907853793928}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.012164447973360906, 'reg__num_leaves': 39, 'reg__subsample': 0.5684855611160954, 'reg__colsample_bytree': 0.7986129547084247, 'reg__min_child_samples': 28, 'reg__reg_alpha': 8.883549256429092, 'reg__reg_lambda': 12.976195697831558, 'reg__min_split_gain': 4.733048211032517, 'reg__path_smooth': 4.213384523333875}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.07879505434361649, 'reg__num_leaves': 58, 'reg__subsample': 0.5479301145606894, 'reg__colsample_bytree': 0.8129672652031376, 'reg__min_child_samples': 22, 'reg__reg_alpha': 8.93600013213734, 'reg__reg_lambda': 32.93571260344295, 'reg__min_split_gain': 4.994992142330687, 'reg__path_smooth': 4.511250982593304}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.018803351062379736, 'reg__num_leaves': 38, 'reg__subsample': 0.5246437084600596, 'reg__colsample_bytree': 0.8475509728883217, 'reg__min_child_samples': 38, 'reg__reg_alpha': 8.600768560901688, 'reg__reg_lambda': 13.795780571660751, 'reg__min_split_gain': 4.636990704063729, 'reg__path_smooth': 3.3439529795936815}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.01829978269638991, 'reg__num_leaves': 40, 'reg__subsample': 0.5287472159220981, 'reg__colsample_bytree': 0.8386995793337089, 'reg__min_child_samples': 31, 'reg__reg_alpha': 8.358199004548101, 'reg__reg_lambda': 20.967164369067387, 'reg__min_split_gain': 4.604228734280683, 'reg__path_smooth': 4.5623552960499785}

Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.010153938608418043, 'reg__num_leaves': 28, 'reg__subsample': 0.5448284694806217, 'reg__colsample_bytree': 0.8068579366850935, 'reg__min_child_samples': 34, 'reg__reg_alpha': 7.030443529862483, 'reg__reg_lambda': 18.010039885295278, 'reg__min_split_gain': 4.968282546559053, 'reg__path_smooth': 3.292436675666096}

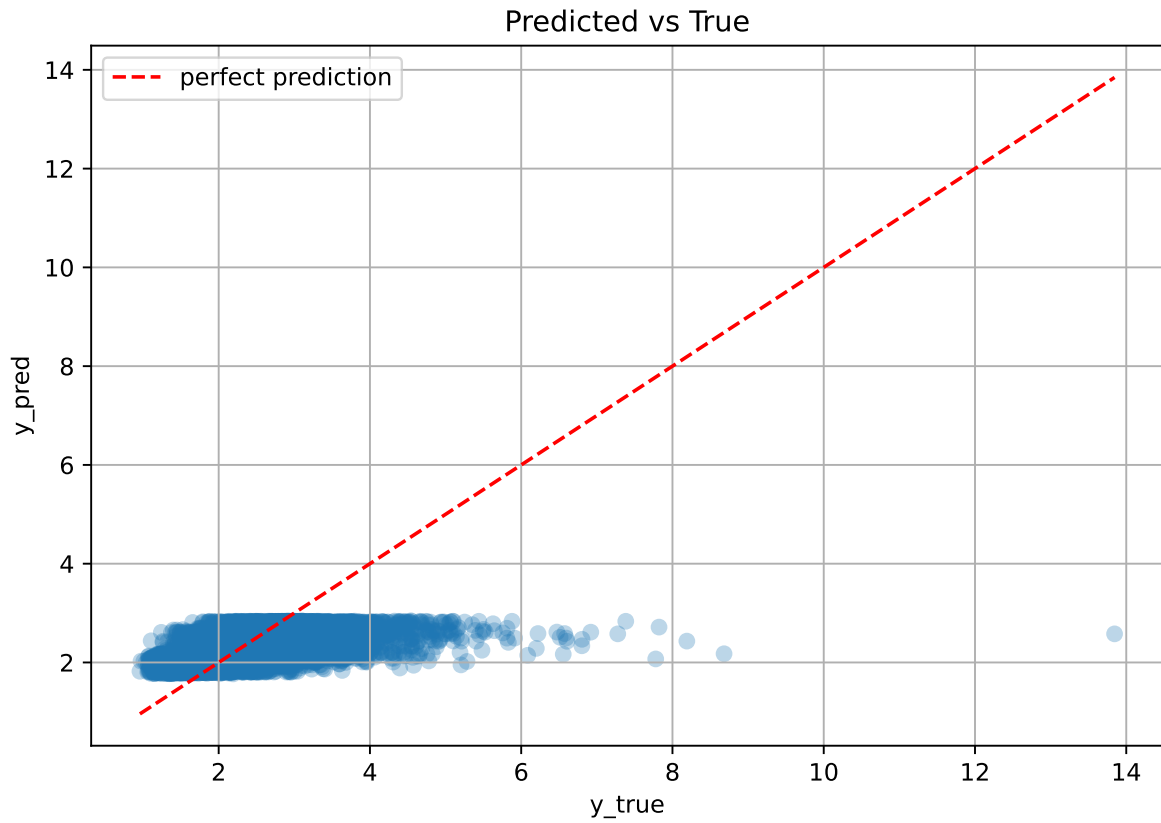
Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.09869299729477107, 'reg__num_leaves': 62, 'reg__subsample': 0.6070869046404893, 'reg__colsample_bytree': 0.6958725786892174, 'reg__min_child_samples': 24, 'reg__reg_alpha': 9.739388437118357, 'reg__reg_lambda': 28.990391738168206, 'reg__min_split_gain': 4.223630510942828, 'reg__path_smooth': 1.9516930186424362}

Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.010881577930566306, 'reg__num_leaves': 33, 'reg__subsample': 0.5018405182913562,

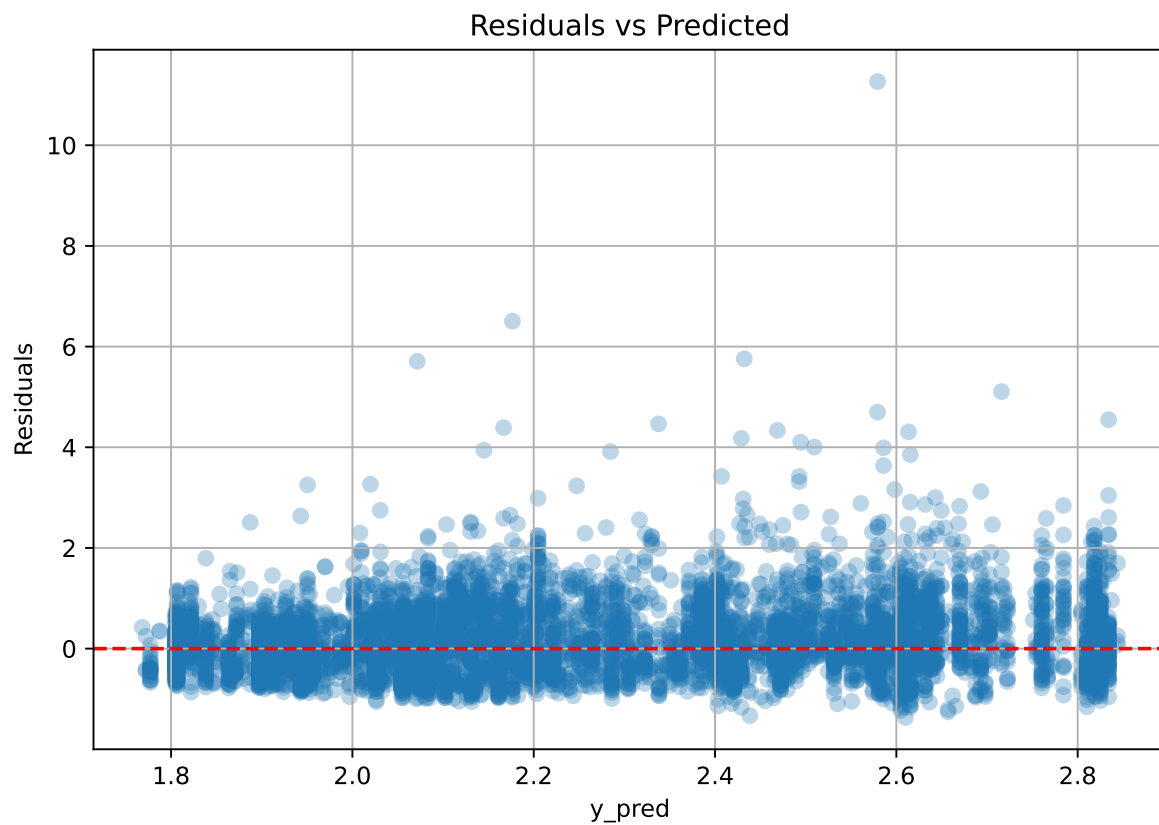
'reg__colsample_bytree': 0.7326758752815664, 'reg__min_child_samples': 34, 'reg__reg_alpha': 6.862261543343005, 'reg__reg_lambda': 17.82170333370101, 'reg__min_split_gain': 4.911262688250859, 'reg__path_smooth': 4.244684461550833}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.01956521459643554, 'reg__num_leaves': 24, 'reg__subsample': 0.5258510436445203, 'reg__colsample_bytree': 0.7899237902069431, 'reg__min_child_samples': 31, 'reg__reg_alpha': 8.422203313145365, 'reg__reg_lambda': 13.669791705483714, 'reg__min_split_gain': 4.589351450213025, 'reg__path_smooth': 3.310288546409436}

2.4.6 Scatter



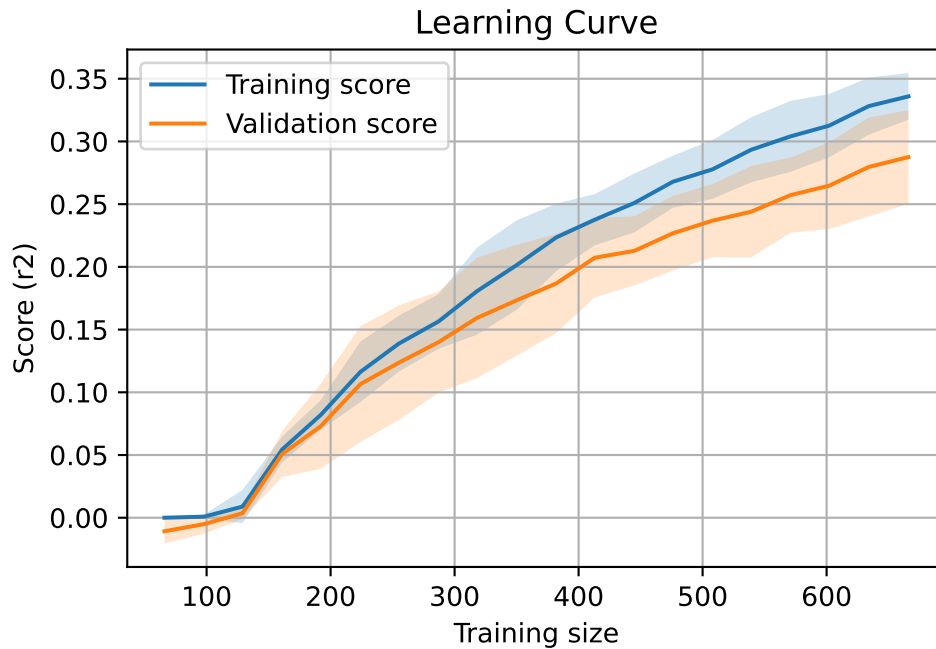
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 4

`Len(group)` : 1360, `Len(training)` : 952,

`Len(training_real)` : 667, `Len(test_of_training)` : 285, `Len(test)` : 408

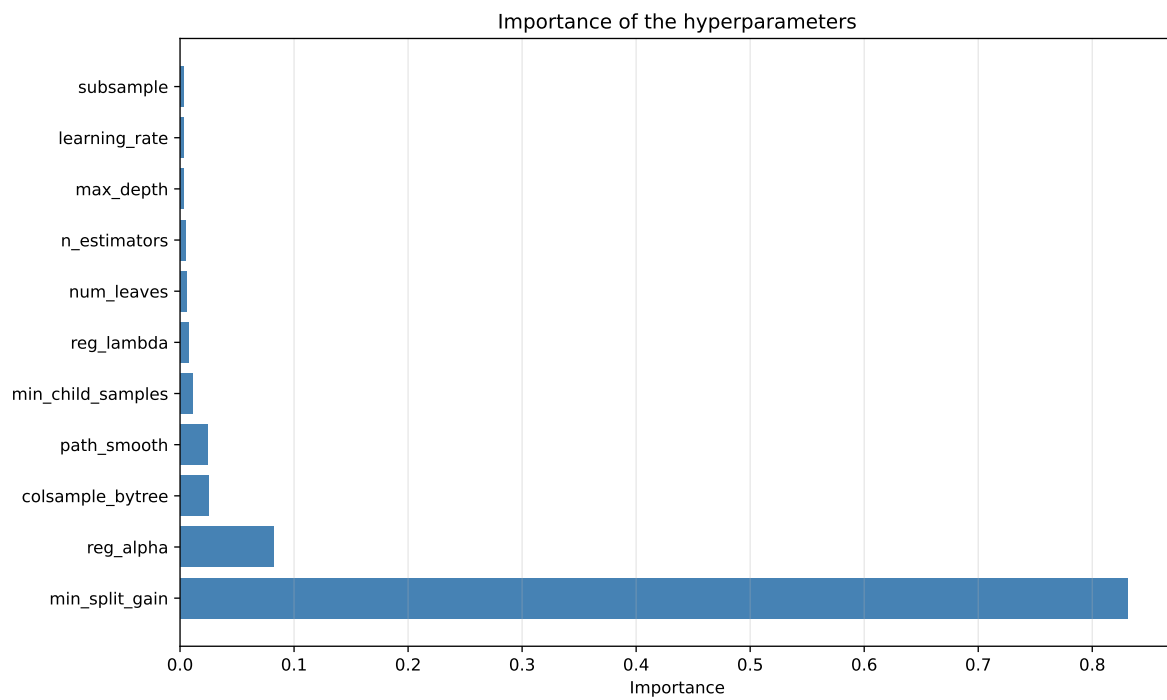
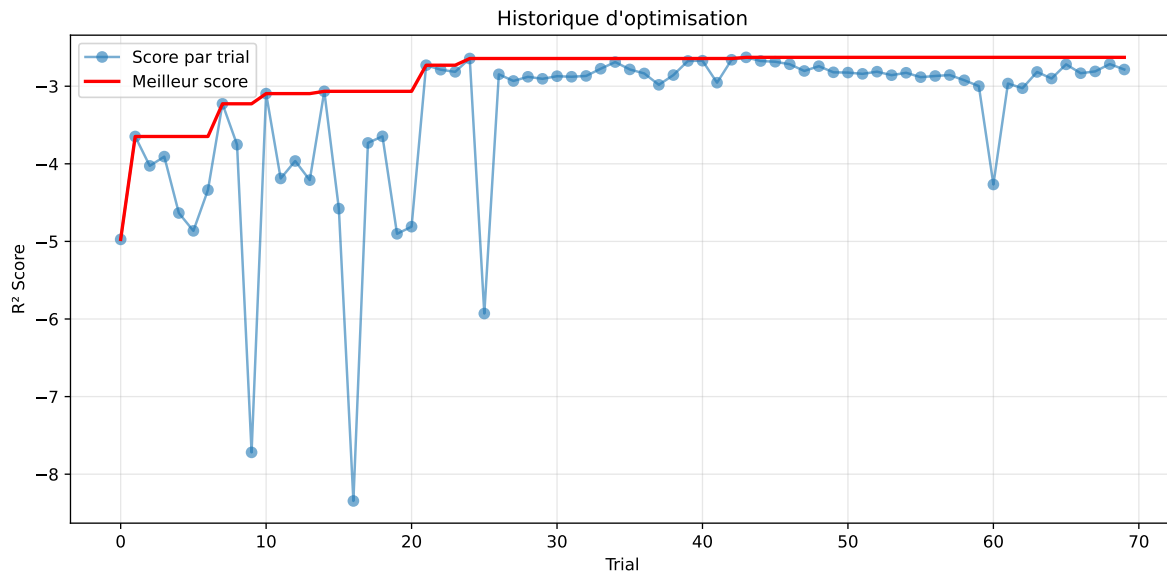


2.4.9 Gridsearch — Group: 5

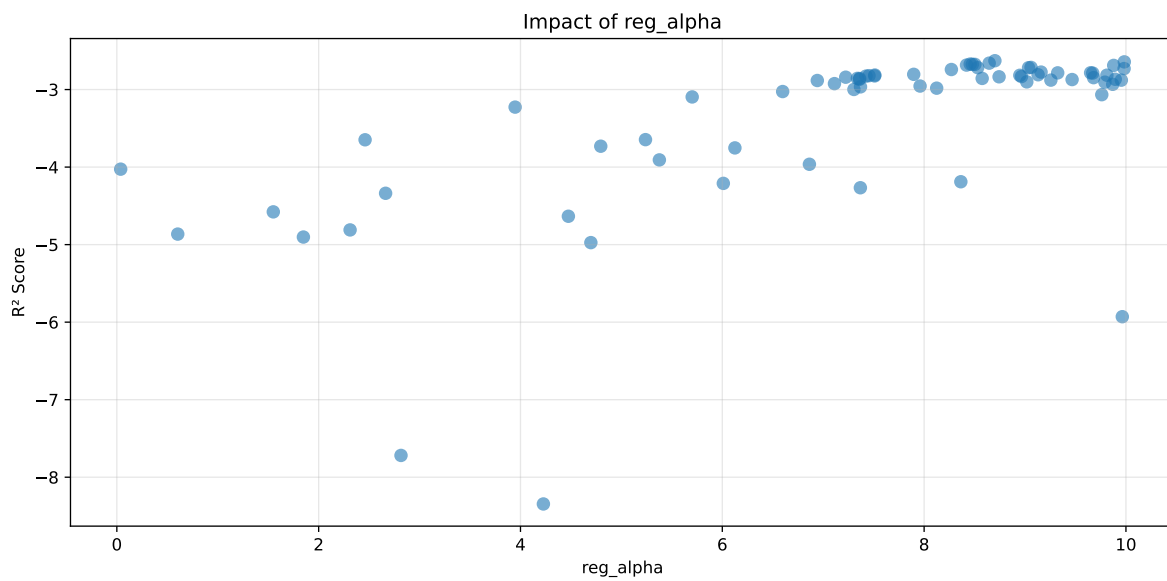
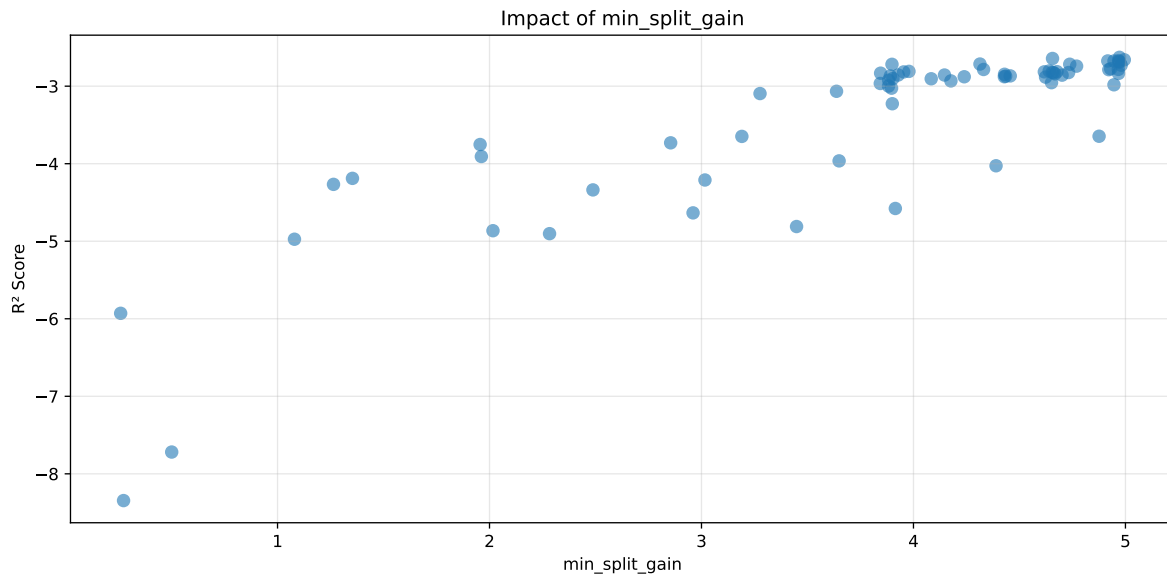
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 278 r_NH3 67 r_temperature 61 r_THI 4 r_CO2 2

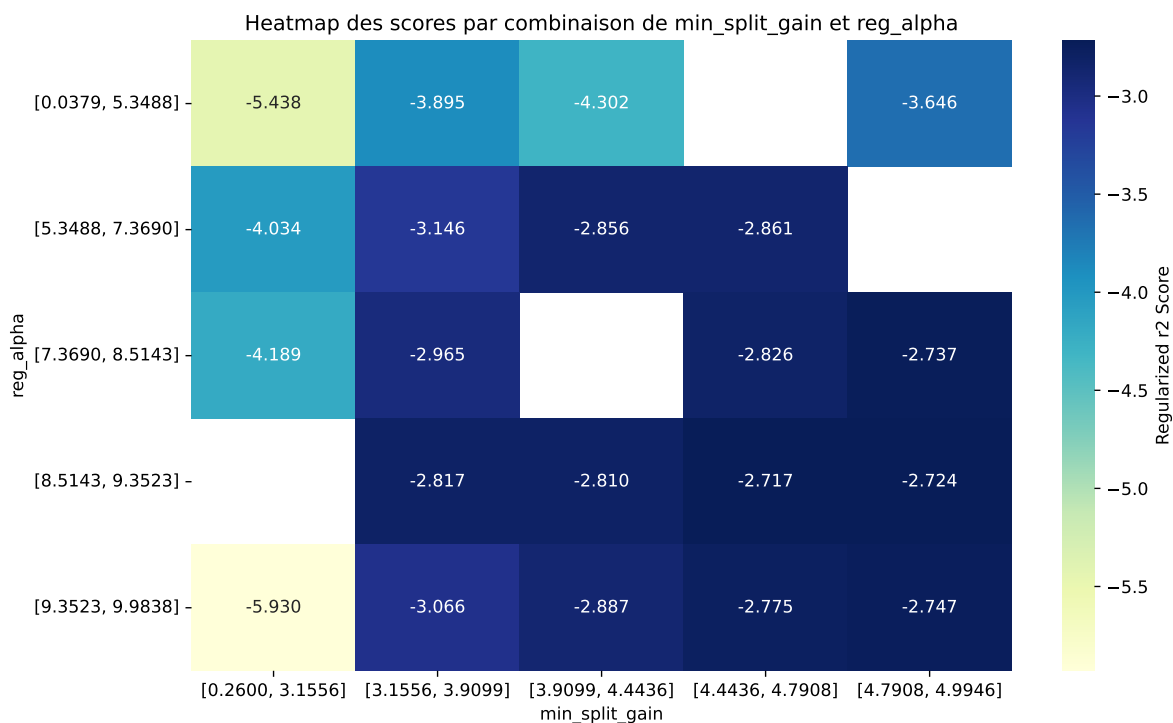


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.412	0.3819	0.4379	0.0216	-2.628	0.4591	0.0773	1.2024	1.0788
2	0.4131	0.3828	0.4387	0.0221	-2.6428	0.4597	0.0769	1.2009	1.079
3	0.4127	0.3823	0.4384	0.0215	-2.6594	0.4589	0.0767	1.2006	1.0796
4	0.4132	0.3827	0.439	0.0217	-2.6712	0.4597	0.0771	1.2014	1.0798
5	0.4138	0.3832	0.4394	0.0219	-2.6745	0.4608	0.0776	1.2025	1.0798
6	0.4137	0.3831	0.4392	0.0221	-2.6752	0.4607	0.0776	1.2027	1.0798
7	0.4137	0.383	0.4394	0.0221	-2.6844	0.4612	0.0781	1.204	1.0801
8	0.409	0.3783	0.4375	0.0224	-2.6885	0.4595	0.0812	1.2146	1.0811
9	0.4162	0.3852	0.4419	0.023	-2.7145	0.4642	0.079	1.2051	1.0805
10	0.4151	0.3841	0.4406	0.0223	-2.7171	0.4623	0.0782	1.2036	1.0807
---	---	---	---	---	---	---	---	---	---
70	0.5543	0.4662	0.5315	0.0246	-8.3451	0.5485	0.0823	1.1766	1.189

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.012346296378940995, 'reg__num_leaves': 63, 'reg__subsample': 0.585980382485345, 'reg__colsample_bytree': 0.742685312696042, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.701398668862716, 'reg__reg_lambda': 19.757196114670244, 'reg__min_split_gain': 4.971744344720629, 'reg__path_smooth': 0.8903184303963092}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.011741301772207018, 'reg__num_leaves': 37, 'reg__subsample': 0.6384379080967583, 'reg__colsample_bytree': 0.7555429155926868, 'reg__min_child_samples': 49, 'reg__reg_alpha': 9.983755109726673, 'reg__reg_lambda': 15.053991129966988, 'reg__min_split_gain': 4.655953913604049, 'reg__path_smooth': 0.0419664526653567}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.010830520329289687, 'reg__num_leaves': 60, 'reg__subsample': 0.5866858467558462, 'reg__colsample_bytree': 0.7384948008119145, 'reg__min_child_samples': 43, 'reg__reg_alpha': 8.644811659910273, 'reg__reg_lambda': 18.228909122535995, 'reg__min_split_gain': 4.994629122877174, 'reg__path_smooth': 0.7677838281466649}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.011649406614080903, 'reg__num_leaves': 62, 'reg__subsample': 0.5825391692448816, 'reg__colsample_bytree': 0.740539624269215, 'reg__min_child_samples': 45, 'reg__reg_alpha': 8.456686044621197, 'reg__reg_lambda': 18.36965995881826, 'reg__min_split_gain': 4.967692259678956, 'reg__path_smooth': 0.8110369938824995}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.01058010694925425, 'reg__num_leaves': 62, 'reg__subsample': 0.5933099866881468, 'reg__colsample_bytree': 0.7357419109980462, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.47782186606547, 'reg__reg_lambda': 18.233813381679244, 'reg__min_split_gain': 4.9166606979656535, 'reg__path_smooth': 0.709867473138493}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.012198507668417362, 'reg__num_leaves': 21, 'reg__subsample': 0.5810640089058821, 'reg__colsample_bytree': 0.7357787982607292, 'reg__min_child_samples': 42, 'reg__reg_alpha': 8.503323839932248, 'reg__reg_lambda': 18.513736137621326, 'reg__min_split_gain': 4.945973426394822, 'reg__path_smooth': 0.8684878451413046}

Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.012567022331850315, 'reg__num_leaves': 20, 'reg__subsample': 0.5889554138951792, 'reg__colsample_bytree': 0.730649972336225, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.42022564866521, 'reg__reg_lambda': 17.442982886868375, 'reg__min_split_gain': 4.968992305995531, 'reg__path_smooth': 0.8356334503564744}

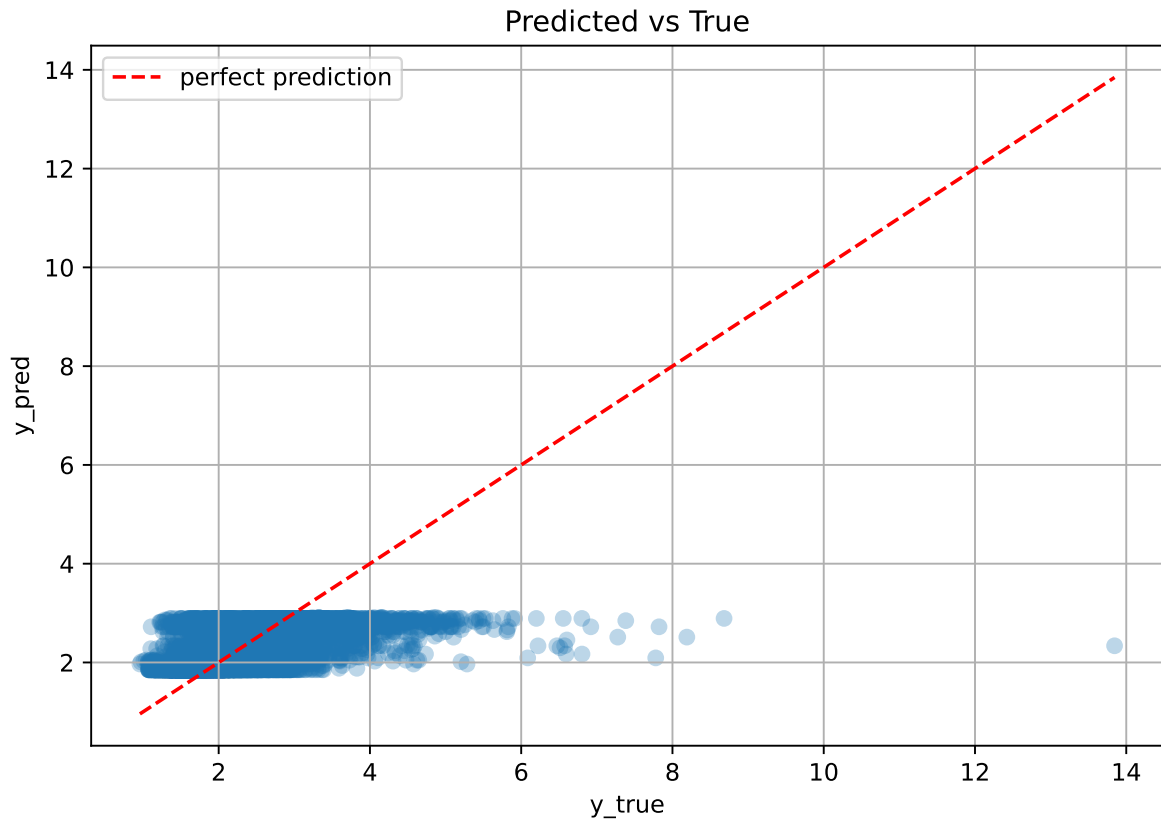
Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.03529583368281901, 'reg__num_leaves': 15, 'reg__subsample': 0.8883211817590364, 'reg__colsample_bytree': 0.6290964155639158, 'reg__min_child_samples': 50, 'reg__reg_alpha': 9.878496413014606, 'reg__reg_lambda': 14.664583069724655, 'reg__min_split_gain': 4.9730669493030275, 'reg__path_smooth': 0.2080671980788465}

Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.01777901475151098, 'reg__num_leaves': 58, 'reg__subsample': 0.6518344419893234,

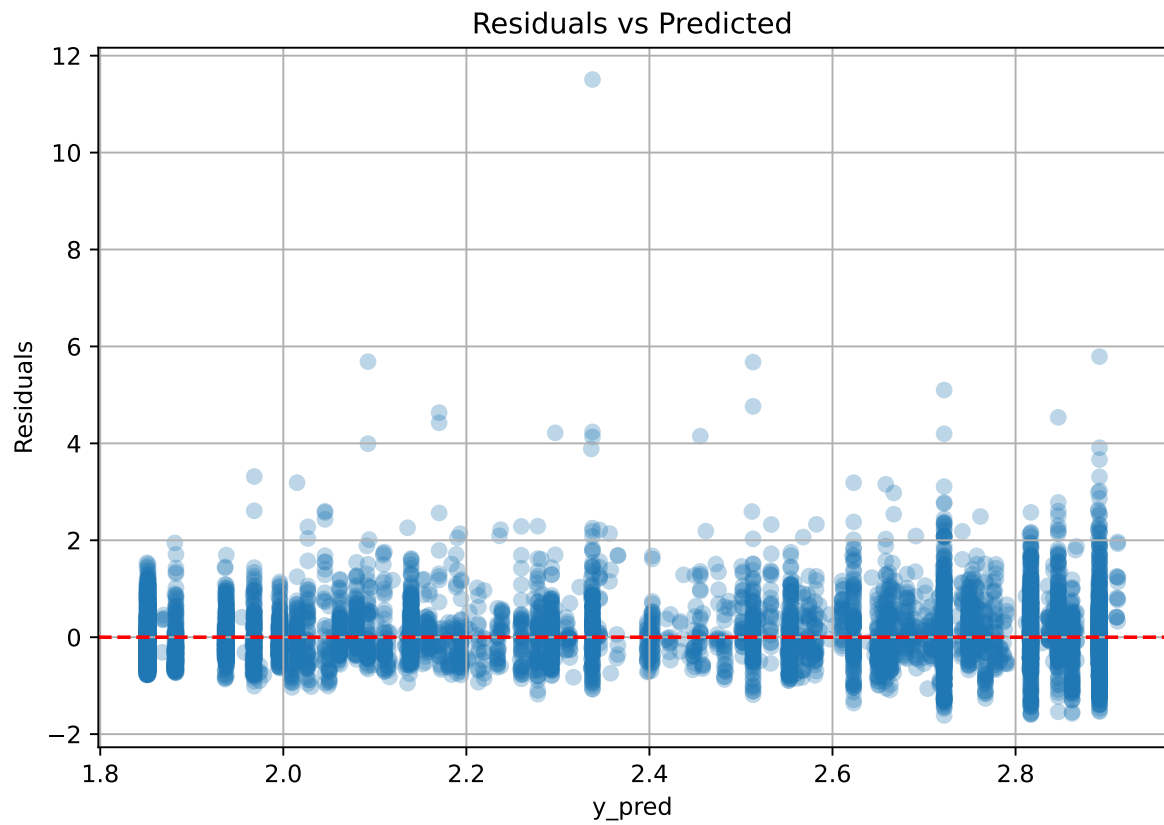
'reg__colsample_bytree': 0.7035693686383327, 'reg__min_child_samples': 34, 'reg__reg_alpha': 9.05949120204991, 'reg__reg_lambda': 23.221913253421935, 'reg__min_split_gain': 4.313698098898418, 'reg__path_smooth': 1.5004923914313526}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.012444986919483989, 'reg__num_leaves': 21, 'reg__subsample': 0.5840931897320187, 'reg__colsample_bytree': 0.7384663392795004, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.530820140241342, 'reg__reg_lambda': 18.912069131569382, 'reg__min_split_gain': 4.737189255673911, 'reg__path_smooth': 0.9184324911235392}

2.4.10 Scatter



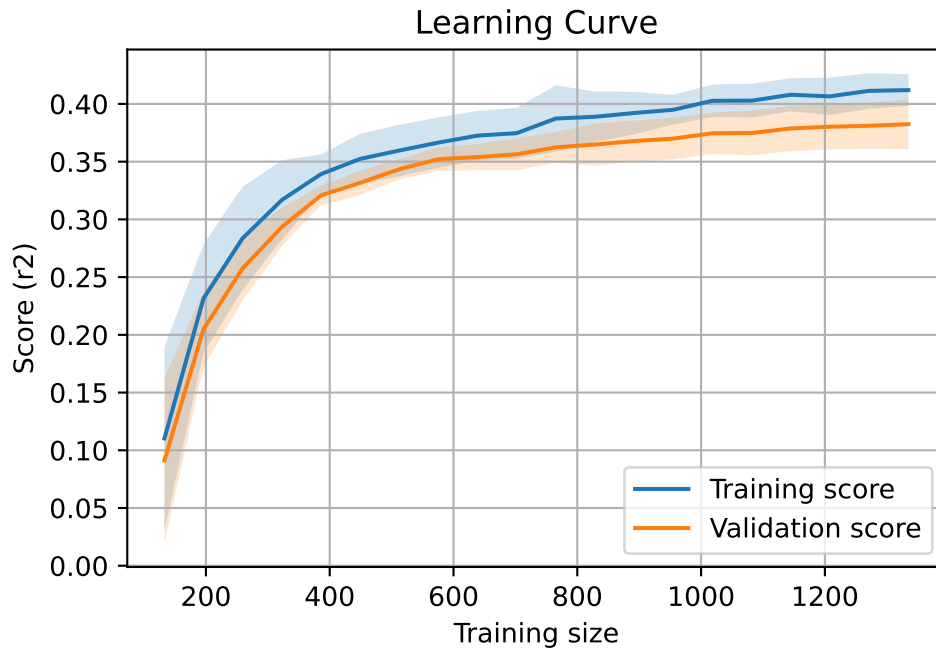
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

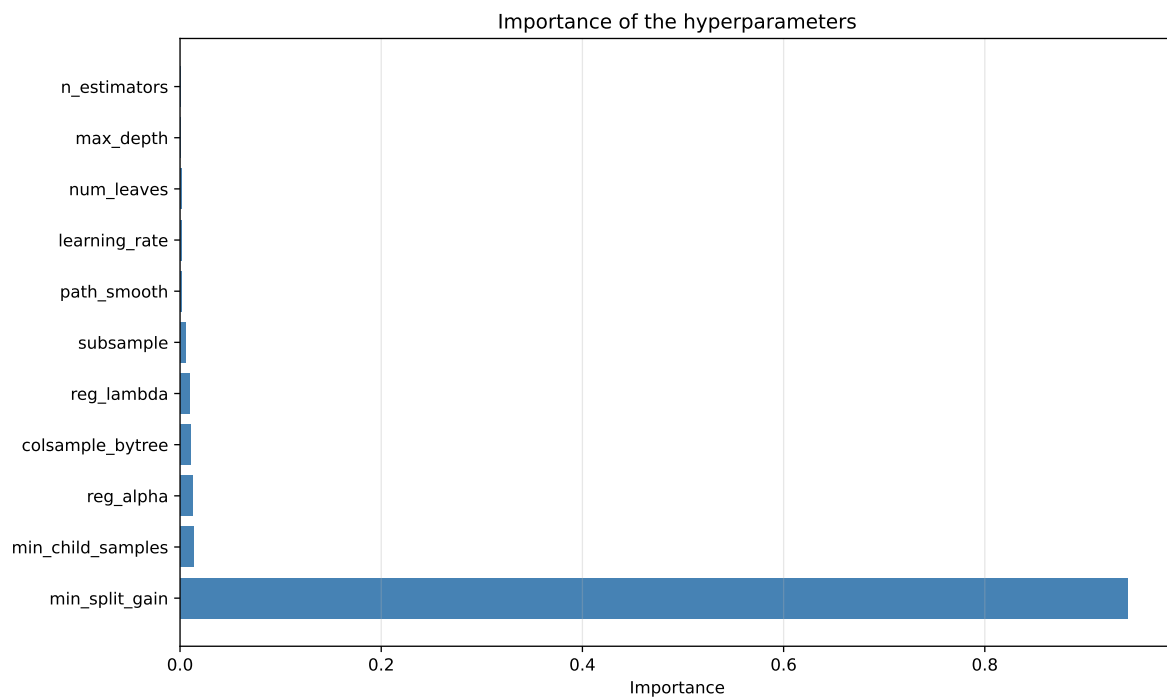
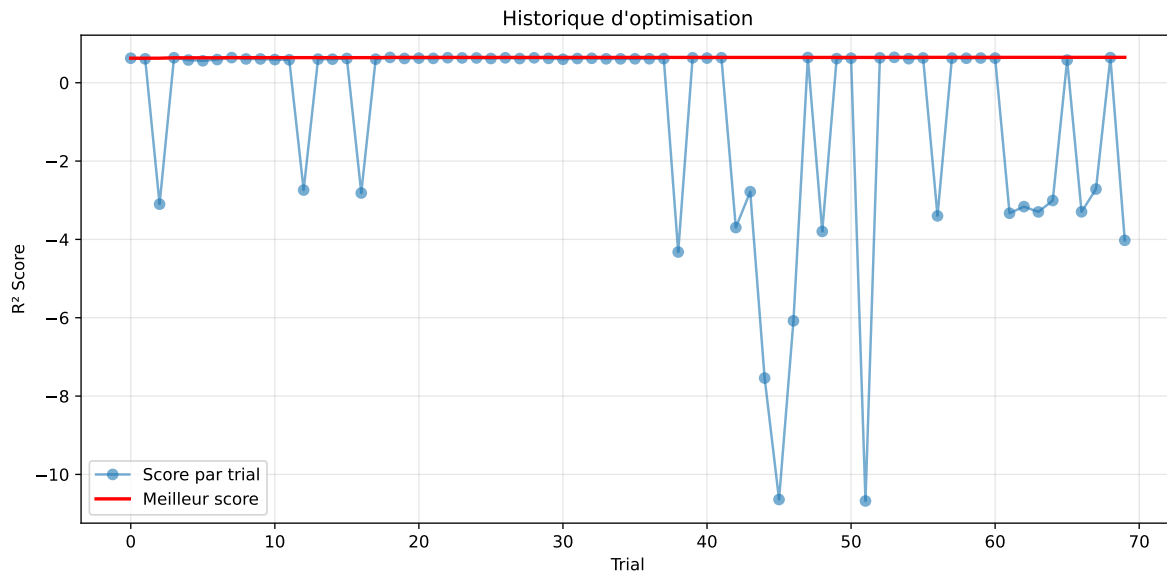


2.4.13 Gridsearch — Group: 11

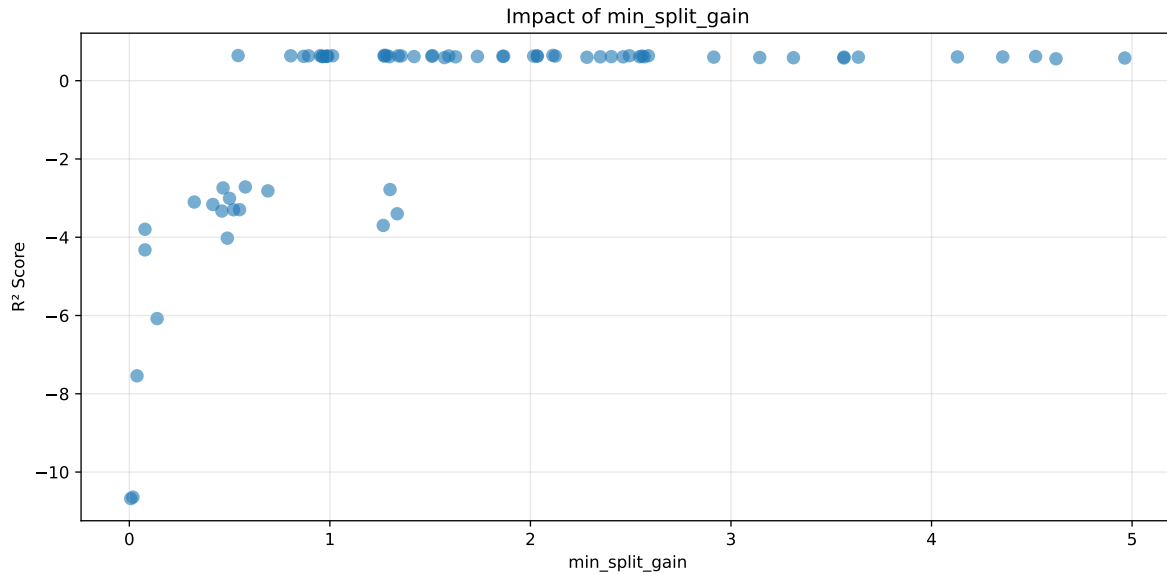
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity 496 r_NH3 484 r_CO2 408 r_temperature 197 r_THI 178



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6776	0.6481	0.5498	0.0121	0.6481	0.5688	-0.0793	0.8776	1.0455
2	0.6774	0.6462	0.5507	0.0129	0.6462	0.5667	-0.0795	0.8769	1.0482
3	0.6759	0.6444	0.5501	0.0133	0.6444	0.569	-0.0754	0.8831	1.0488
4	0.6691	0.6419	0.548	0.0122	0.6419	0.5649	-0.077	0.88	1.0424
5	0.6657	0.6411	0.5444	0.013	0.6411	0.5606	-0.0805	0.8744	1.0383
6	0.6586	0.6389	0.5454	0.0153	0.6389	0.5586	-0.0803	0.8744	1.0309
7	0.6632	0.6383	0.5416	0.0127	0.6383	0.559	-0.0793	0.8757	1.039
8	0.6649	0.637	0.542	0.0123	0.637	0.5612	-0.0758	0.881	1.0438
9	0.6595	0.6369	0.5394	0.0131	0.6369	0.5578	-0.0791	0.8758	1.0355
10	0.6568	0.6363	0.5417	0.0131	0.6363	0.5578	-0.0785	0.8767	1.0322
---	---	---	---	---	---	---	---	---	---
70	0.8096	0.6959	0.5799	0.013	-10.6779	0.5959	-0.1	0.8564	1.1634

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.01896861081172153, 'reg__num_leaves': 43, 'reg__subsample': 0.5516642373123343, 'reg__colsample_bytree': 0.6807201658666423, 'reg__min_child_samples': 44, 'reg__reg_alpha': 0.7642191442184533, 'reg__reg_lambda': 16.899990255868104, 'reg__min_split_gain': 1.2711378626976364, 'reg__path_smooth': 0.9100811497480867}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.041253658304988734, 'reg__num_leaves': 26, 'reg__subsample': 0.6014945819698524,

'reg__colsample_bytree': 0.8679910983969508, 'reg__min_child_samples': 10, 'reg__reg_alpha': 1.190032529998396, 'reg__reg_lambda': 23.083176642559, 'reg__min_split_gain': 2.1117405457295355, 'reg__path_smooth': 2.9117447792712414}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.04767015905949955, 'reg__num_leaves': 21, 'reg__subsample': 0.6912800852393242, 'reg__colsample_bytree': 0.8634960430142502, 'reg__min_child_samples': 11, 'reg__reg_alpha': 4.578737847348739, 'reg__reg_lambda': 20.168096510189052, 'reg__min_split_gain': 1.281482699510496, 'reg__path_smooth': 2.9092896094325646}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.012573782159478554, 'reg__num_leaves': 63, 'reg__subsample': 0.5896068023272449, 'reg__colsample_bytree': 0.5958900268279723, 'reg__min_child_samples': 32, 'reg__reg_alpha': 8.024279840760732, 'reg__reg_lambda': 11.824154992961235, 'reg__min_split_gain': 0.5423977296265672, 'reg__path_smooth': 1.7629535895389807}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.015152191858708879, 'reg__num_leaves': 60, 'reg__subsample': 0.6822192101236729, 'reg__colsample_bytree': 0.8793100835003724, 'reg__min_child_samples': 42, 'reg__reg_alpha': 6.748793923696374, 'reg__reg_lambda': 18.462774414507383, 'reg__min_split_gain': 0.9522521686671159, 'reg__path_smooth': 0.03609559504292925}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.057177104942599054, 'reg__num_leaves': 36, 'reg__subsample': 0.5768792975233522, 'reg__colsample_bytree': 0.5012067084993106, 'reg__min_child_samples': 35, 'reg__reg_alpha': 0.8046520915249964, 'reg__reg_lambda': 31.399495692194414, 'reg__min_split_gain': 1.5121726038040555, 'reg__path_smooth': 0.7025610447522315}

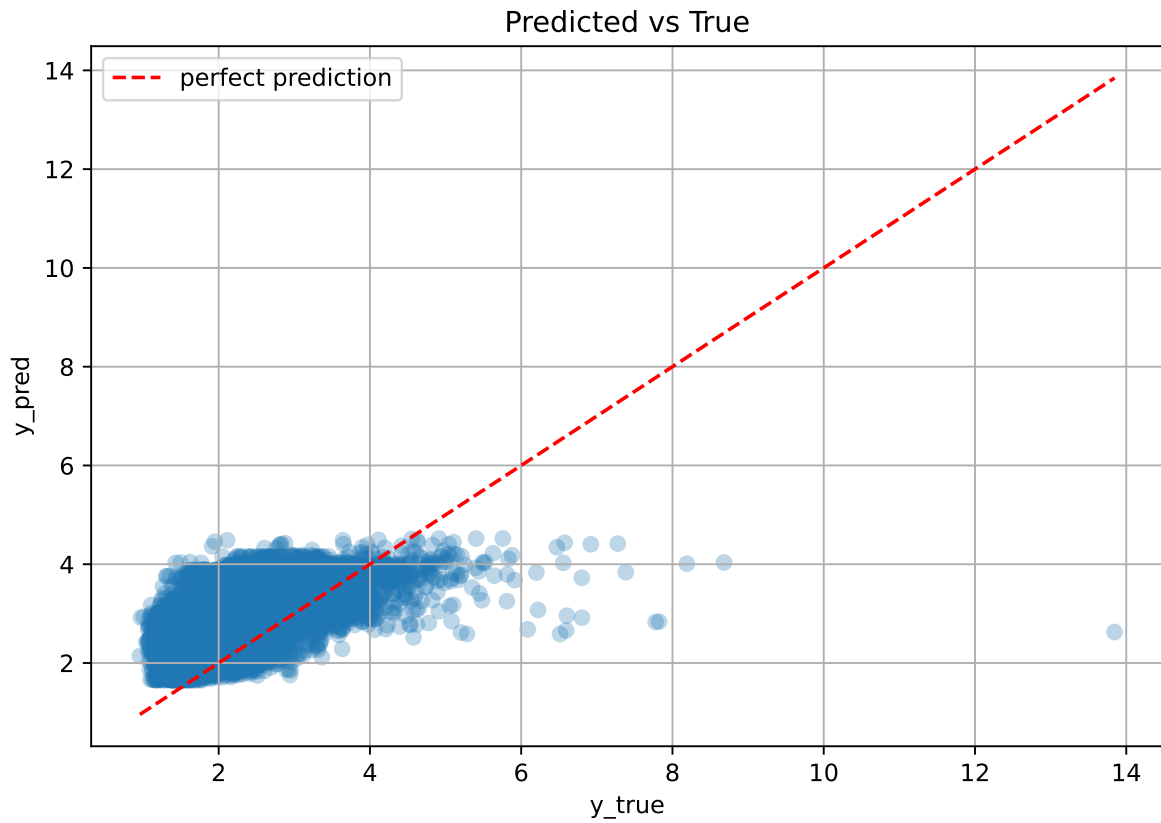
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.04678469362333684, 'reg__num_leaves': 43, 'reg__subsample': 0.5444426298996253, 'reg__colsample_bytree': 0.8548001072859901, 'reg__min_child_samples': 38, 'reg__reg_alpha': 0.9074413494562205, 'reg__reg_lambda': 20.258984852578802, 'reg__min_split_gain': 2.4937212317107686, 'reg__path_smooth': 1.1468612556546338}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.018847073061502576, 'reg__num_leaves': 24, 'reg__subsample': 0.7782500525903702, 'reg__colsample_bytree': 0.7082795846081176, 'reg__min_child_samples': 20, 'reg__reg_alpha': 9.007441318926357, 'reg__reg_lambda': 5.090234840891688, 'reg__min_split_gain': 1.0135025449317658, 'reg__path_smooth': 4.725582514236173}

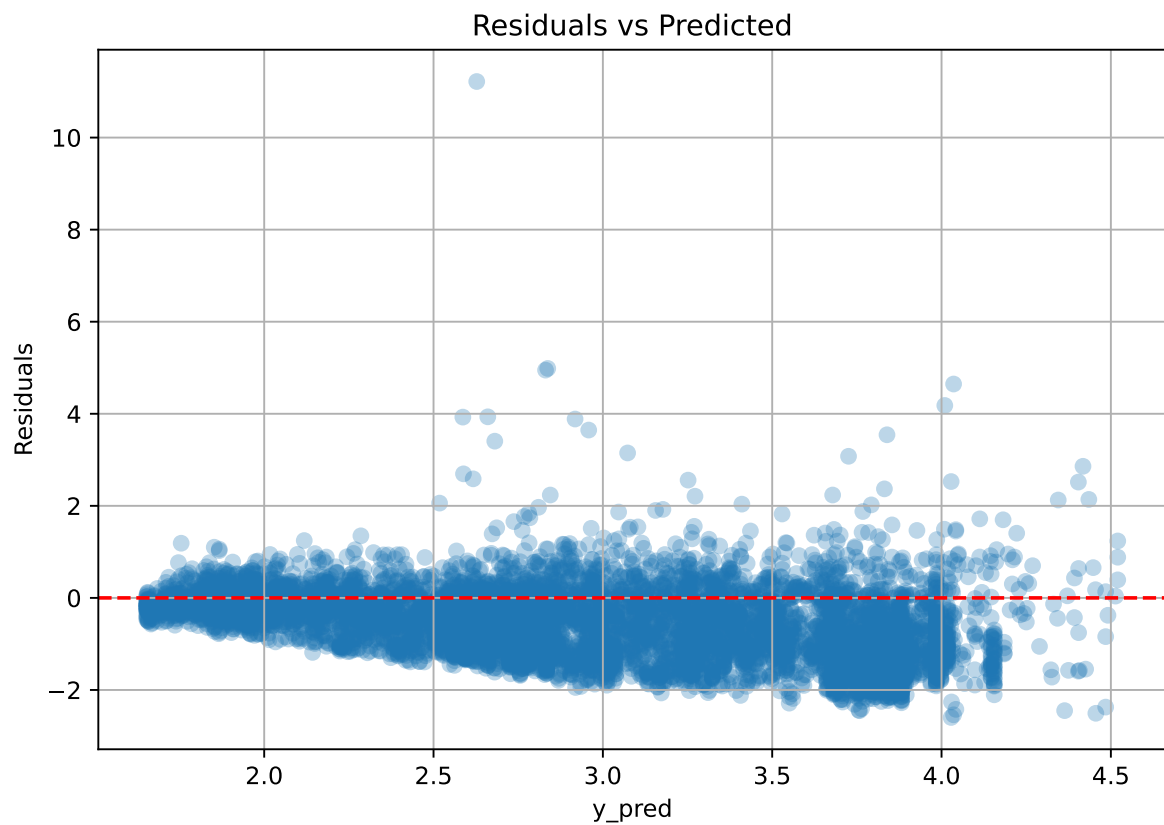
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.047391796494739576, 'reg__num_leaves': 29, 'reg__subsample': 0.5258220581949413, 'reg__colsample_bytree': 0.8598572236158503, 'reg__min_child_samples': 40, 'reg__reg_alpha': 0.9523078106269454, 'reg__reg_lambda': 20.268436420901985, 'reg__min_split_gain': 2.5881463594849916, 'reg__path_smooth': 1.3669550169328584}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.024670752902789285, 'reg__num_leaves': 32, 'reg__subsample': 0.8400677778688526, 'reg__colsample_bytree': 0.8169802877149993, 'reg__min_child_samples': 48, 'reg__reg_alpha': 6.181770072039951, 'reg__reg_lambda': 27.686482036591507, 'reg__min_split_gain': 0.8050335188409607, 'reg__path_smooth': 0.2386684334395235}

2.4.14 Scatter



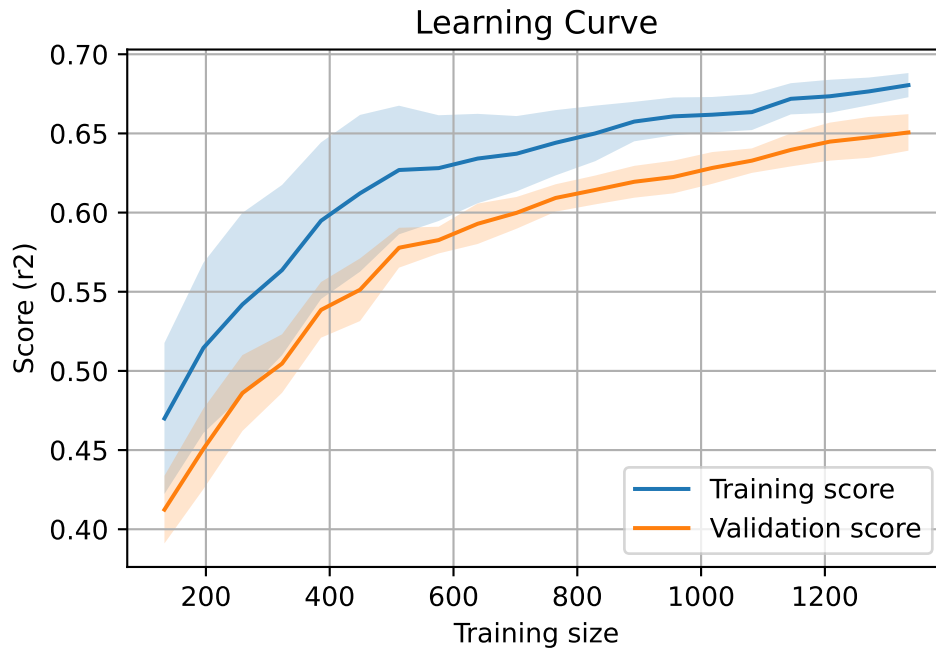
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



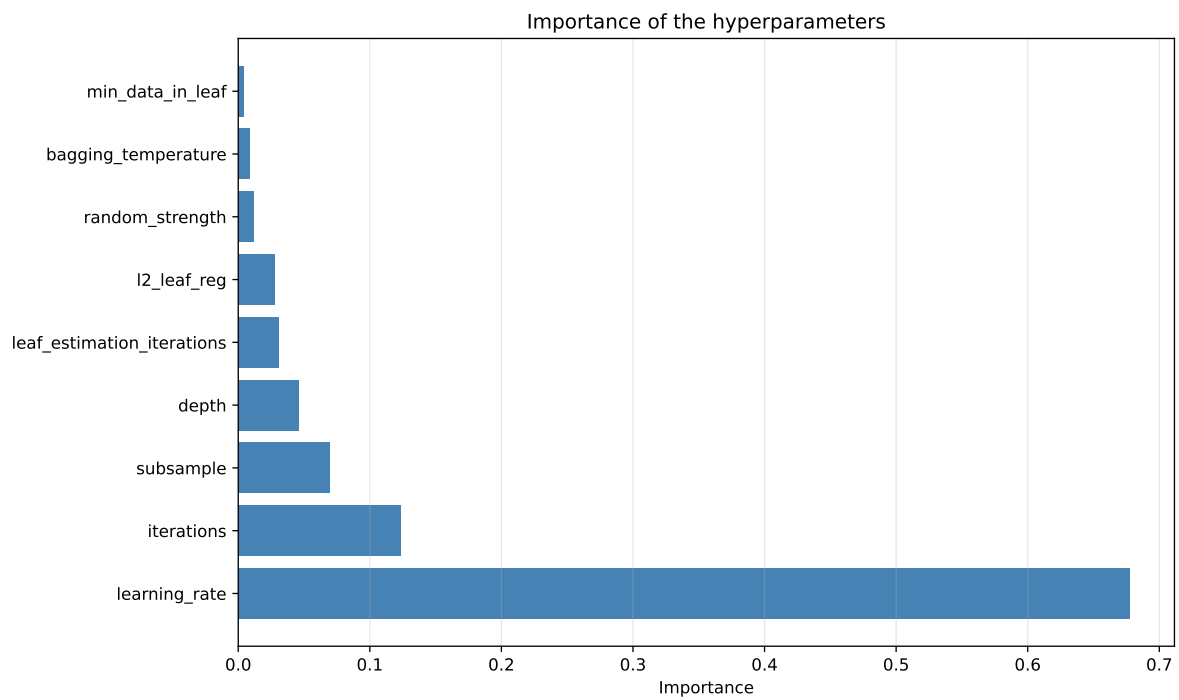
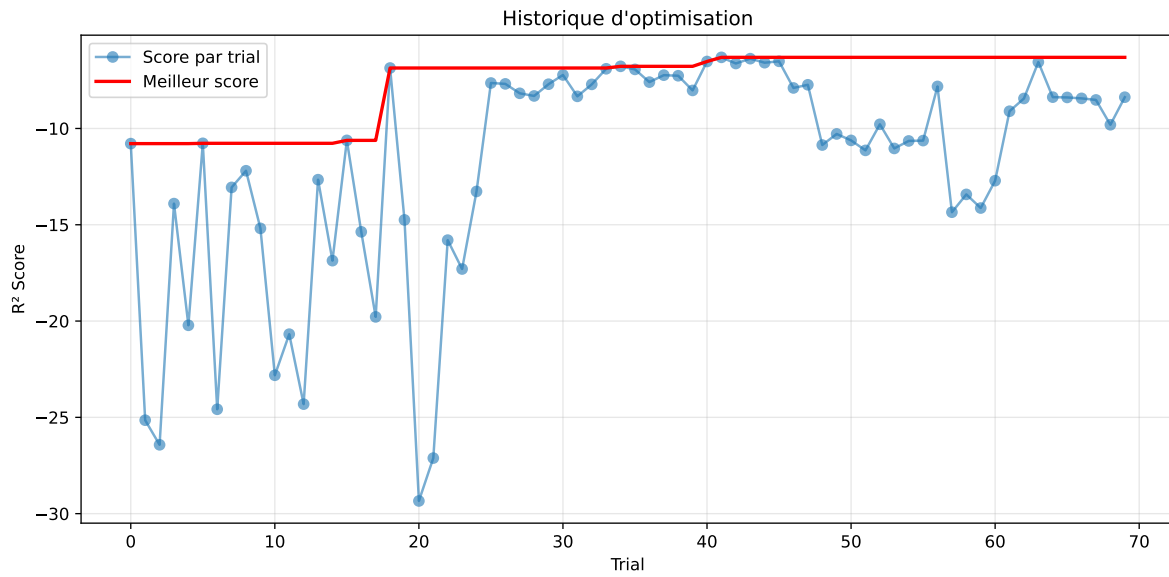
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 3

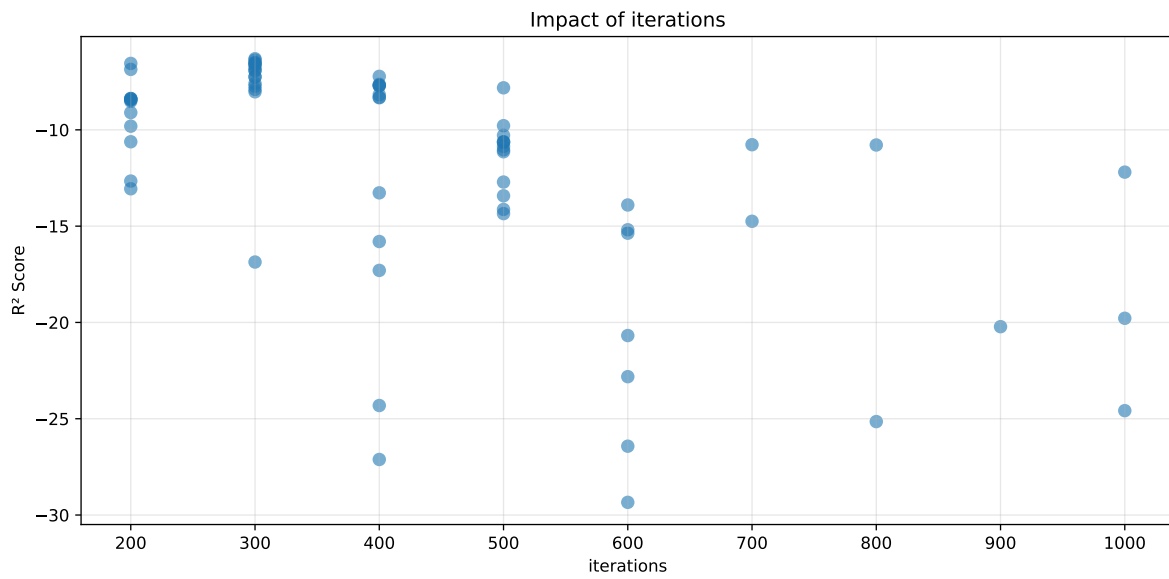
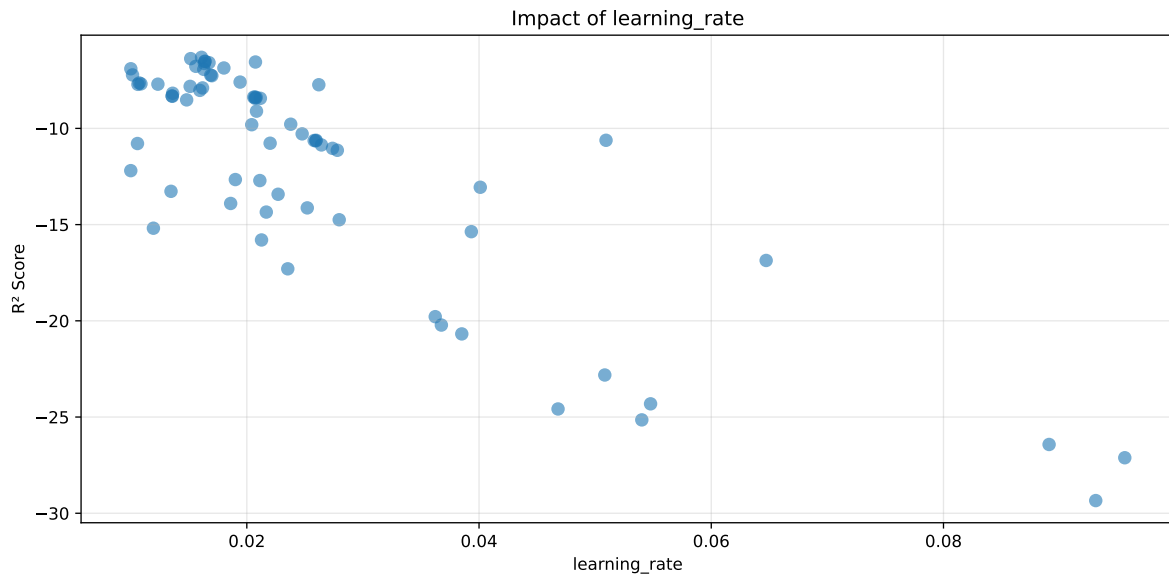
Len(group) : 1356, Len(training) : 950, Len(test) : 406

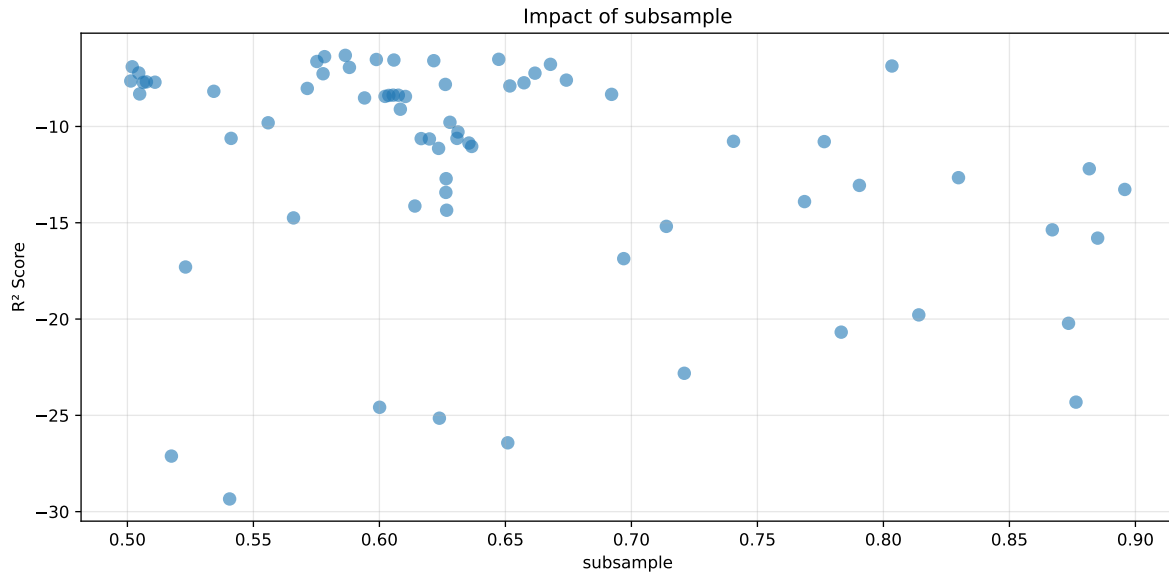
Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086
y_test: mean=2.318, std=0.740, min=1.068, max=5.447

===== TOP Importance FEATURES =====
feature importance r_umidity 62.550276 r_NH3 11.599070 r_THI 11.016326 r_temperature
8.637990 r_CO2 6.196338



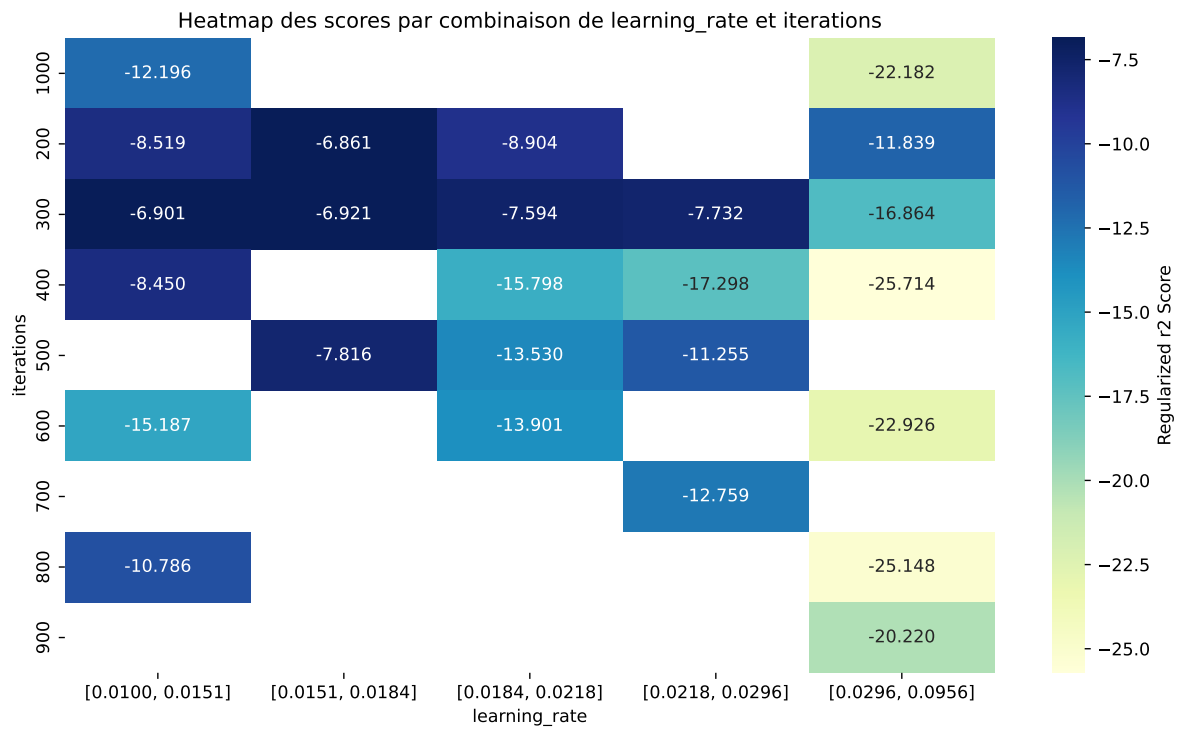
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'subsample']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6768	0.6077	0.6906	0.0378	-6.3074	0.6964	0.0888	1.1461	1.1138
2	0.68	0.6101	0.6922	0.0379	-6.3754	0.6962	0.0861	1.1411	1.1145
3	0.6857	0.6144	0.6967	0.0375	-6.513	0.6996	0.0852	1.1386	1.116
4	0.6816	0.6103	0.6943	0.0365	-6.5225	0.6999	0.0896	1.1468	1.1169
5	0.6712	0.5997	0.6848	0.0369	-6.5503	0.6872	0.0875	1.1459	1.1192
6	0.687	0.615	0.6969	0.038	-6.5848	0.7002	0.0852	1.1386	1.1171
7	0.6845	0.6121	0.6951	0.0373	-6.6265	0.6994	0.0872	1.1425	1.1183
8	0.6884	0.6146	0.6977	0.0375	-6.7739	0.7027	0.0881	1.1434	1.1202
9	0.6795	0.6049	0.6893	0.0365	-6.8606	0.6937	0.0888	1.1468	1.1234
10	0.6793	0.6043	0.6848	0.0347	-6.9013	0.6897	0.0854	1.1414	1.1242
---	---	---	---	---	---	---	---	---	---
70	0.9857	0.6855	0.7436	0.0164	-29.3415	0.7634	0.0779	1.1137	1.438

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.016099098854508824, 'reg__l2_leaf_reg': 45.07145376697486, 'reg__bagging_temperature': 2.6967130636700753, 'reg__random_strength': 2.6323683176251245, 'reg__subsample': 0.5864735252969792, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 6}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.015167455134796752, 'reg__l2_leaf_reg': 27.865169513516335, 'reg__bagging_temperature': 2.8225733910829747, 'reg__random_strength': 2.390287905988081, 'reg__subsample': 0.5782462470951114, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 9}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01638574554220784, 'reg__l2_leaf_reg': 28.82216853417995, 'reg__bagging_temperature': 1.9599180414032422, 'reg__random_strength': 2.4329756988059277, 'reg__subsample': 0.6473570834674631, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 9}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.016398225823513725, 'reg__l2_leaf_reg': 37.146567905526915, 'reg__bagging_temperature': 2.8365604690698314, 'reg__random_strength': 2.3980640152945414, 'reg__subsample': 0.5987851709493242, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 6}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.020744681491765463, 'reg__l2_leaf_reg': 20.234114667137547, 'reg__bagging_temperature': 3.2501100388543493, 'reg__random_strength': 2.9434137162086547, 'reg__subsample': 0.6057431265550328, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 10}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01674516159262022, 'reg__l2_leaf_reg': 27.825721536968437, 'reg__bagging_temperature': 2.8206793347068513, 'reg__random_strength': 2.469576177298847, 'reg__subsample': 0.6215751846872468, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 9}

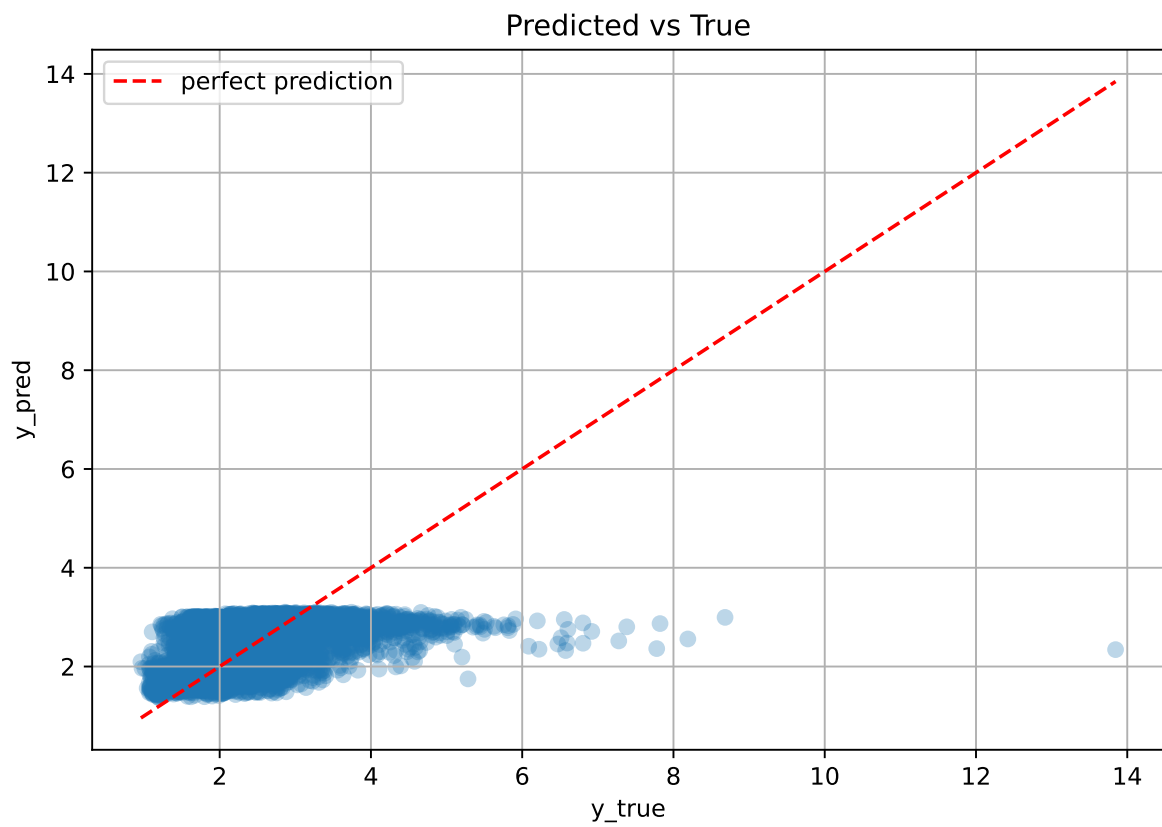
Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.016344705309263442, 'reg__l2_leaf_reg': 28.970569467527657, 'reg__bagging_temperature': 2.9200661410202207, 'reg__random_strength': 2.4499988744557837, 'reg__subsample': 0.5751796239593879, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 9}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.015609794350901266, 'reg__l2_leaf_reg': 30.72586119188999, 'reg__bagging_temperature': 2.526469341422401, 'reg__random_strength': 1.761090953980195, 'reg__subsample': 0.6678728489089996, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 9}

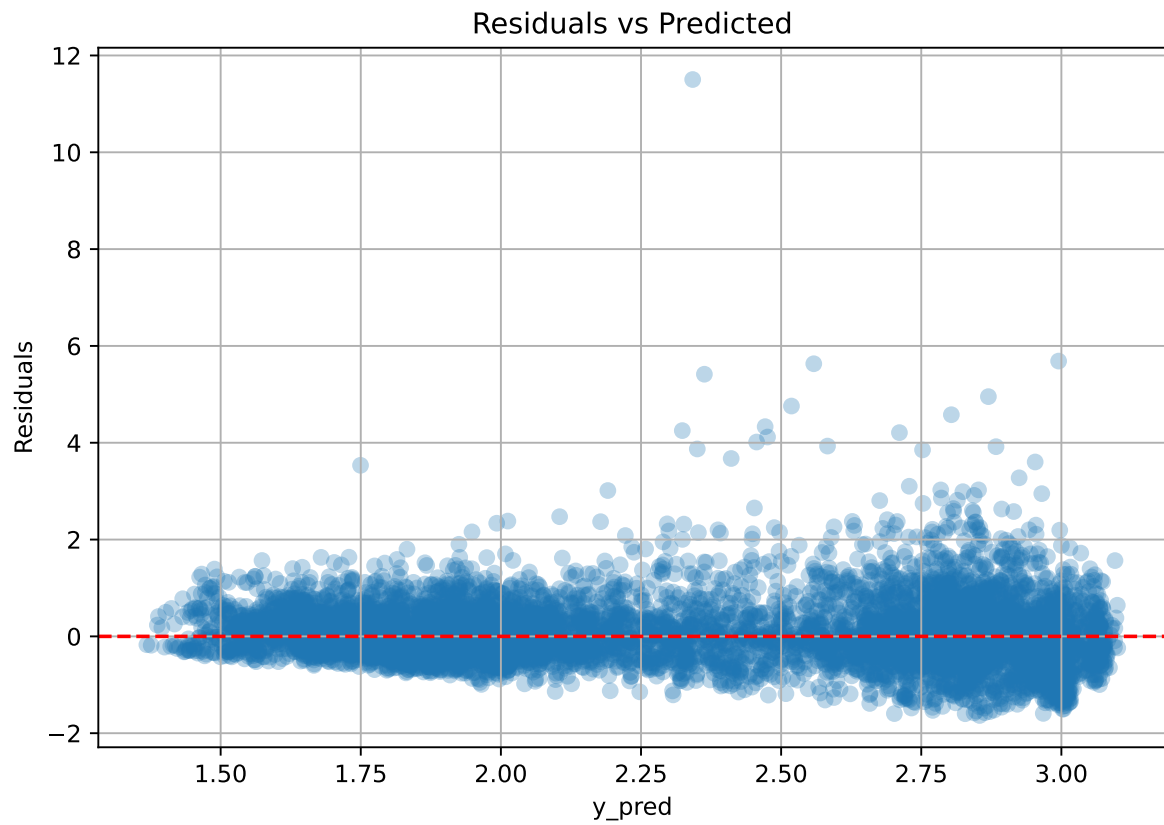
Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.018024893231014413, 'reg__l2_leaf_reg': 35.391075466752206, 'reg__bagging_temperature': 0.05483662676666334, 'reg__random_strength': 1.012332677347643, 'reg__subsample': 0.803377201796075, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 7}

Rank 10: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.010011047467795001, 'reg__l2_leaf_reg': 30.608906766780624, 'reg__bagging_temperature': 2.5953769665721858, 'reg__random_strength': 1.763083507612142, 'reg__subsample': 0.5019240185370939, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 9}

2.5.2 Scatter



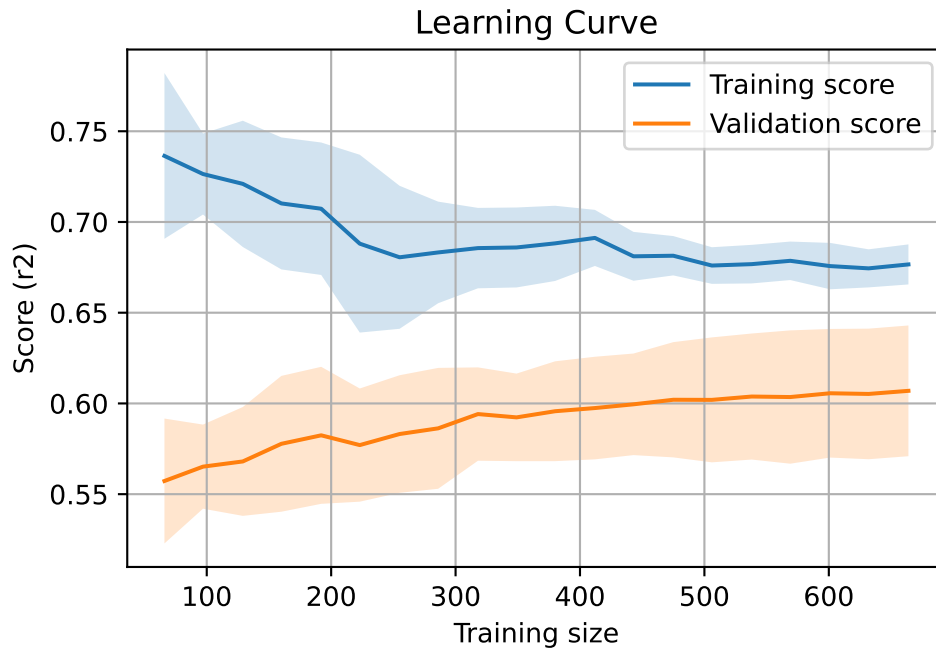
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 3

`Len(group)` : 1356, `Len(training)` : 950,

`Len(training_real)` : 665, `Len(test_of_training)` : 285, `Len(test)` : 406

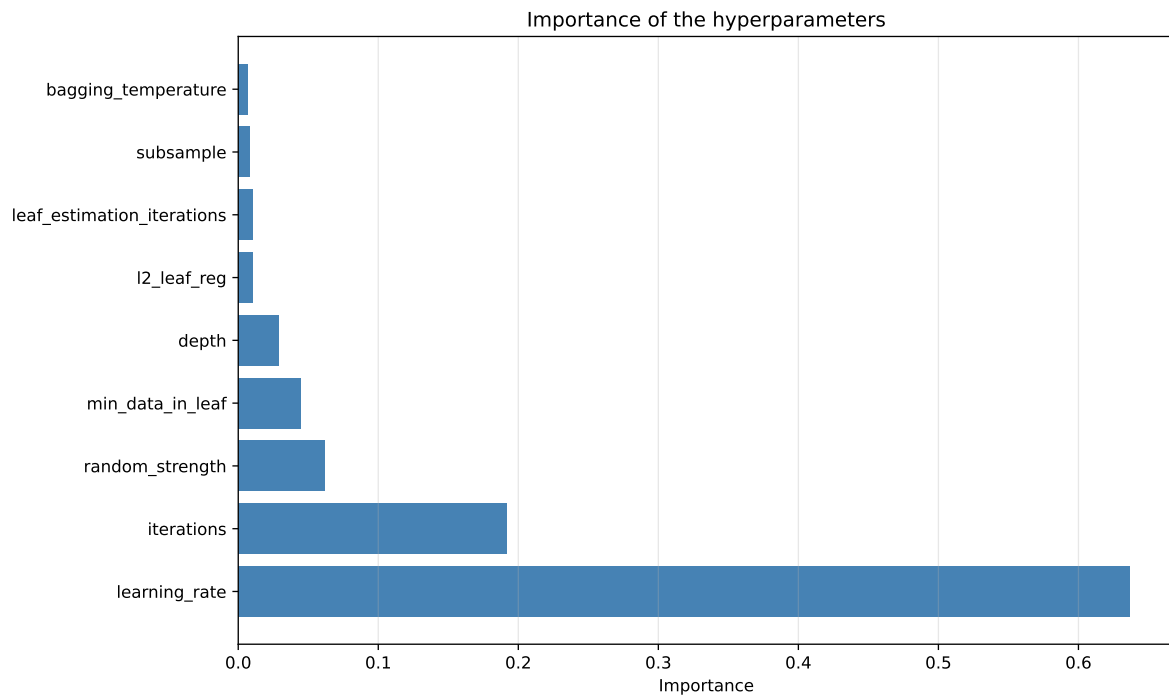
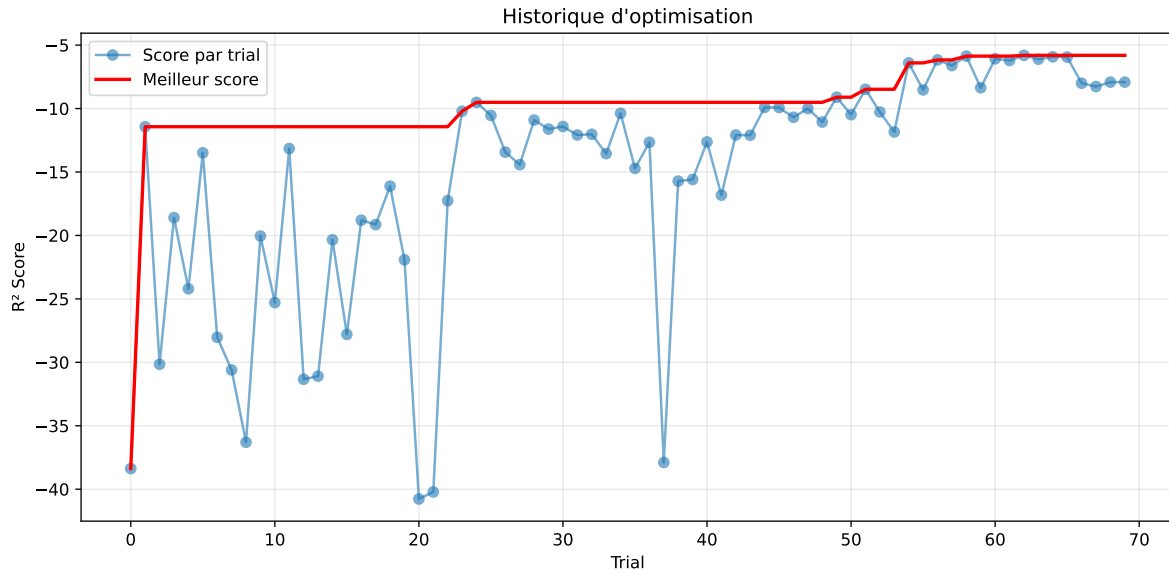


2.5.5 Gridsearch — Group: 4

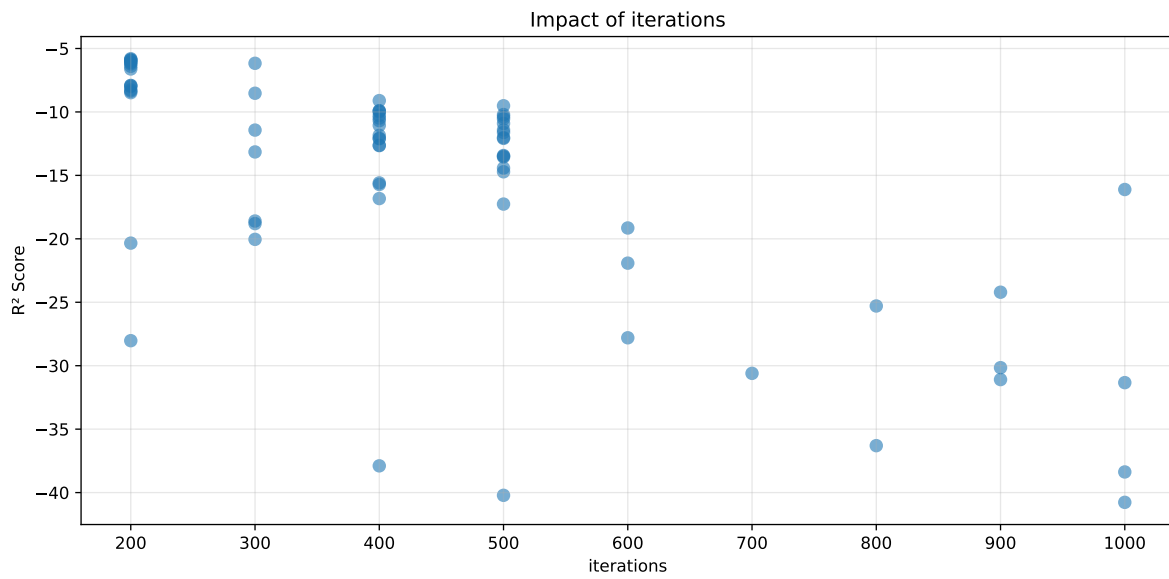
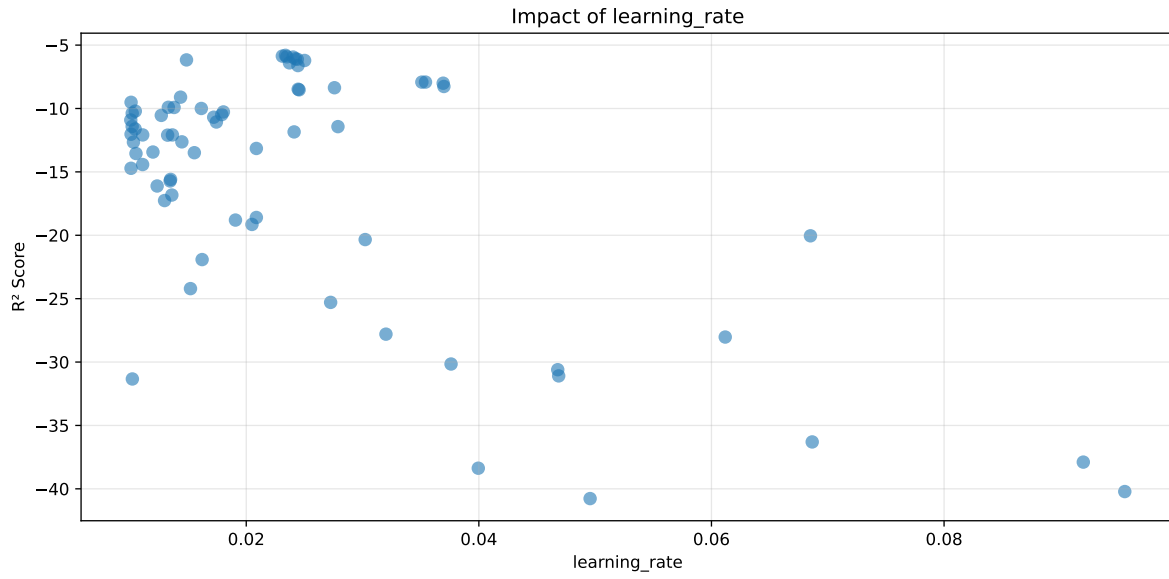
Len(group) : 1360, Len(training) : 952, Len(test) : 408

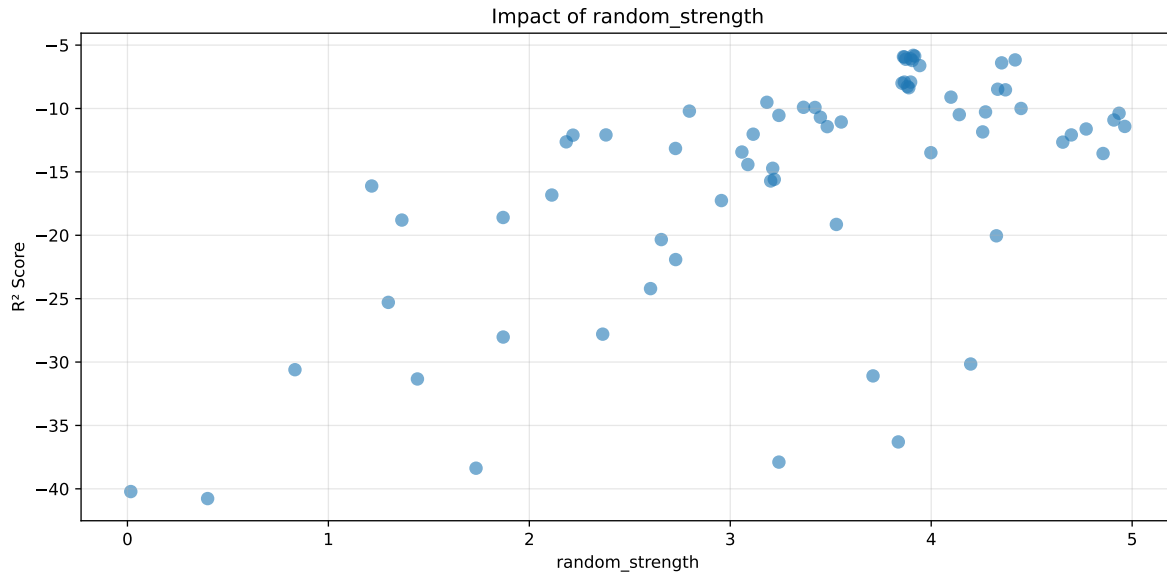
Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383
y_test: mean=2.202, std=0.791, min=1.062, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 49.335497 r_THI 26.547604 r_CO2 10.934654 r_NH3 8.819544
r_temperature 4.362702



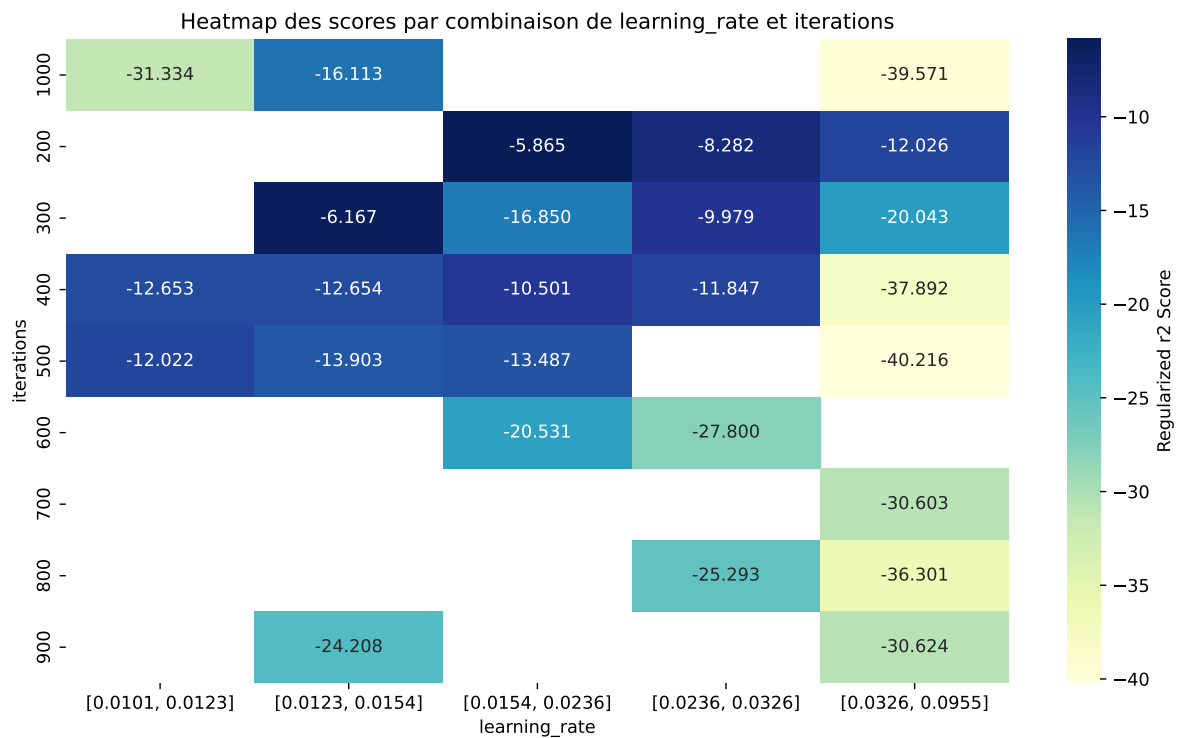
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'random_strength']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4567	0.3947	0.4328	0.0362	-5.8066	0.4475	0.0529	1.1339	1.1571
2	0.4589	0.3963	0.4347	0.0365	-5.868	0.4497	0.0534	1.1348	1.1581
3	0.4586	0.3954	0.434	0.0356	-5.9206	0.448	0.0526	1.1329	1.1597
4	0.4601	0.3967	0.4348	0.0353	-5.9426	0.4523	0.0556	1.1402	1.1598
5	0.4649	0.4002	0.4385	0.036	-6.0752	0.4559	0.0558	1.1393	1.1618
6	0.4668	0.4017	0.4405	0.0363	-6.1149	0.4568	0.0551	1.1372	1.1622
7	0.4723	0.4065	0.4529	0.0393	-6.1671	0.4645	0.0579	1.1425	1.1617
8	0.4689	0.4028	0.4419	0.0363	-6.2122	0.4566	0.0539	1.1337	1.1642
9	0.4797	0.4116	0.4585	0.0392	-6.4005	0.4641	0.0525	1.1275	1.1655
10	0.4899	0.4196	0.4654	0.0369	-6.6141	0.4714	0.0519	1.1236	1.1676
---	---	---	---	---	---	---	---	---	---
70	0.9484	0.5353	0.5575	0.0378	-40.7692	0.583	0.0476	1.089	1.7716

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.0233738401021788, 'reg__l2_leaf_reg': 23.452261737899605, 'reg__bagging_temperature': 0.8995839010048343, 'reg__random_strength': 3.9105237138331757, 'reg__subsample': 0.7930236267910952, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.023102618390804266, 'reg__l2_leaf_reg': 21.174766860697158, 'reg__bagging_temperature': 1.5642553360243976, 'reg__random_strength': 3.9177706528638168, 'reg__subsample': 0.7999593138033118, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 1}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.023489929832357176, 'reg__l2_leaf_reg': 23.168530646781253, 'reg__bagging_temperature': 1.3094189412383785, 'reg__random_strength': 3.861864032543035, 'reg__subsample': 0.7964426278417758, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.024031021416873723, 'reg__l2_leaf_reg': 24.015129695300622, 'reg__bagging_temperature': 1.1804181337652426, 'reg__random_strength': 3.8668629581667218, 'reg__subsample': 0.798789860810462, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.024218479858409803, 'reg__l2_leaf_reg': 21.979956432286016, 'reg__bagging_temperature': 1.3373154413549235, 'reg__random_strength': 3.897092274055034, 'reg__subsample': 0.8006448824193051, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 1}

Rank 6: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.02438922968869723, 'reg__l2_leaf_reg': 21.77543286351754, 'reg__bagging_temperature': 0.8996960327437808, 'reg__random_strength': 3.8733683022997063, 'reg__subsample': 0.801678224502643, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 1}

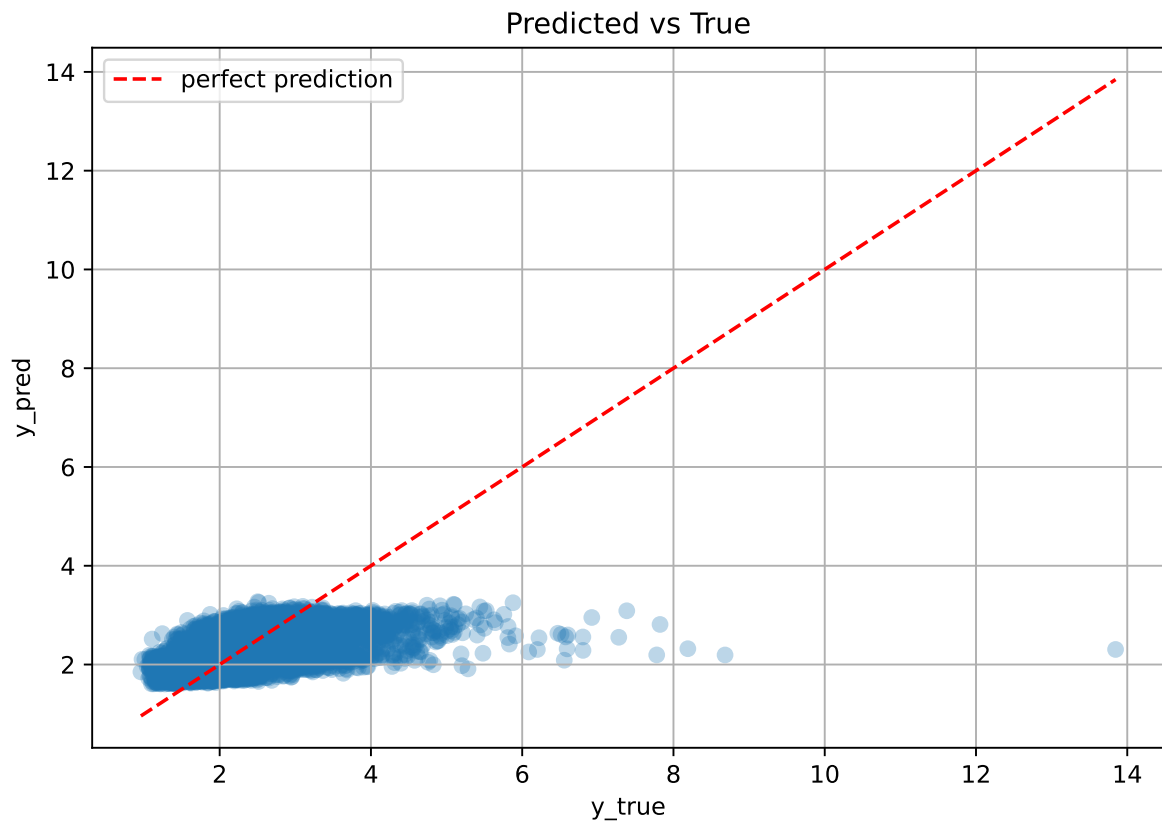
Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.0148626501511495, 'reg__l2_leaf_reg': 24.207400069146193, 'reg__bagging_temperature': 2.580104471622376, 'reg__random_strength': 4.417704593775429, 'reg__subsample': 0.7976583518441633, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 2}

Rank 8: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.025018191078831672, 'reg__l2_leaf_reg': 22.241136751325314, 'reg__bagging_temperature': 1.5180451300246856, 'reg__random_strength': 3.9055047393033018, 'reg__subsample': 0.8072322032535807, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 1}

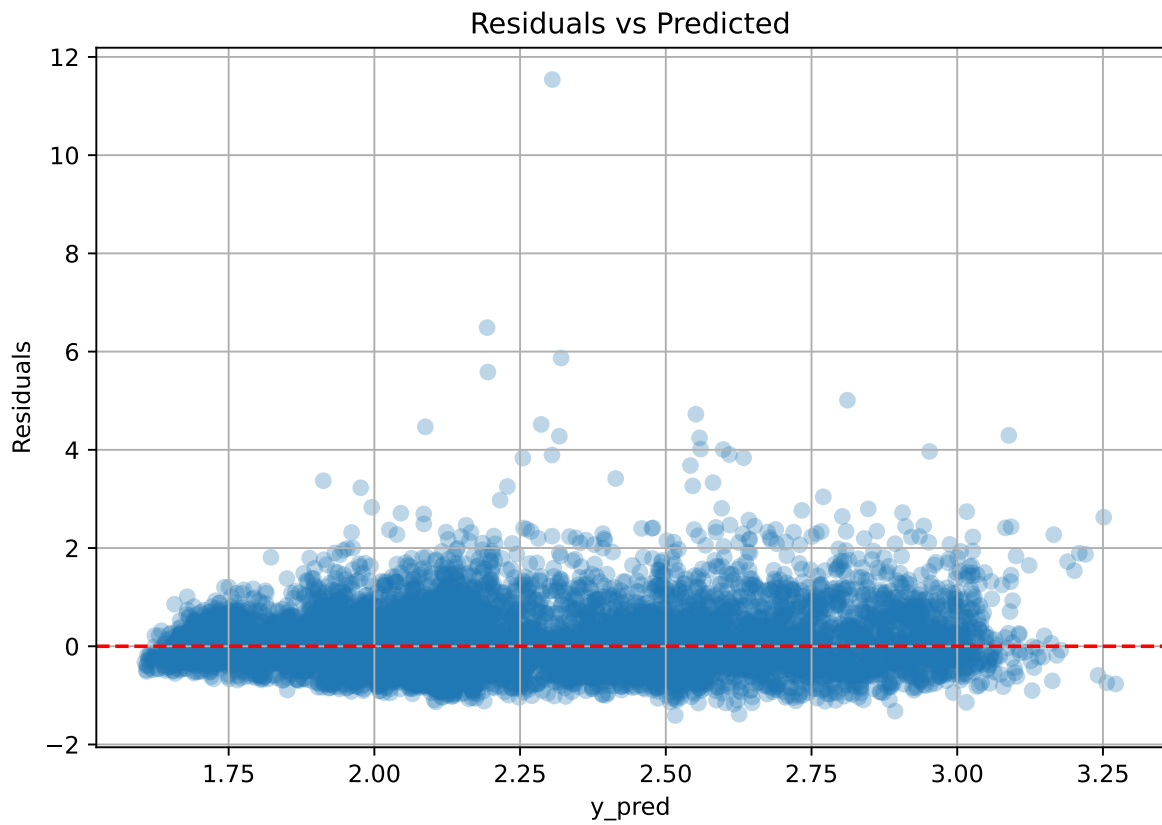
Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.02371578595477248, 'reg__l2_leaf_reg': 23.1570569253024, 'reg__bagging_temperature': 2.572815322373269, 'reg__random_strength': 4.3505156117425114, 'reg__subsample': 0.791516696647502, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 2}

Rank 10: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.02444406499799404, 'reg__l2_leaf_reg': 23.65358774814734, 'reg__bagging_temperature': 1.4268636272123019, 'reg__random_strength': 3.94295513888267, 'reg__subsample': 0.7944647927709592, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 2}

2.5.6 Scatter



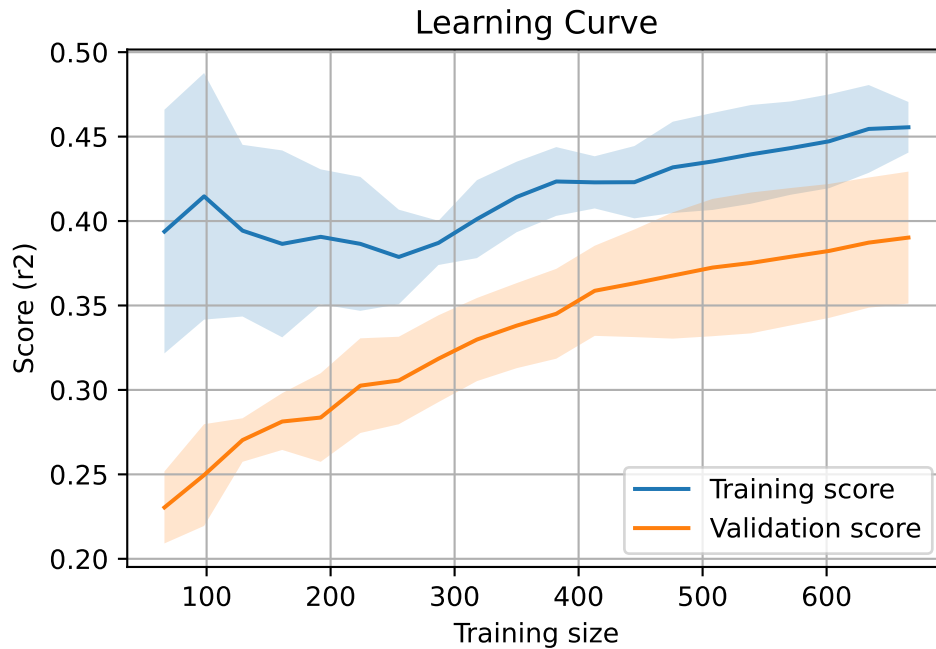
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 4

`Len(group)` : 1360, `Len(training)` : 952,

`Len(training_real)` : 667, `Len(test_of_training)` : 285, `Len(test)` : 408

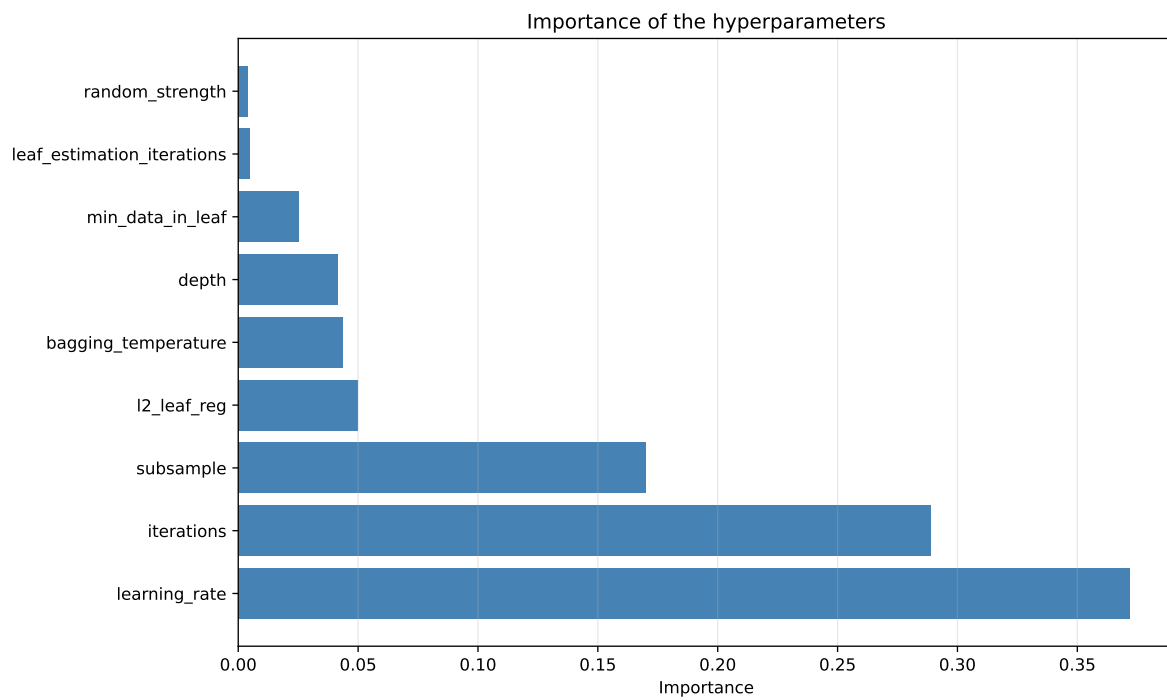
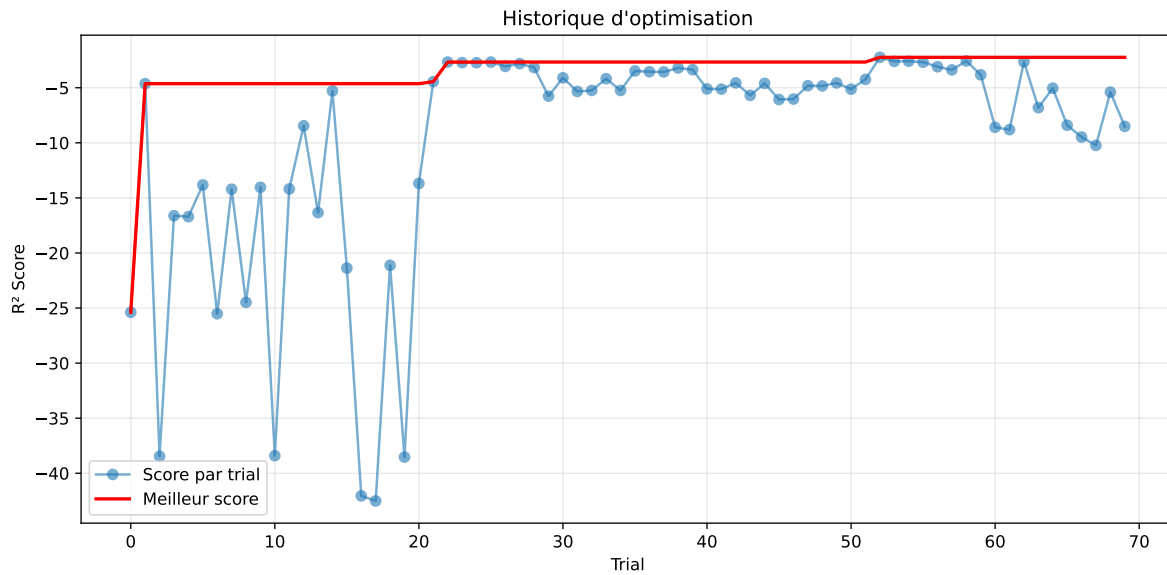


2.5.9 Gridsearch — Group: 5

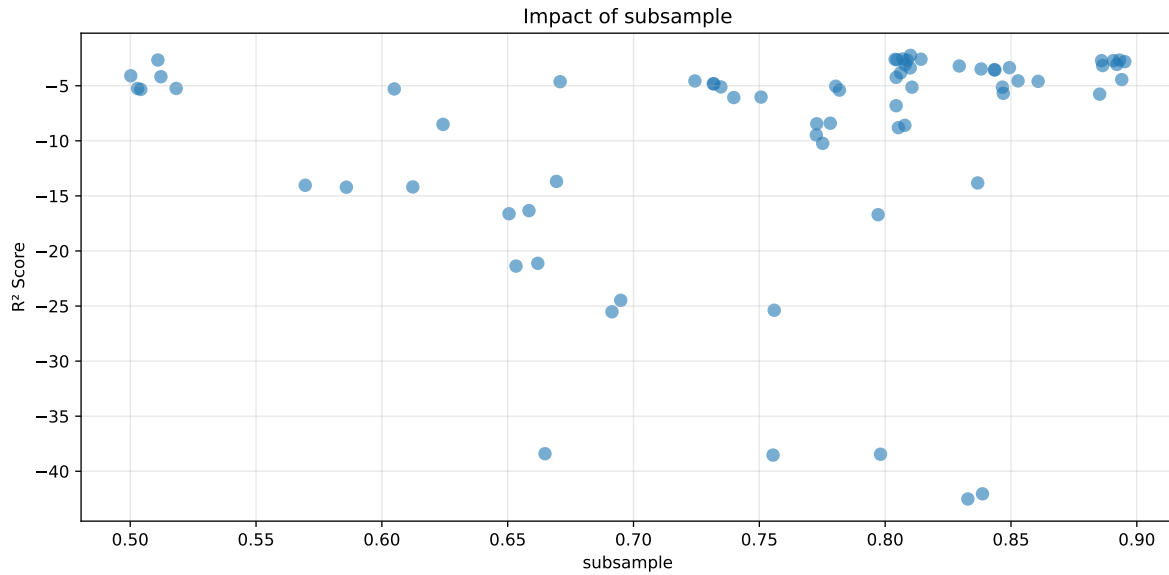
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 66.873540 r_temperature 11.216050 r_CO2 8.632141 r_NH3
7.635942 r_THI 5.642326

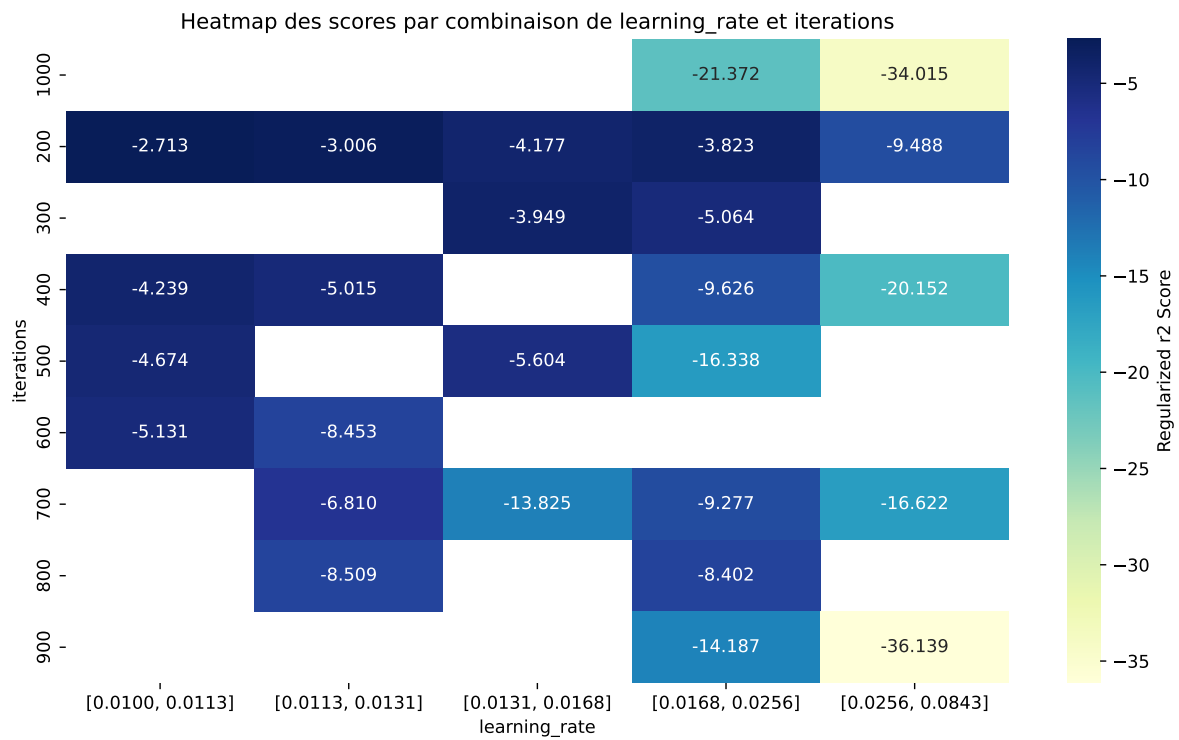


Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'subsample']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4375	0.411	0.4671	0.0193	-2.24	0.4714	0.0604	1.147	1.0645
2	0.4531	0.4232	0.4805	0.0215	-2.5667	0.4874	0.0642	1.1517	1.0707
3	0.4527	0.4225	0.4802	0.0199	-2.5901	0.4851	0.0625	1.148	1.0713
4	0.4538	0.4235	0.4814	0.0195	-2.6059	0.4887	0.0652	1.154	1.0715
5	0.4548	0.4241	0.4823	0.02	-2.6379	0.4897	0.0656	1.1547	1.0722
6	0.451	0.4201	0.4783	0.019	-2.6654	0.481	0.0609	1.1451	1.0734
7	0.4519	0.4211	0.4794	0.0191	-2.6677	0.4841	0.063	1.1496	1.0734
8	0.4562	0.4252	0.4832	0.0199	-2.6794	0.4877	0.0625	1.147	1.073
9	0.4555	0.4241	0.4829	0.0193	-2.7191	0.4864	0.0623	1.1468	1.0741
10	0.4543	0.4229	0.4817	0.0192	-2.7222	0.4862	0.0634	1.1498	1.0744
---	---	---	---	---	---	---	---	---	---
70	0.9803	0.5496	0.6007	0.0293	-42.5153	0.6308	0.0812	1.1477	1.7835

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.010100828813095694, 'reg__l2_leaf_reg': 35.70513371698317, 'reg__bagging_temperature': 4.3350916437439615, 'reg__random_strength': 3.576884120461096, 'reg__subsample': 0.8100704652759421, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 2}

Rank 2: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011474673125062517, 'reg__l2_leaf_reg': 40.88055246948379, 'reg__bagging_temperature': 3.714680218296053, 'reg__random_strength': 2.9807009449494037, 'reg__subsample': 0.8070123811192216, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 2}

Rank 3: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011312887730146796, 'reg__l2_leaf_reg': 34.38773186278362, 'reg__bagging_temperature': 3.7215018013988845, 'reg__random_strength': 3.0481683217066715, 'reg__subsample': 0.8142280972052752, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2}

Rank 4: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011426221606030266, 'reg__l2_leaf_reg': 36.44495440724847, 'reg__bagging_temperature': 4.994550454758436, 'reg__random_strength': 2.92869927361679, 'reg__subsample': 0.8039454333461753, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2}

Rank 5: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011876362210721369, 'reg__l2_leaf_reg': 44.246928701562354, 'reg__bagging_temperature': 3.0571715129273063, 'reg__random_strength': 2.881967188187188, 'reg__subsample': 0.8046440815509462, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2}

Rank 6: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01011930847969786, 'reg__l2_leaf_reg': 32.26821077309489, 'reg__bagging_temperature': 4.575201161406915, 'reg__random_strength': 3.804954310495402, 'reg__subsample': 0.5109957887985037, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 4}

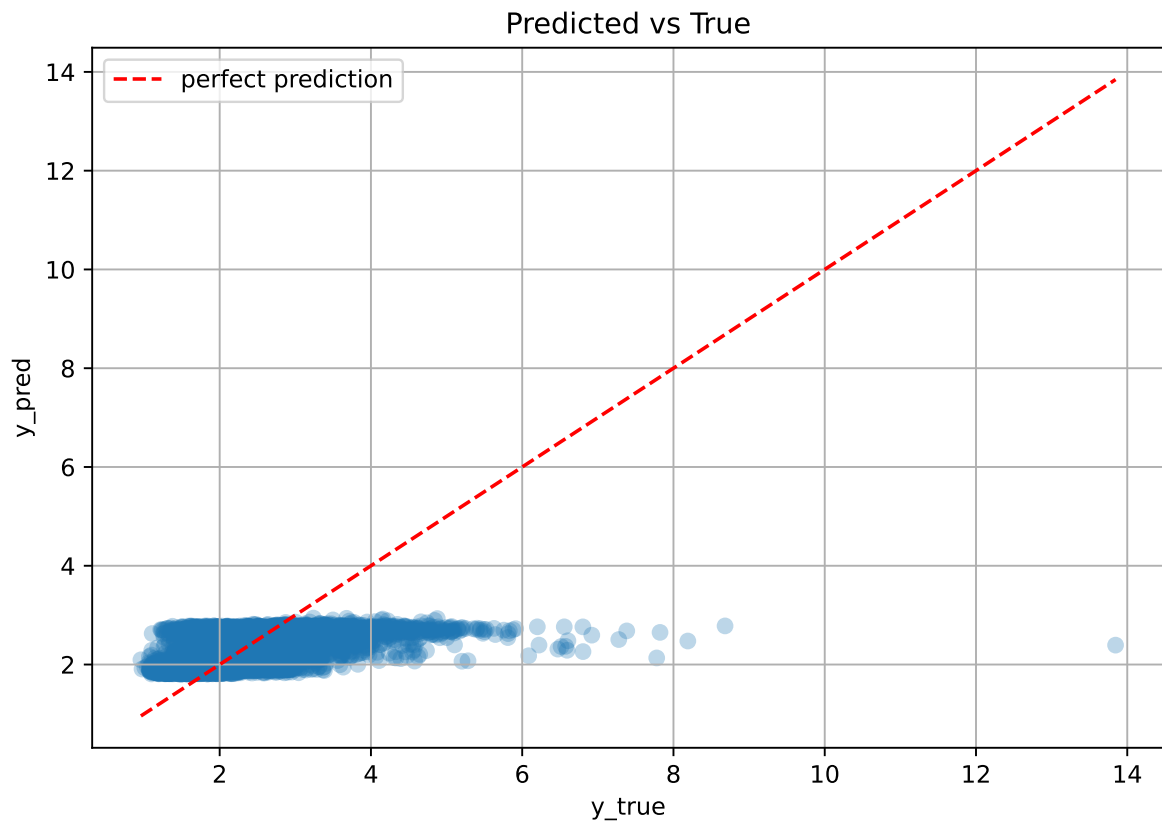
Rank 7: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.010205322811028067, 'reg__l2_leaf_reg': 31.601412298704723, 'reg__bagging_temperature': 4.969633058435132, 'reg__random_strength': 3.806000211132115, 'reg__subsample': 0.8931168537974743, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 4}

Rank 8: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011507321568725403, 'reg__l2_leaf_reg': 33.84974127826968, 'reg__bagging_temperature': 3.7341832636004835, 'reg__random_strength': 2.8658281719860534, 'reg__subsample': 0.8087703153945077, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 2}

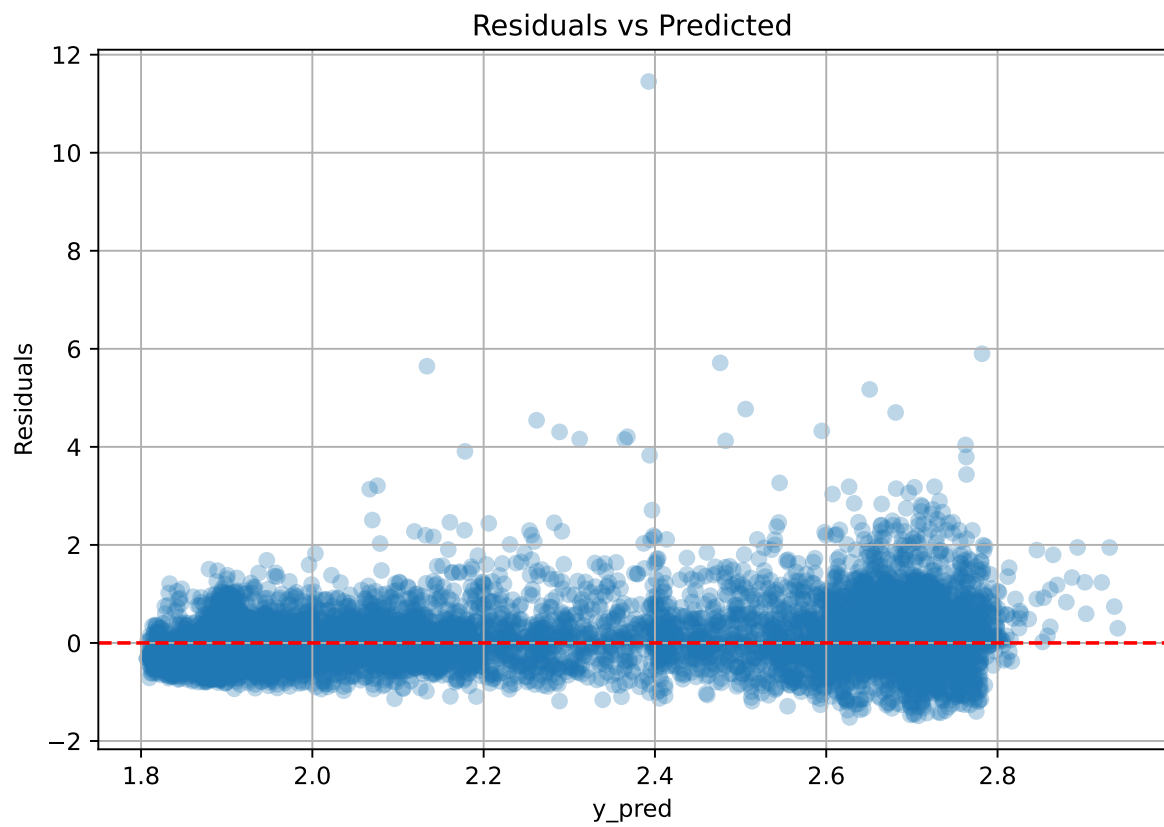
Rank 9: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.010573165738088164, 'reg__l2_leaf_reg': 33.499541874963526, 'reg__bagging_temperature': 4.937943157705574, 'reg__random_strength': 3.6725532785951076, 'reg__subsample': 0.8906836469189192, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 4}

Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.010590927508252407, 'reg__l2_leaf_reg': 34.53988220468703, 'reg__bagging_temperature': 4.532138026626728, 'reg__random_strength': 3.839019031656579, 'reg__subsample': 0.8859939495668007, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 4}

2.5.10 Scatter



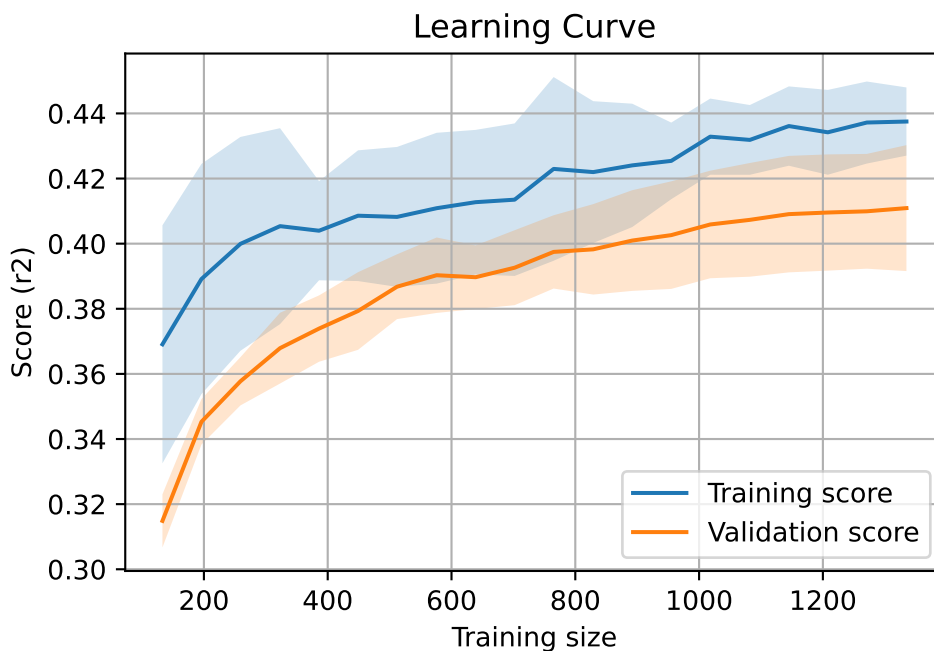
2.5.11 Scatter residuals



2.5.12 Learning curve — Group: 5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

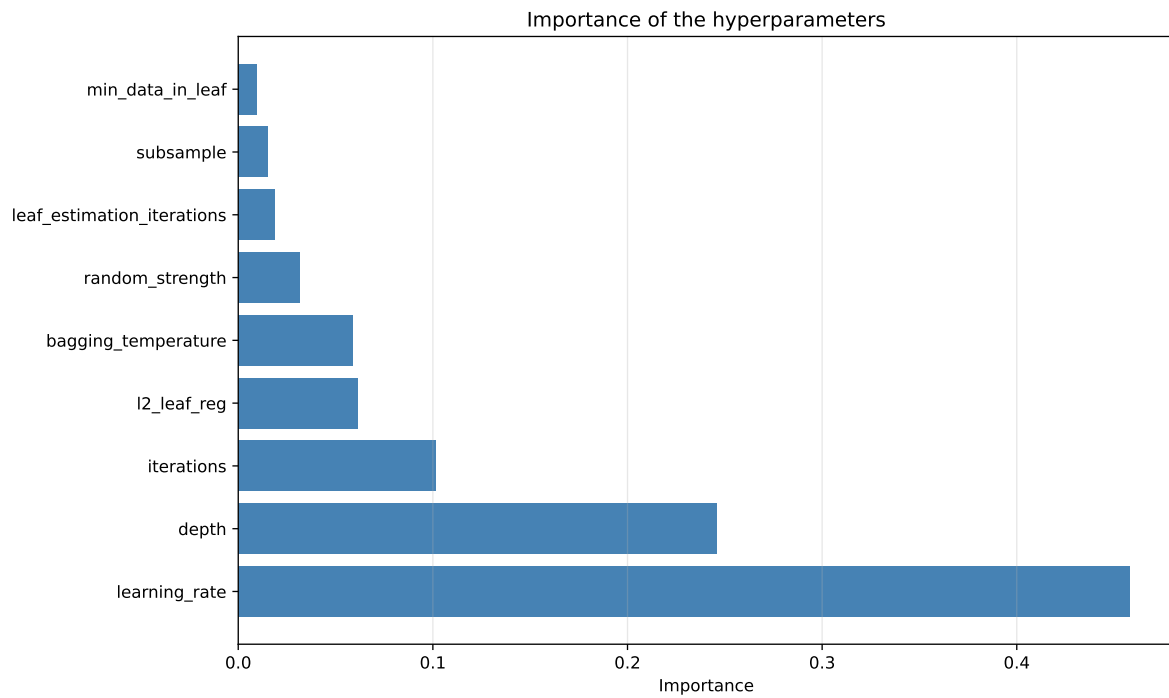
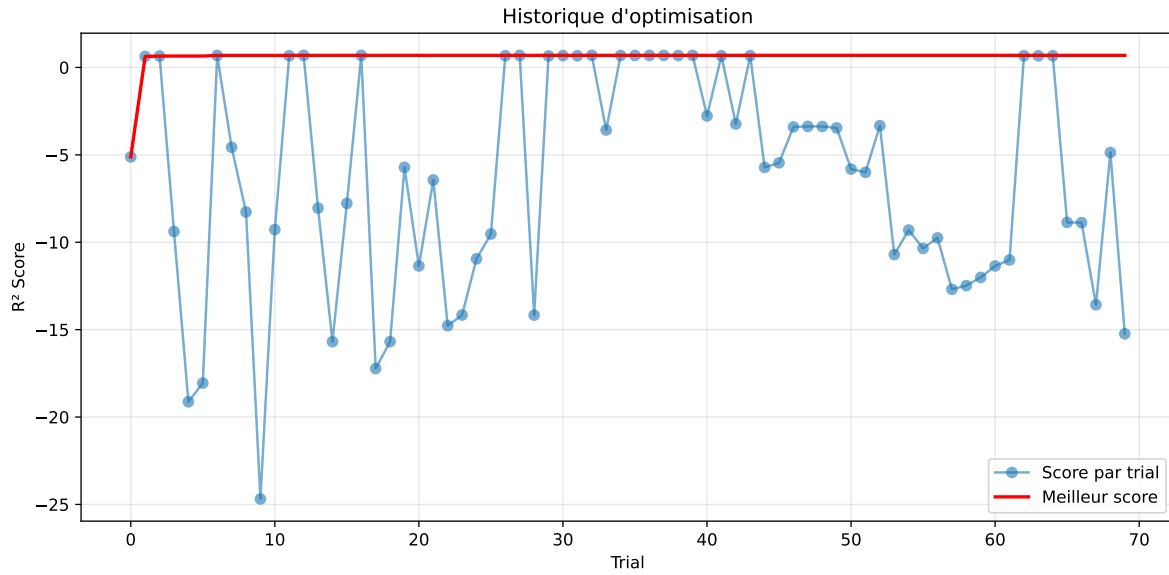


2.5.13 Gridsearch — Group: 11

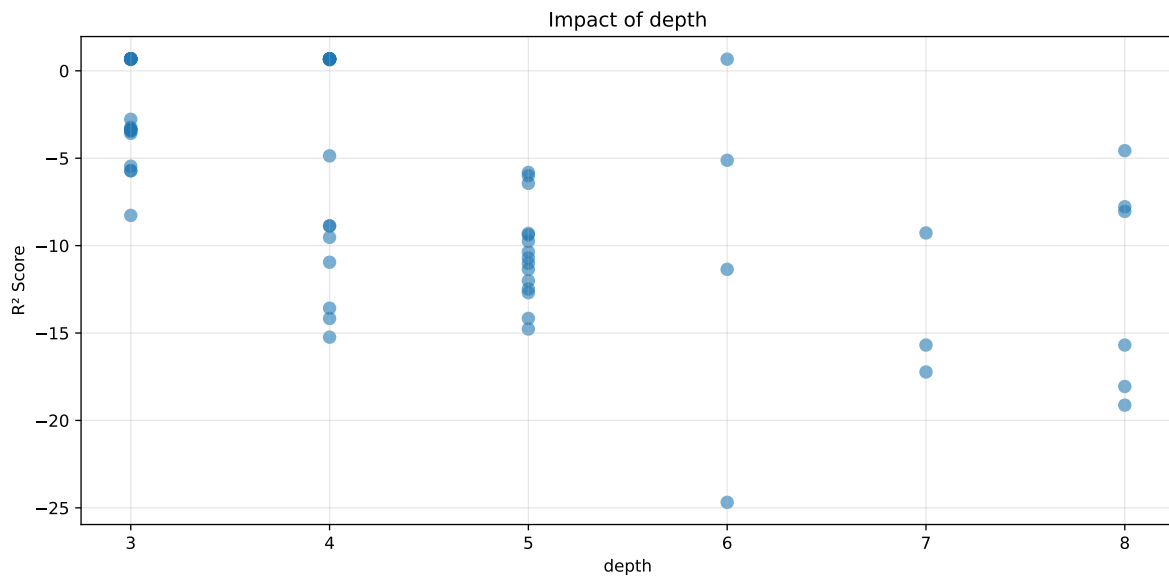
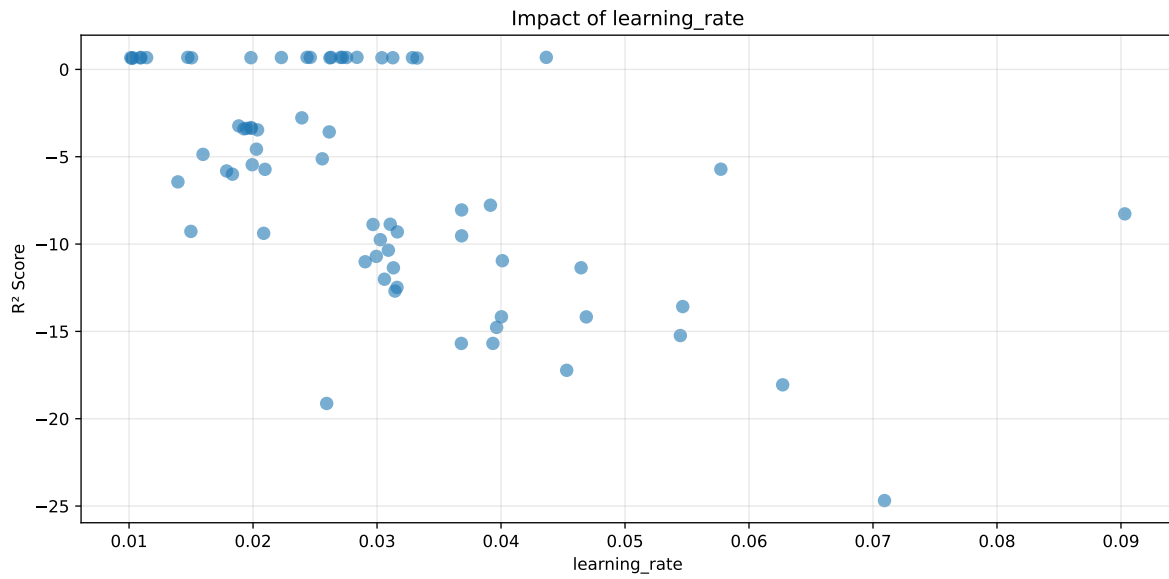
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

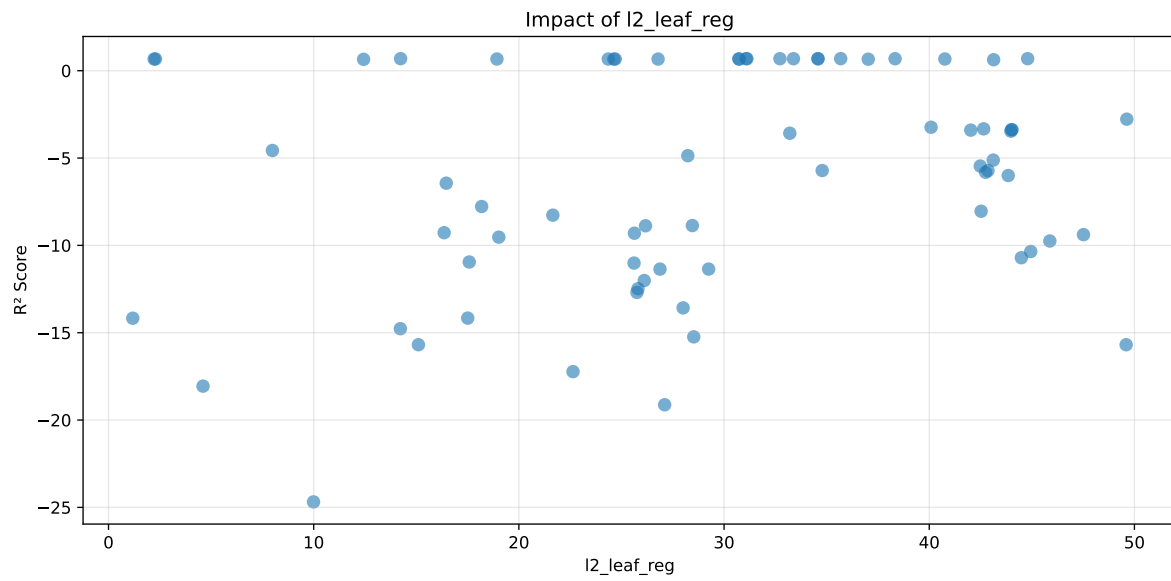
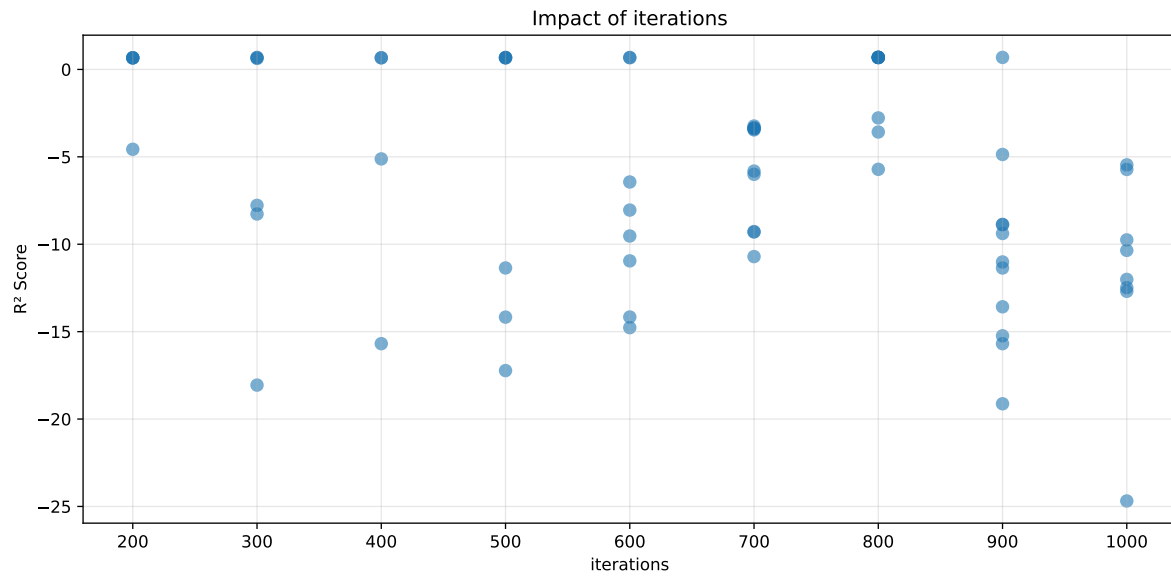
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

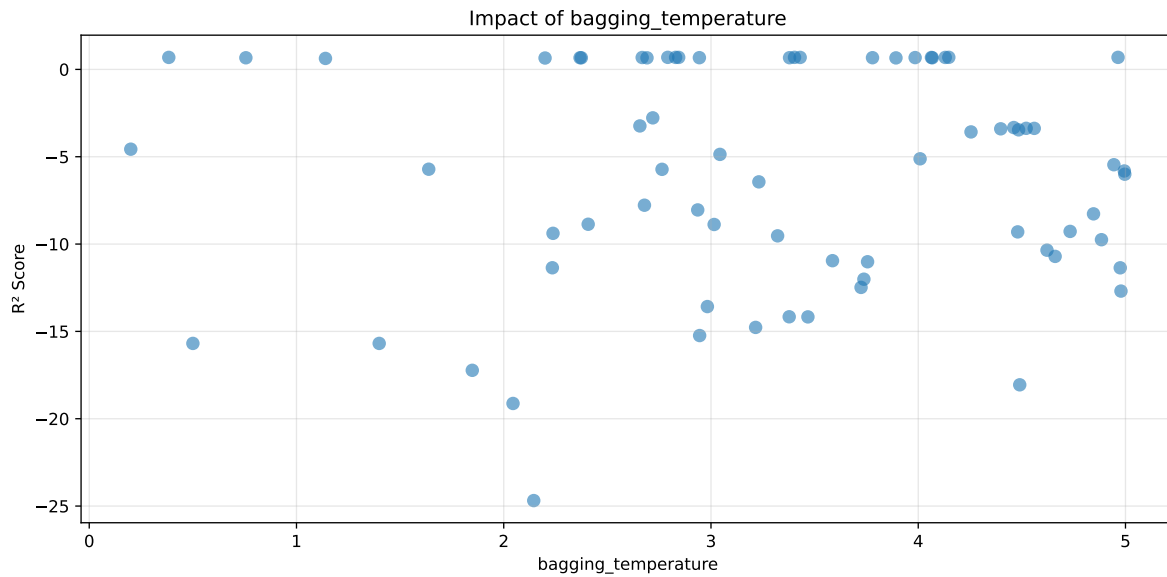
===== TOP Importance FEATURES =====
feature importance r_umidity 36.807564 r_NH3 32.840634 r_CO2 12.150442 r_temperature
12.055569 r_THI 6.145791



Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations', 'l2_leaf_reg', 'bagging_temperature']

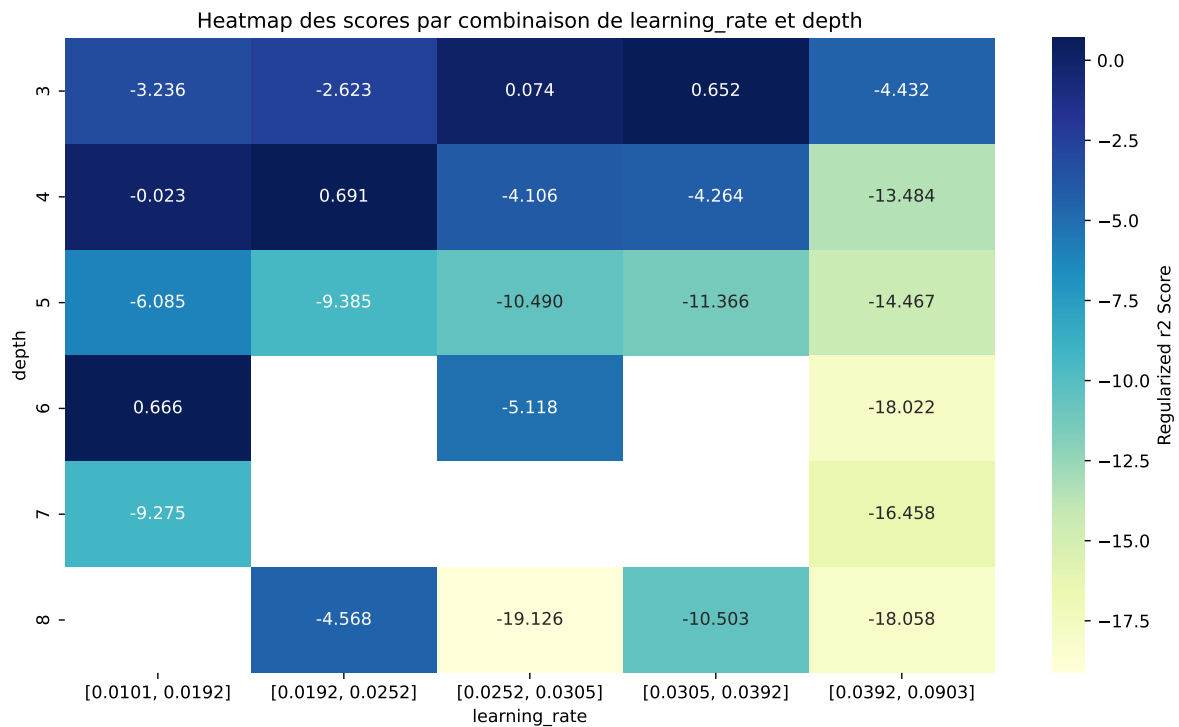






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7241	0.691	0.5919	0.0163	0.691	0.5956	-0.0954	0.8619	1.0479
2	0.7249	0.6905	0.5936	0.0178	0.6905	0.596	-0.0945	0.8631	1.0499
3	0.7207	0.6897	0.594	0.0166	0.6897	0.5964	-0.0933	0.8647	1.0451
4	0.7195	0.6891	0.5947	0.0182	0.6891	0.5975	-0.0917	0.8669	1.0441
5	0.7203	0.6877	0.5912	0.0166	0.6877	0.5952	-0.0925	0.8655	1.0474
6	0.7183	0.6875	0.5912	0.0174	0.6875	0.5933	-0.0942	0.8629	1.0448
7	0.7201	0.6873	0.5916	0.0182	0.6873	0.5952	-0.0921	0.866	1.0478
8	0.7173	0.6865	0.5907	0.0167	0.6865	0.5945	-0.092	0.866	1.0448
9	0.7172	0.6861	0.5913	0.0171	0.6861	0.5966	-0.0895	0.8695	1.0454
10	0.714	0.683	0.5873	0.0157	0.683	0.5932	-0.0898	0.8685	1.0454
---	---	---	---	---	---	---	---	---	---
70	0.9629	0.709	0.5898	0.0195	-24.6864	0.5991	-0.1099	0.845	1.3582

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.024355247090792818, 'reg__l2_leaf_reg': 35.68475318056527, 'reg__bagging_temperature': 4.147463039779848, 'reg__random_strength': 2.8920982131086204, 'reg__subsample': 0.8269845829051746, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02838374058114295, 'reg__l2_leaf_reg': 31.115145597201156, 'reg__bagging_temperature': 2.7915314095678987, 'reg__random_strength': 2.278959945859076, 'reg__subsample': 0.6394117608639126, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 1}

Rank 3: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.01473584931072599, 'reg__l2_leaf_reg': 14.236330326529353, 'reg__bagging_temperature': 3.403029267344003, 'reg__random_strength': 2.5161936092511334, 'reg__subsample': 0.8789975536605735, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 7}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.043651766078543285, 'reg__l2_leaf_reg': 44.7963077465661, 'reg__bagging_temperature': 4.96405190683105, 'reg__random_strength': 2.5746738304265064, 'reg__subsample': 0.6639598680918596, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 4}

Rank 5: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.027213800248633, 'reg__l2_leaf_reg': 32.72306406006474, 'reg__bagging_temperature': 2.829290689814754, 'reg__random_strength': 2.3389221617811455, 'reg__subsample': 0.6773944317508163, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 1}

Rank 6: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02629636156897537, 'reg__l2_leaf_reg': 31.0749112129573, 'reg__bagging_temperature': 2.844150938188808, 'reg__random_strength': 2.282319075782437, 'reg__subsample': 0.6838091927141098, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 1}

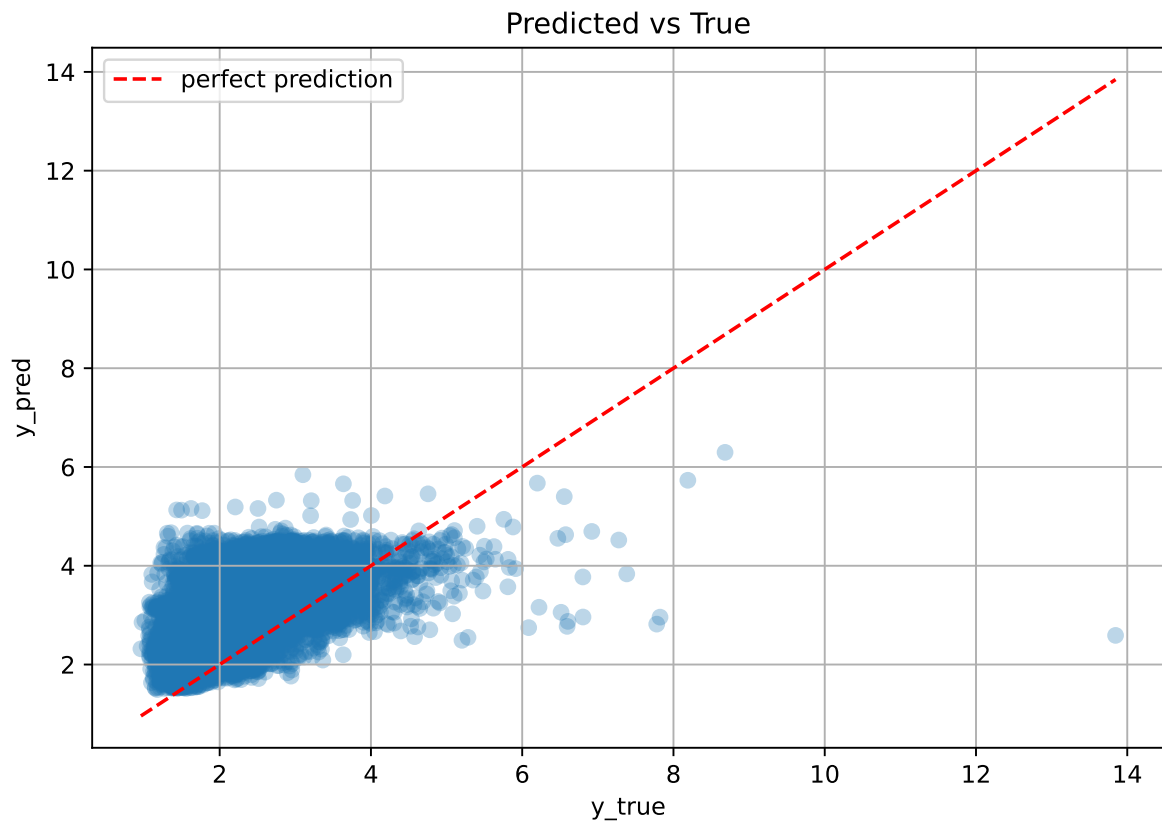
Rank 7: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.024616928150438512, 'reg__l2_leaf_reg': 33.38300484427415, 'reg__bagging_temperature': 0.38379298003907547, 'reg__random_strength': 2.302843440387515, 'reg__subsample': 0.6792675033351674, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 1}

Rank 8: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02706243846832317, 'reg__l2_leaf_reg': 34.57949300642236, 'reg__bagging_temperature': 4.130378876575685, 'reg__random_strength': 2.3126946094910963, 'reg__subsample': 0.6819618493220297, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 1}

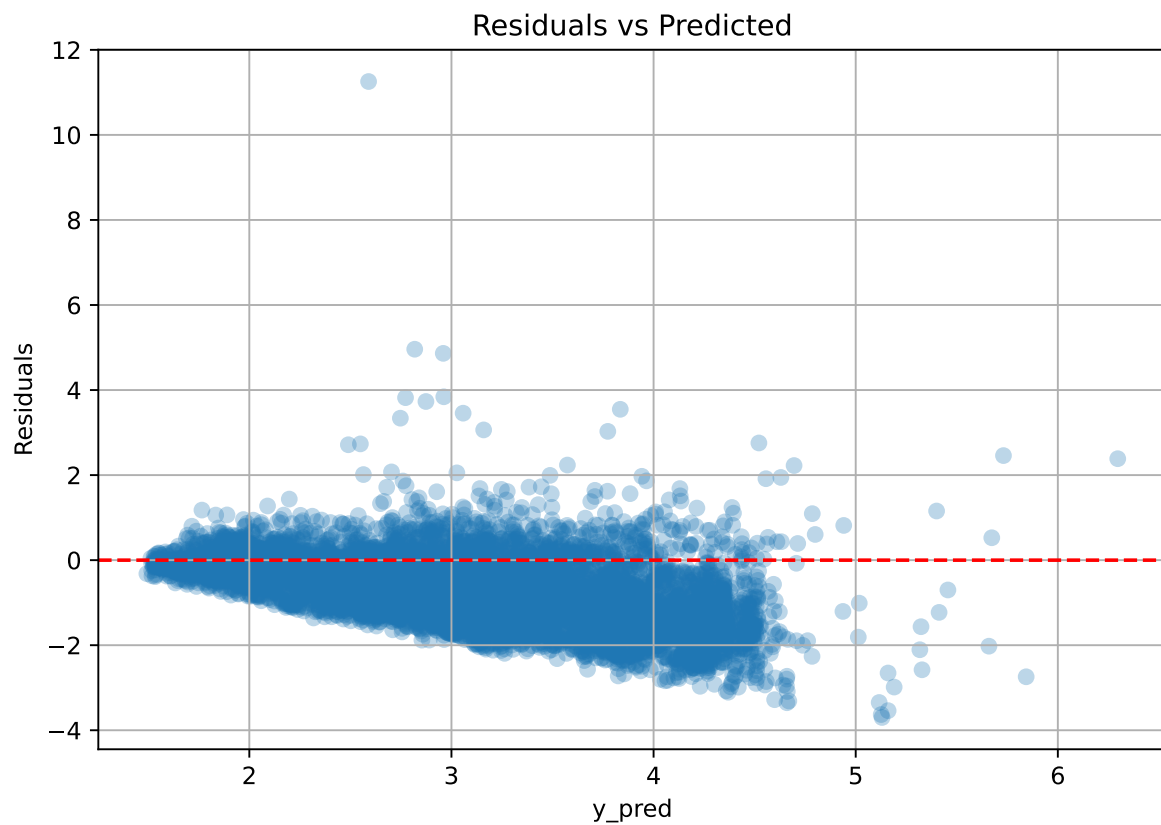
Rank 9: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.010942394424147165, 'reg__l2_leaf_reg': 38.33469314695624, 'reg__bagging_temperature': 3.430313874778954, 'reg__random_strength': 0.8236013592917064, 'reg__subsample': 0.8640363634446274, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 5}

Rank 10: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02754130773154027, 'reg__l2_leaf_reg': 34.57566659419786, 'reg__bagging_temperature': 4.063595289076424, 'reg__random_strength': 2.336296158502987, 'reg__subsample': 0.8280443616808583, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 1}

2.5.14 Scatter



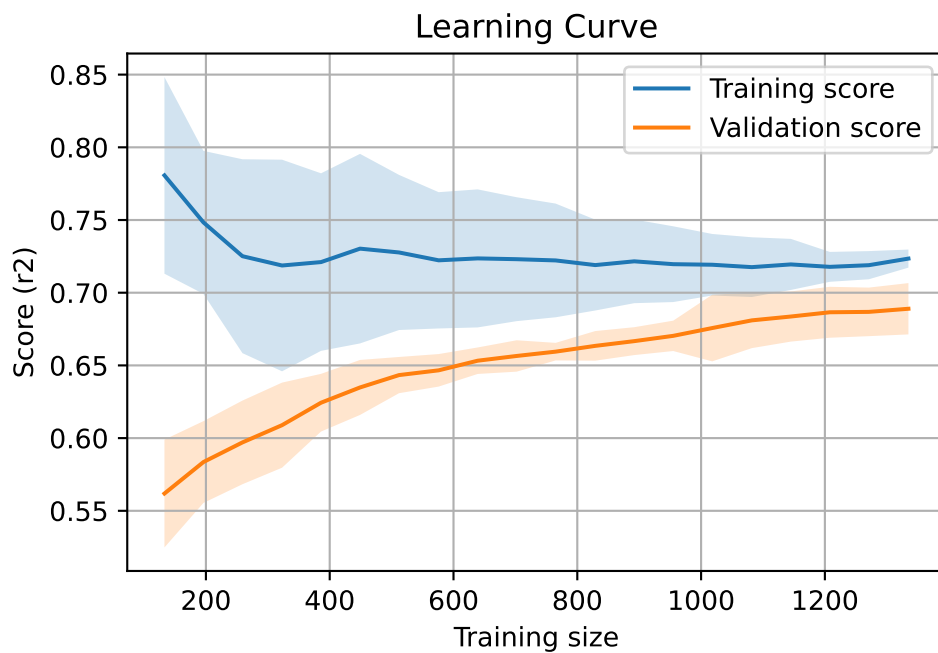
2.5.15 Scatter residuals



2.5.16 Learning curve — Group: 11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	3	950	406	1356	0.596	0.55	0.04	0.64	0.641	0.091	1.165	1.083	nan
Linear	4	952	408	1360	0.302	0.278	0.049	0.276	0.277	-0.001	0.997	1.087	nan
Linear	5	1909	817	2726	0.431	0.404	0.026	0.457	0.457	0.054	1.133	1.069	nan
Linear	11	1909	817	2726	0.463	0.463	0.017	0.394	0.394	-0.069	0.852	1.0	nan
Linear	MEAN	5720	2448	8168	0.448	0.424	0.033	0.442	0.442	0.019	1.037	1.06	nan
LGBM	3	950	406	1356	0.558	0.5	0.04	0.566	0.588	0.088	1.177	1.118	-5.3741
LGBM	4	952	408	1360	0.335	0.288	0.038	0.298	0.347	0.058	1.203	1.164	-4.4275
LGBM	5	1909	817	2726	0.412	0.382	0.022	0.438	0.459	0.077	1.202	1.079	-2.628
LGBM	11	1909	817	2726	0.678	0.648	0.012	0.55	0.569	-0.079	0.878	1.046	0.6481
LGBM	MEAN	5720	2448	8168	0.496	0.454	0.028	0.463	0.491	0.036	1.115	1.101	-2.945
CatBoost	3	950	406	1356	0.677	0.608	0.038	0.691	0.696	0.089	1.146	1.114	-6.3074
CatBoost	4	952	408	1360	0.457	0.395	0.036	0.433	0.448	0.053	1.134	1.157	-5.8066
CatBoost	5	1909	817	2726	0.438	0.411	0.019	0.467	0.471	0.06	1.147	1.064	-2.24
CatBoost	11	1909	817	2726	0.724	0.691	0.016	0.592	0.596	-0.095	0.862	1.048	0.691
CatBoost	MEAN	5720	2448	8168	0.574	0.526	0.027	0.546	0.553	0.027	1.072	1.096	-3.416

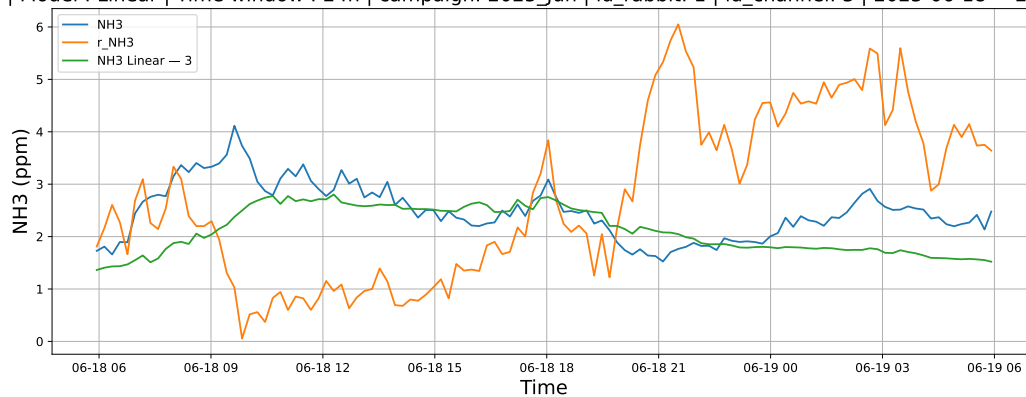
3 Plot

3.1 Standalone plots

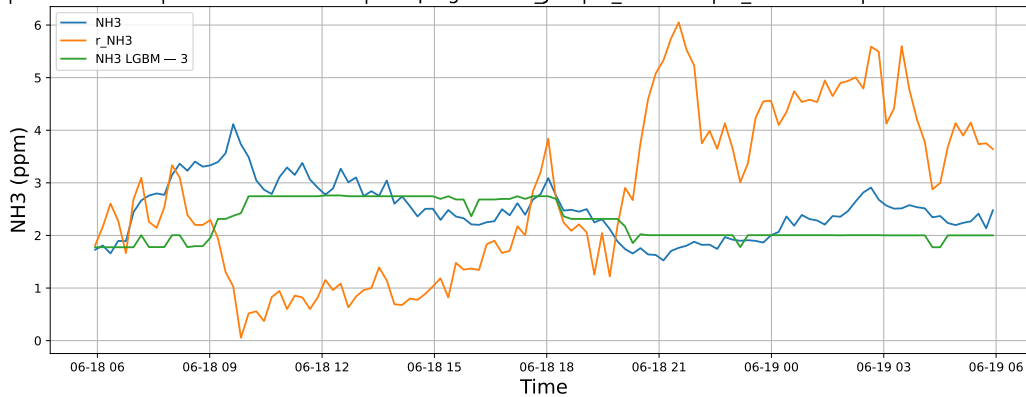
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

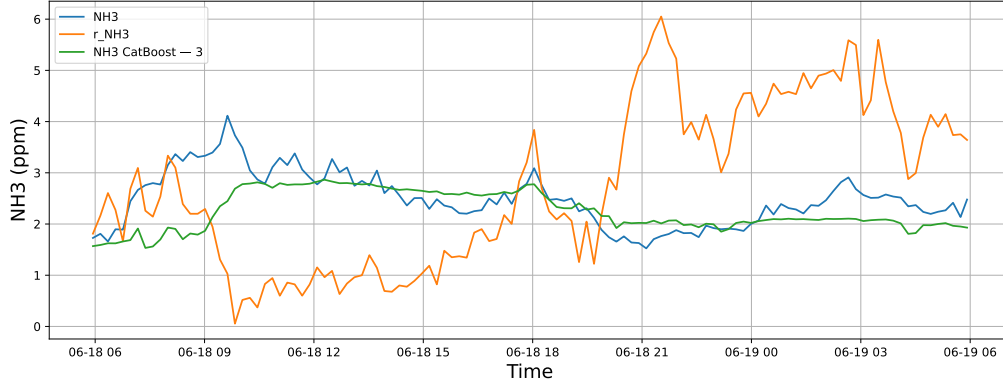
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

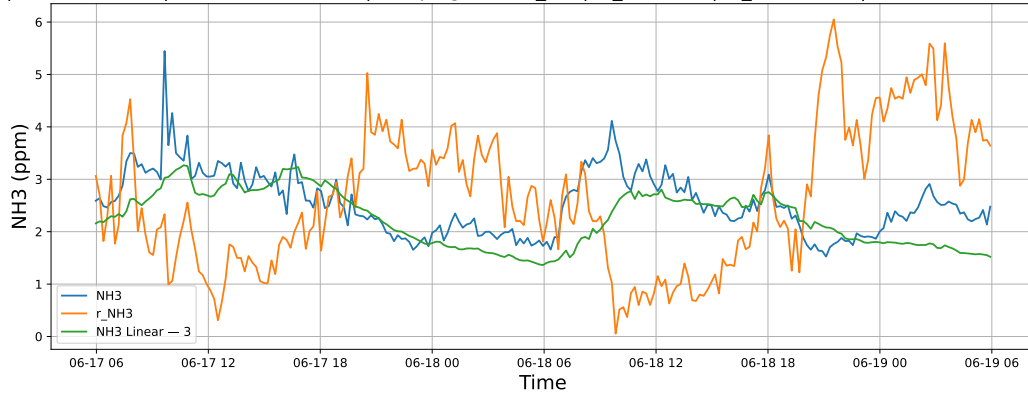


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

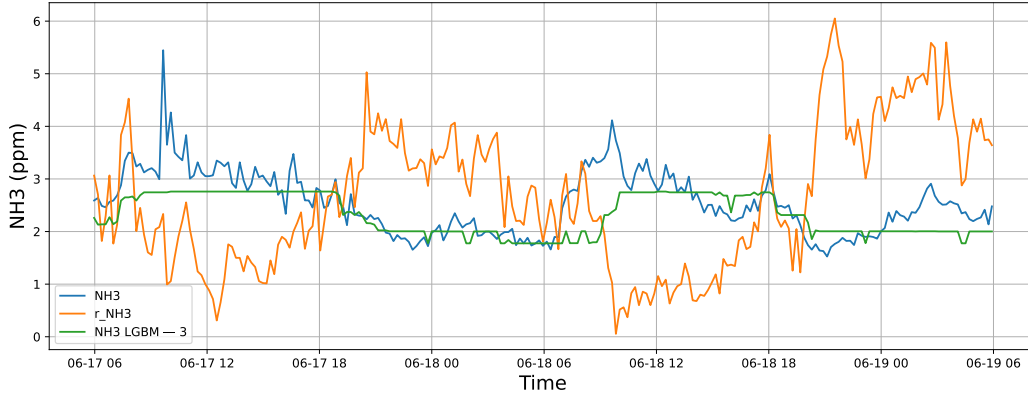


3.1.1.2 Time window : 48

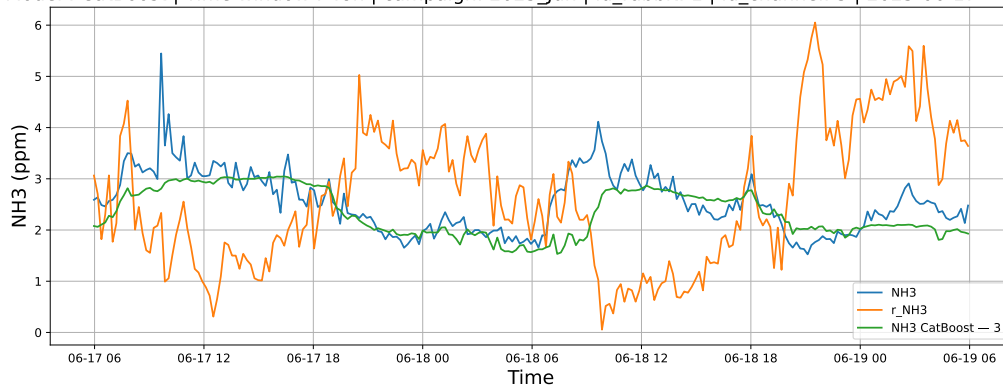
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

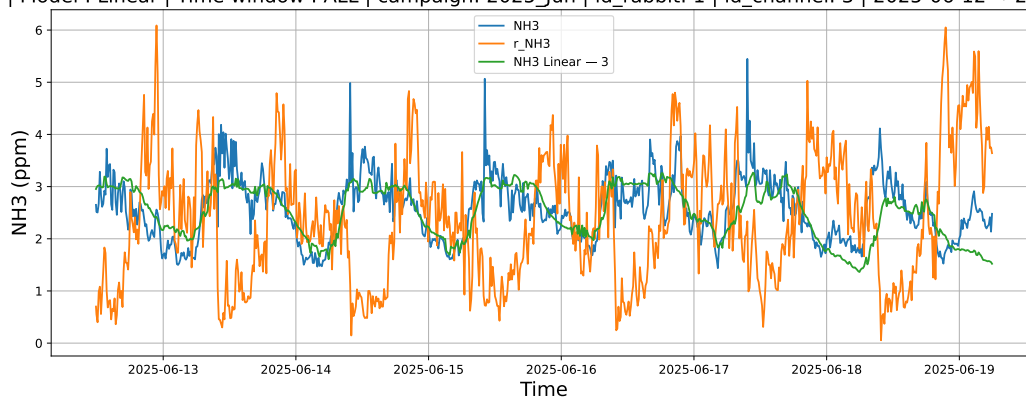


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

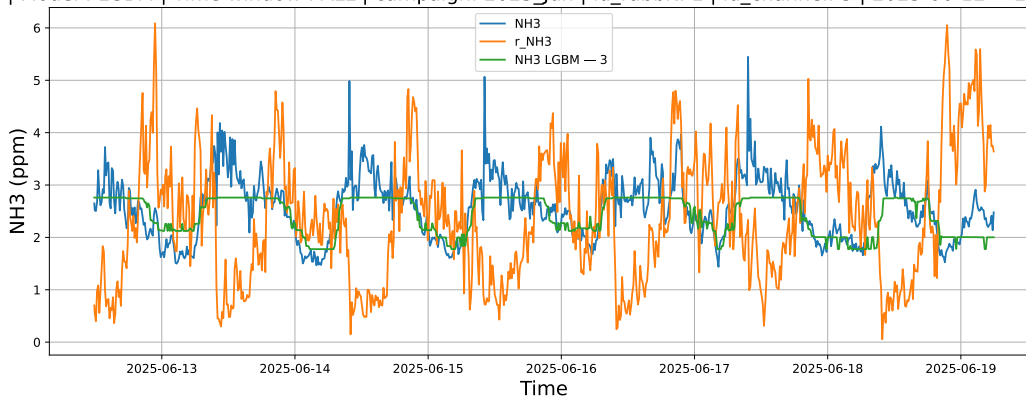


3.1.1.3 Time window : ALL

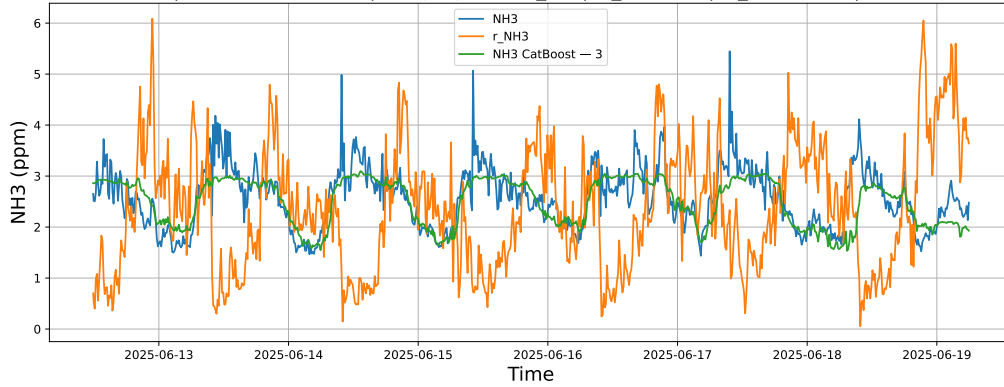
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



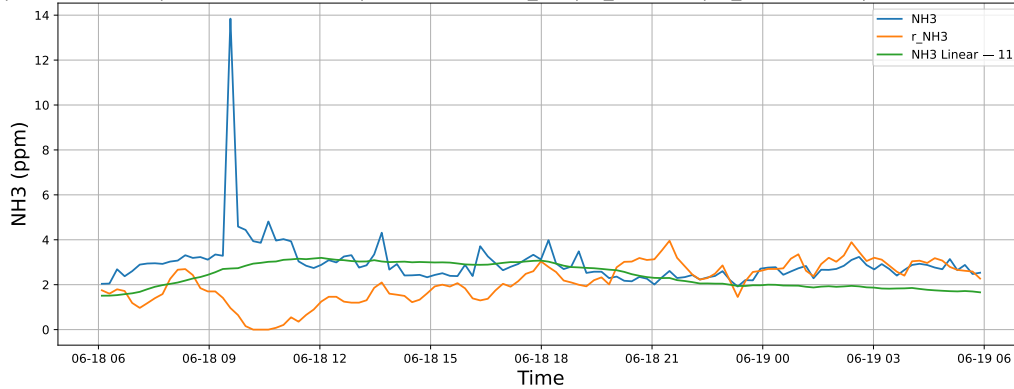
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



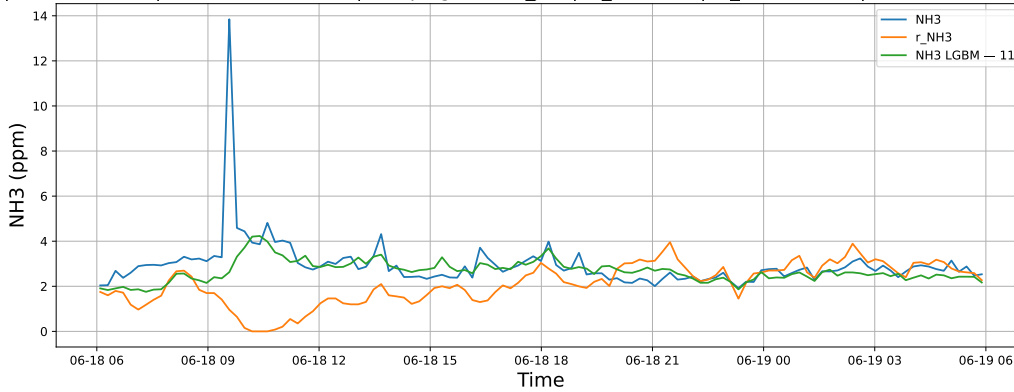
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

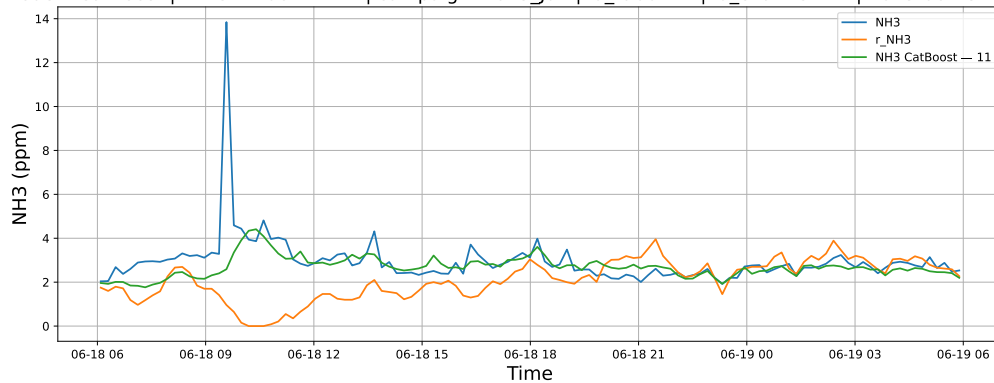
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

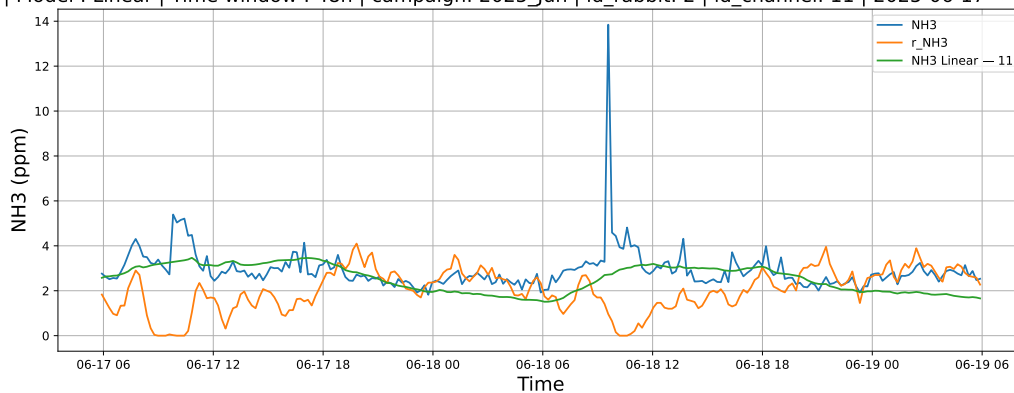


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

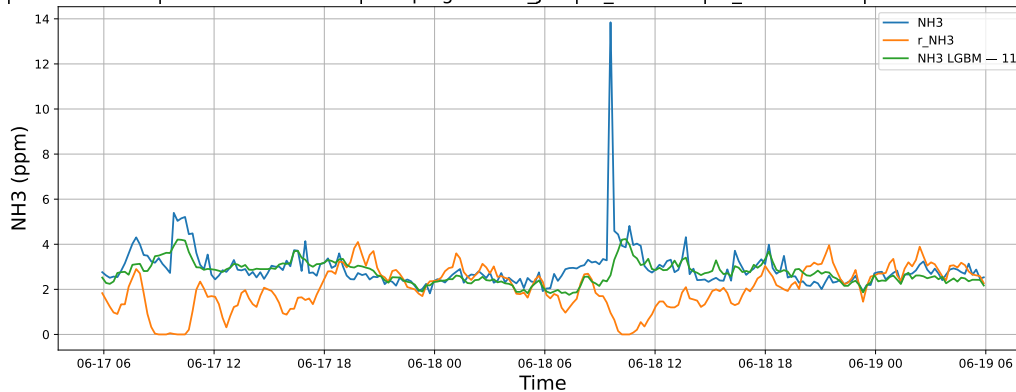


3.1.2.2 Time window : 48

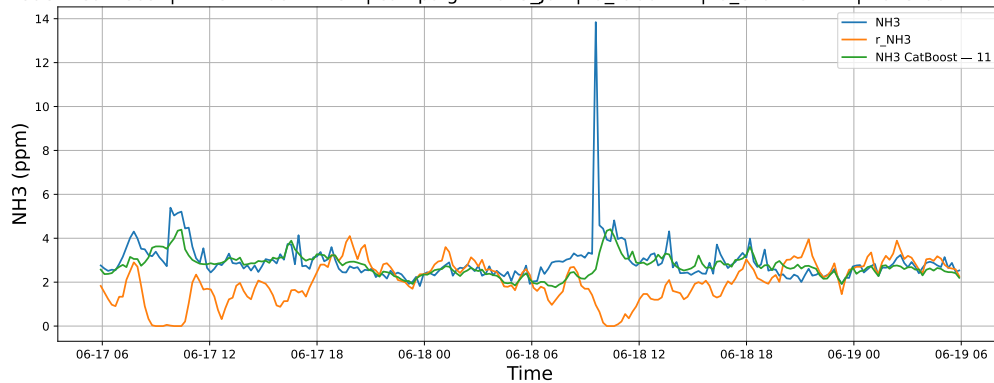
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

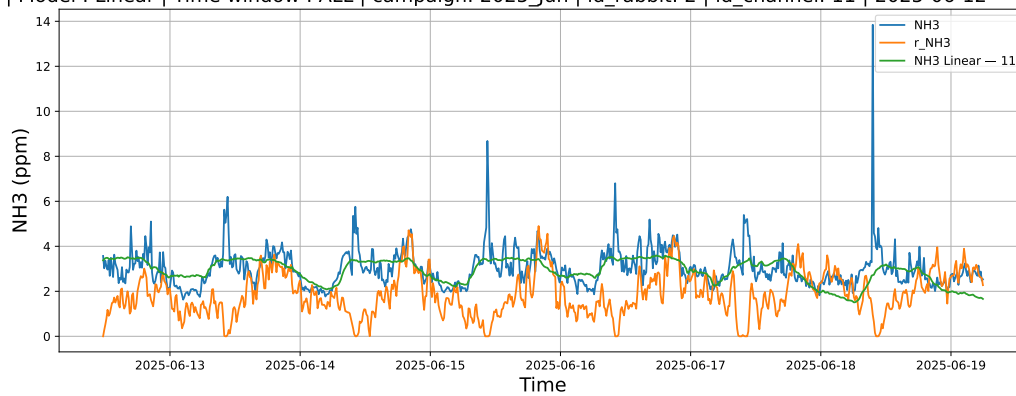


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

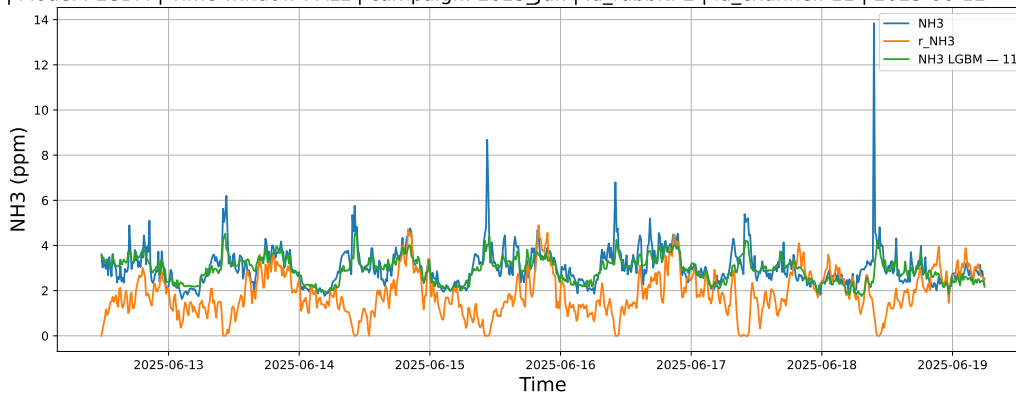


3.1.2.3 Time window : ALL

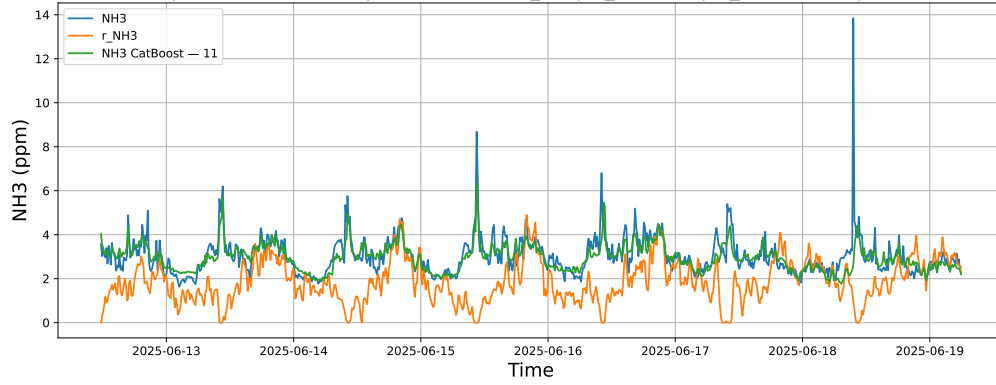
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



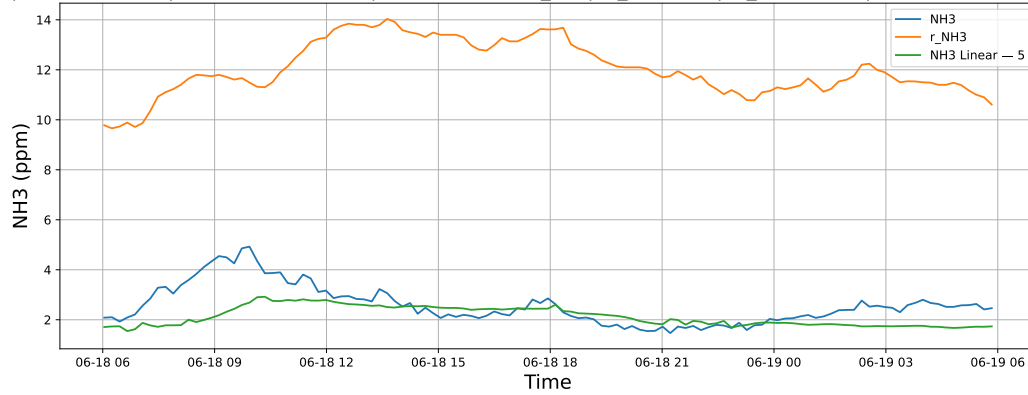
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



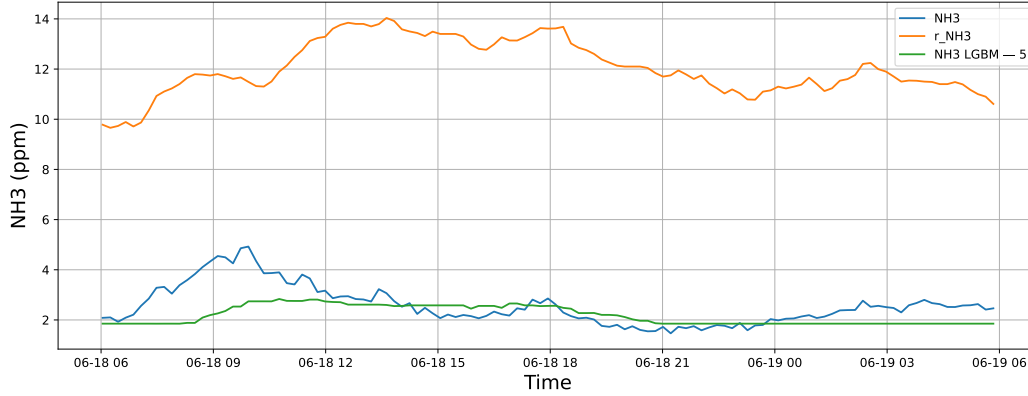
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

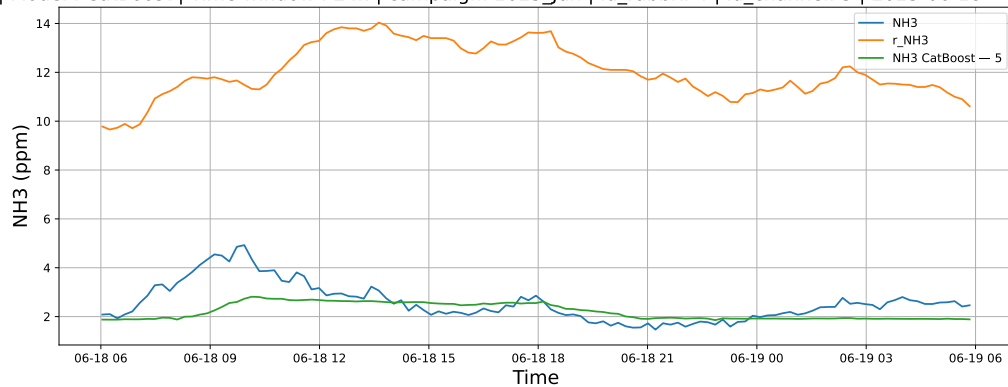
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

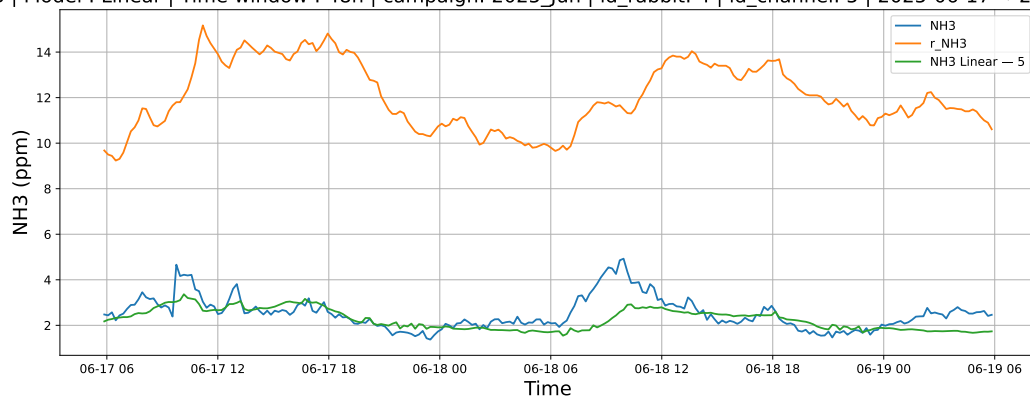


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

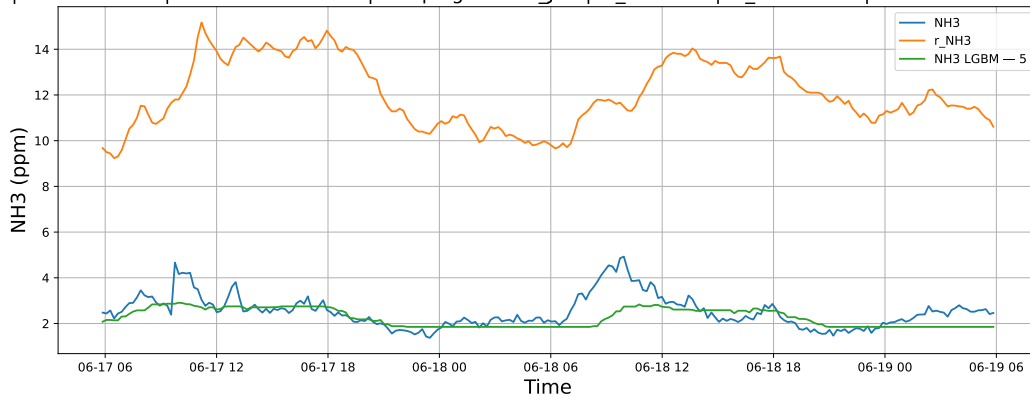


3.1.3.2 Time window : 48

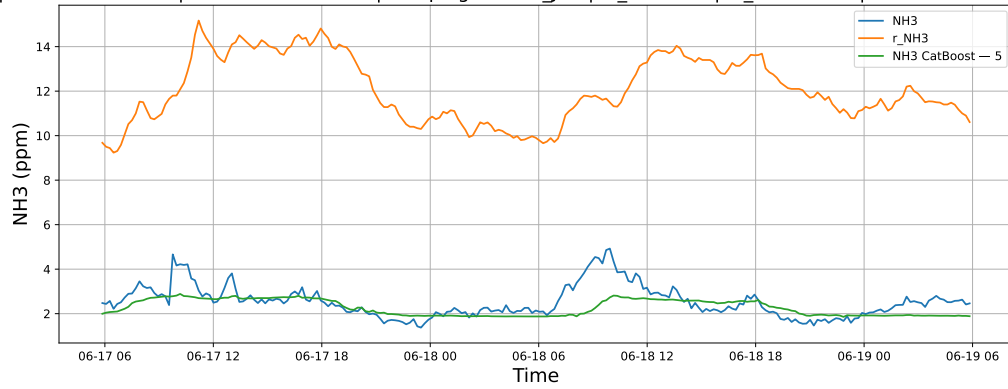
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

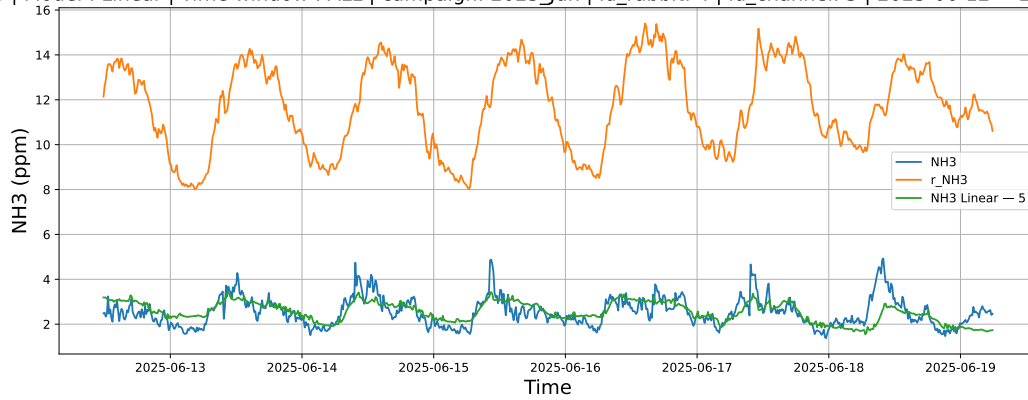


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

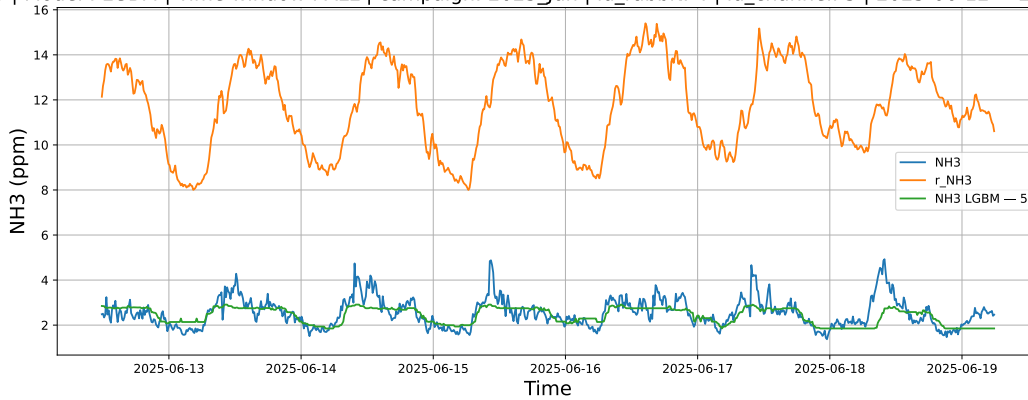


3.1.3.3 Time window : ALL

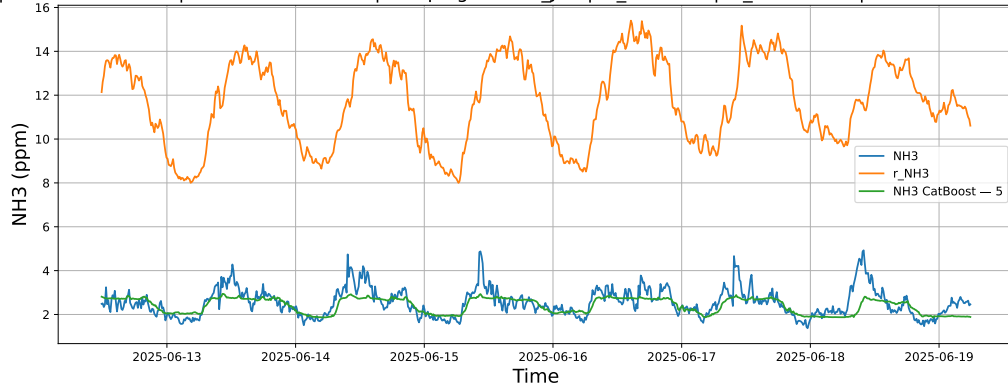
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



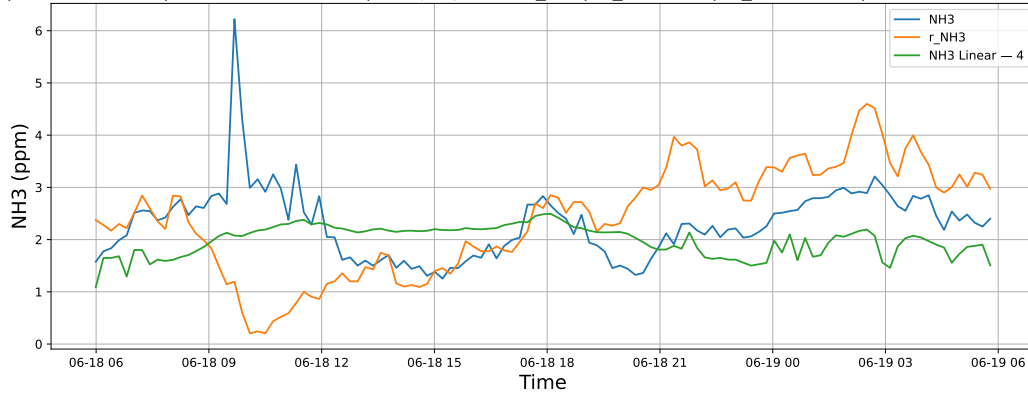
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



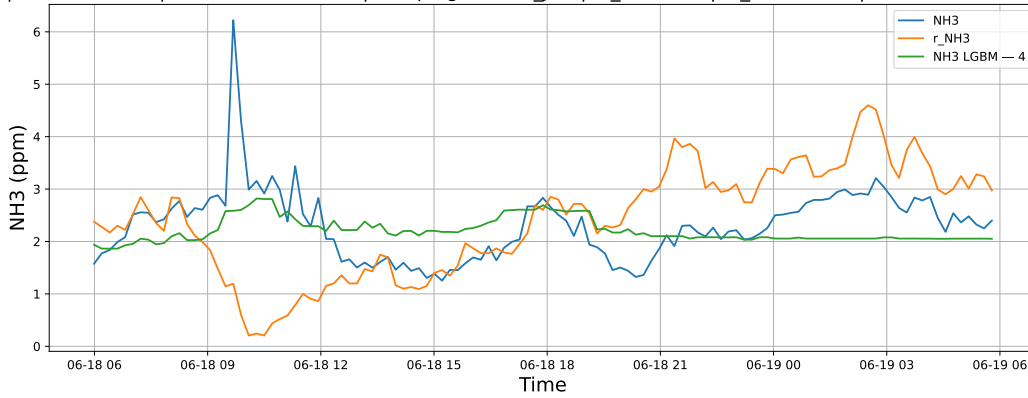
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

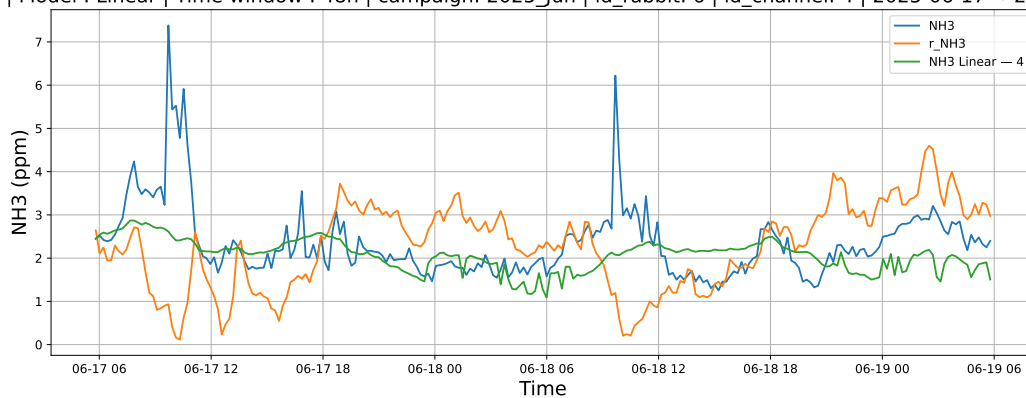


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

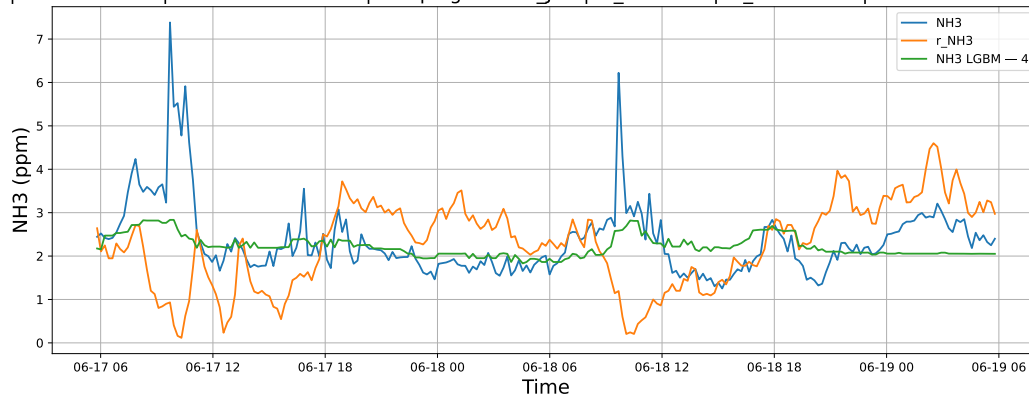


3.1.4.2 Time window : 48

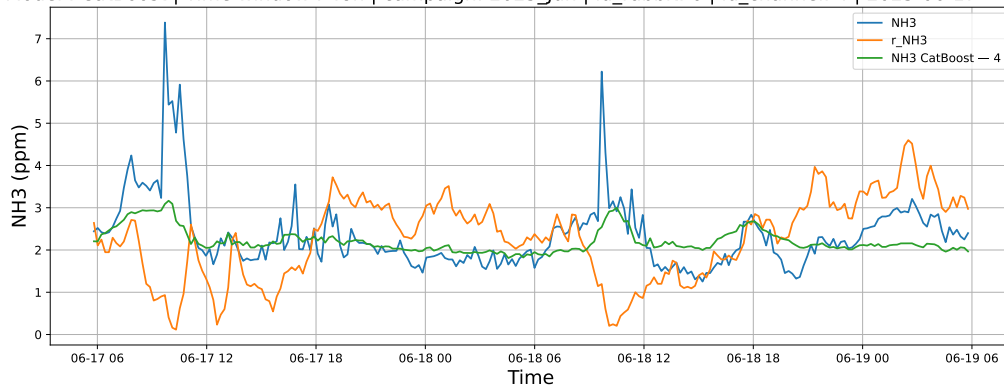
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

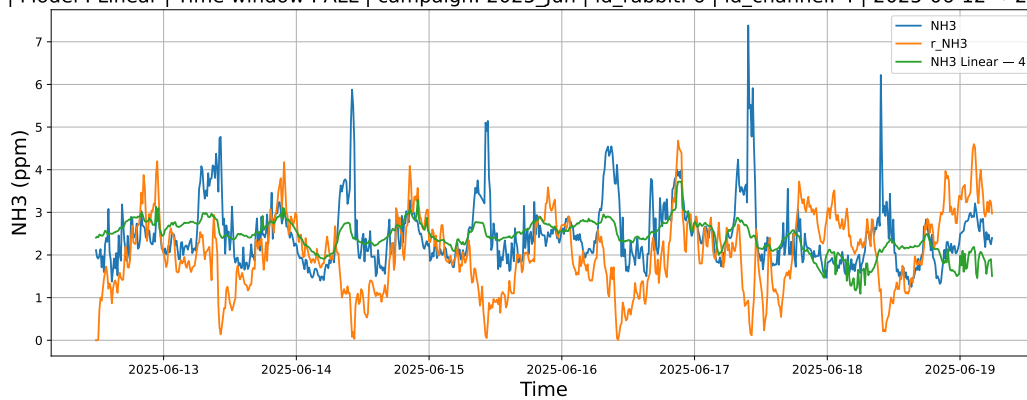


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

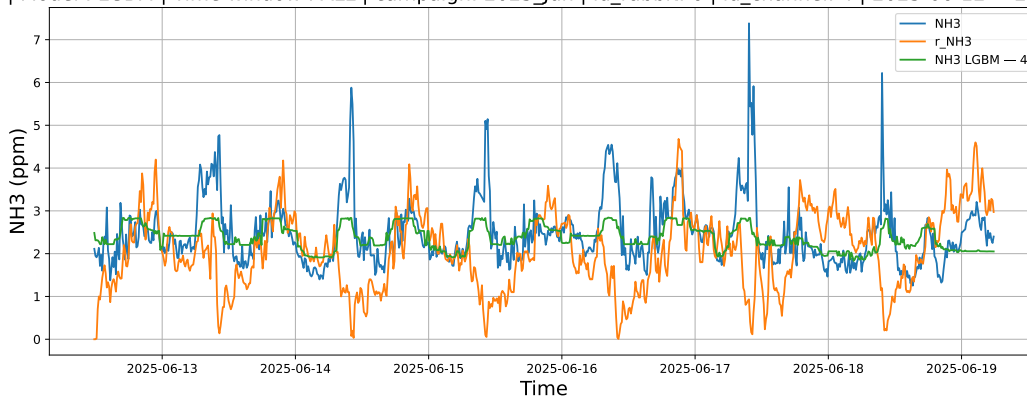


3.1.4.3 Time window : ALL

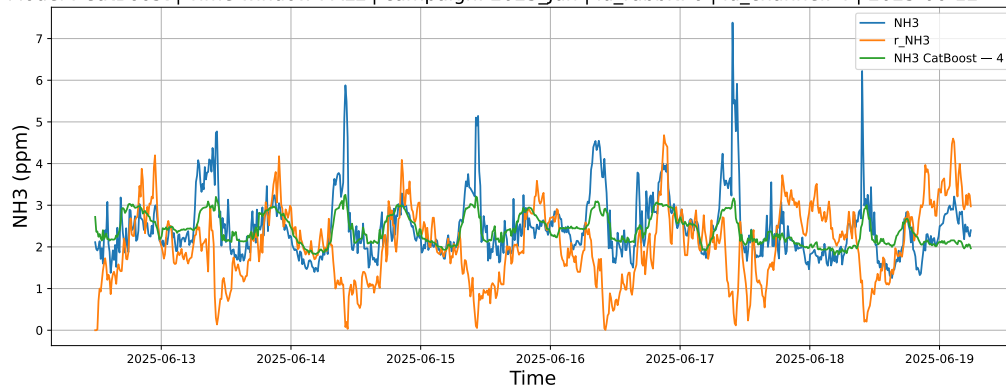
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



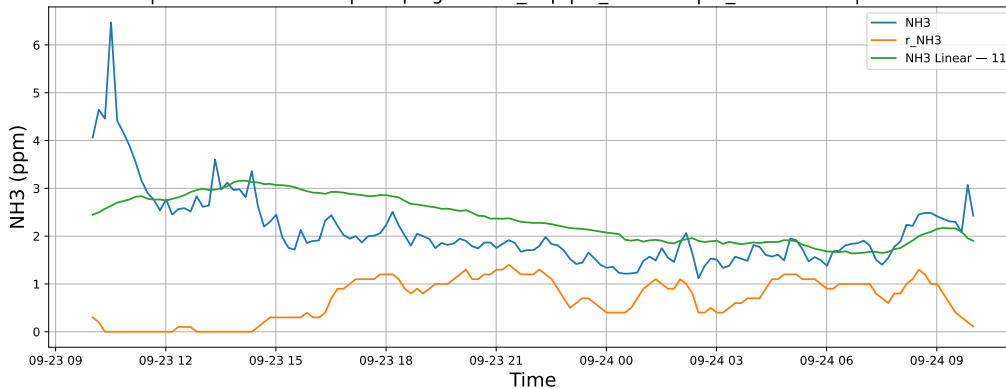
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



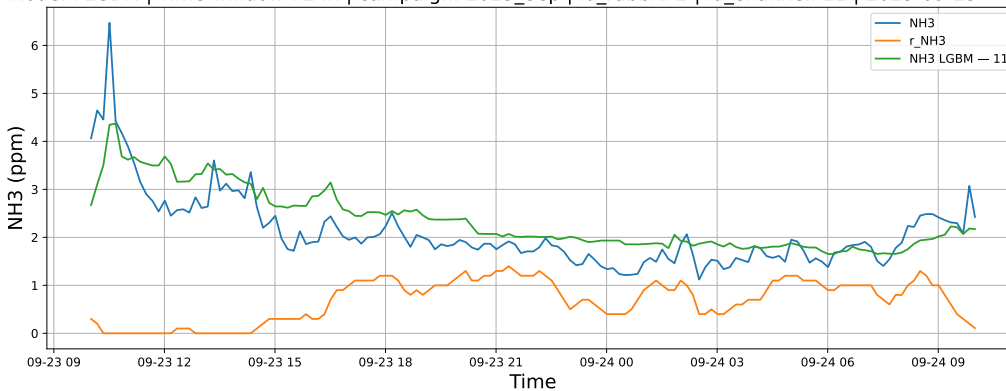
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

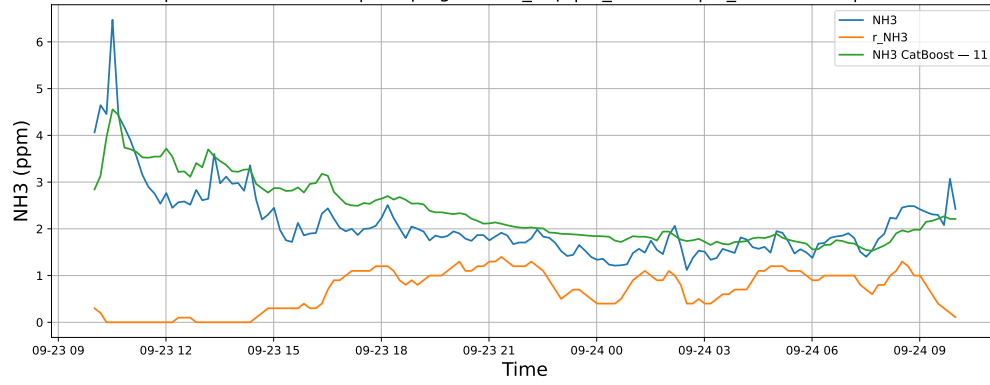
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

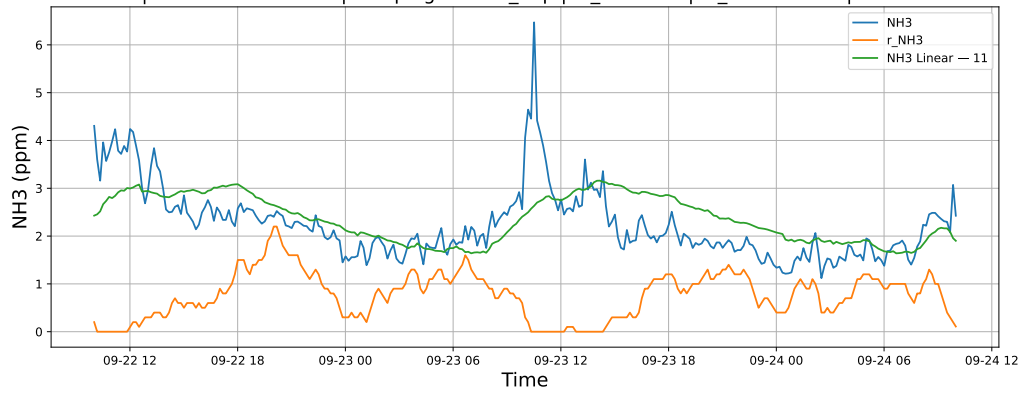


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

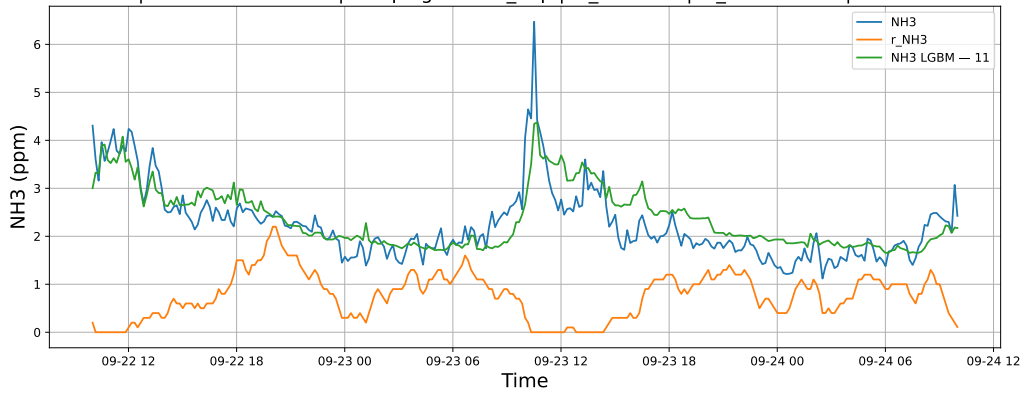


3.1.5.2 Time window : 48

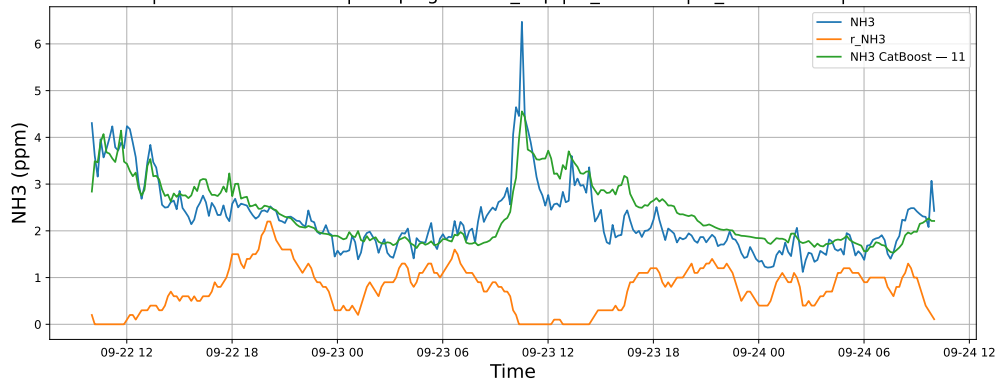
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

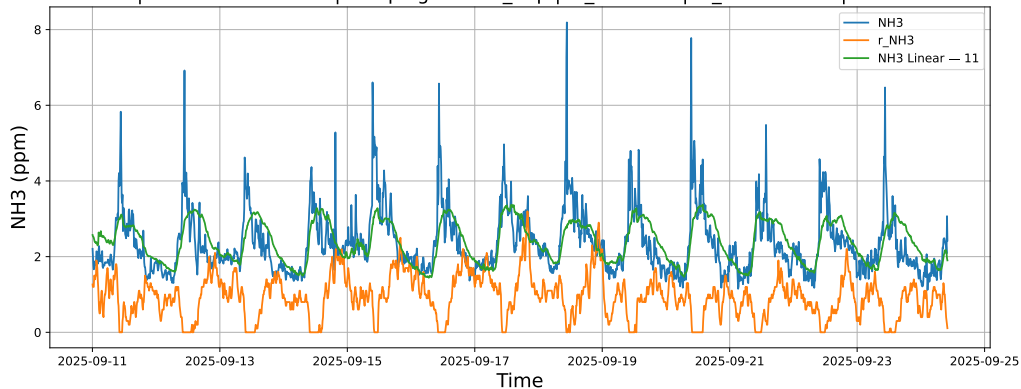


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

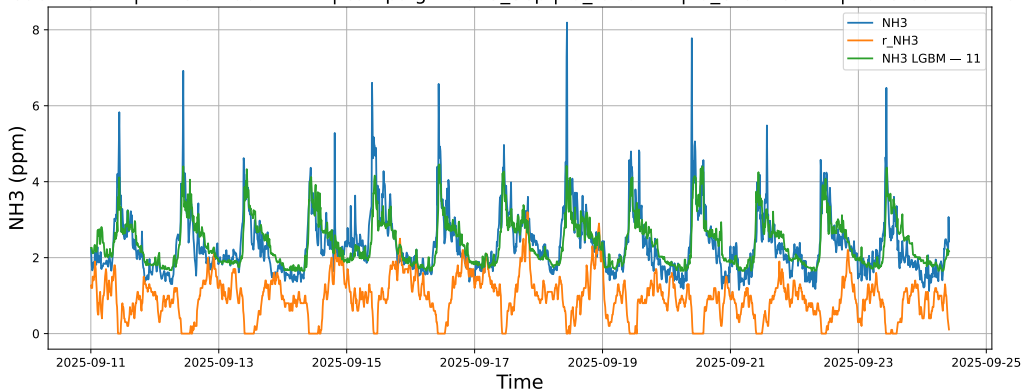


3.1.5.3 Time window : ALL

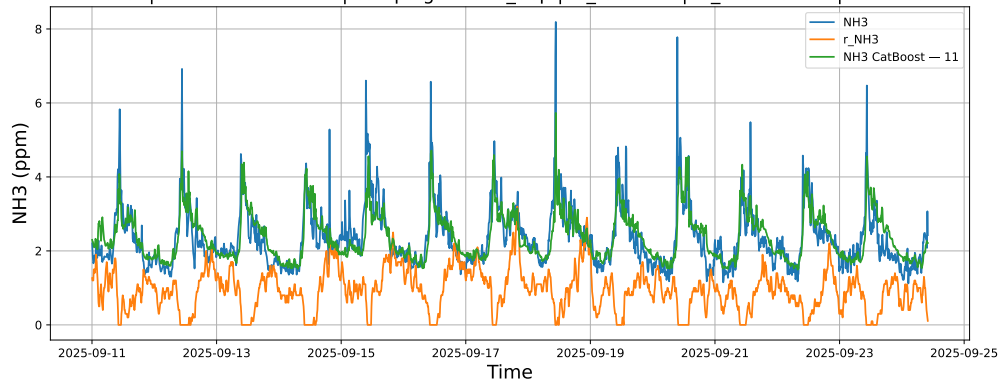
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



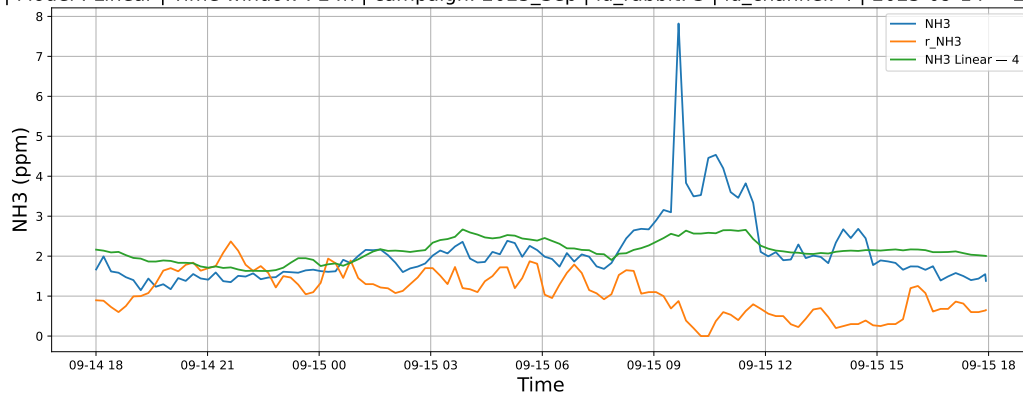
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



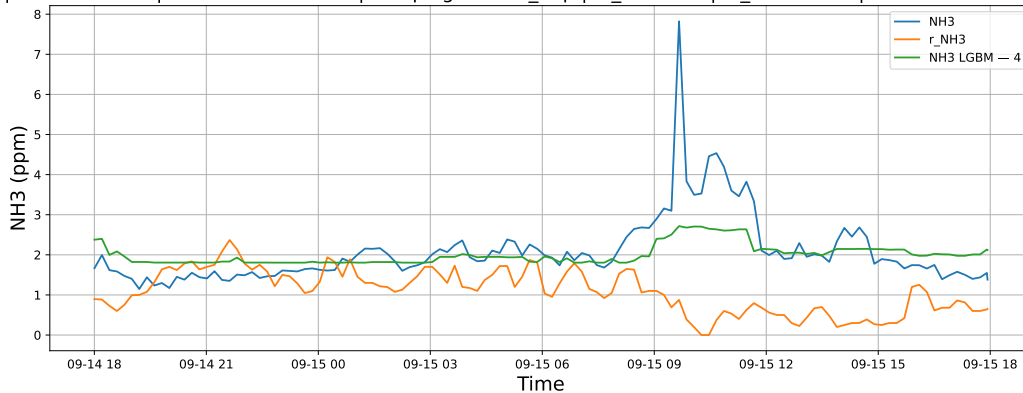
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

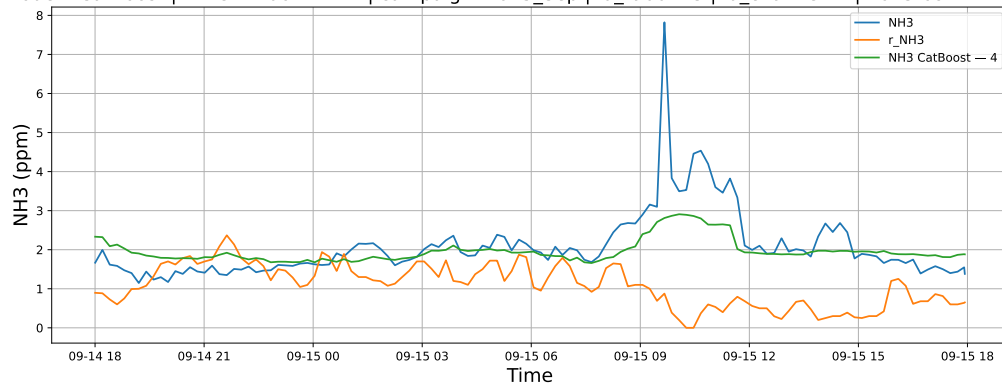
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

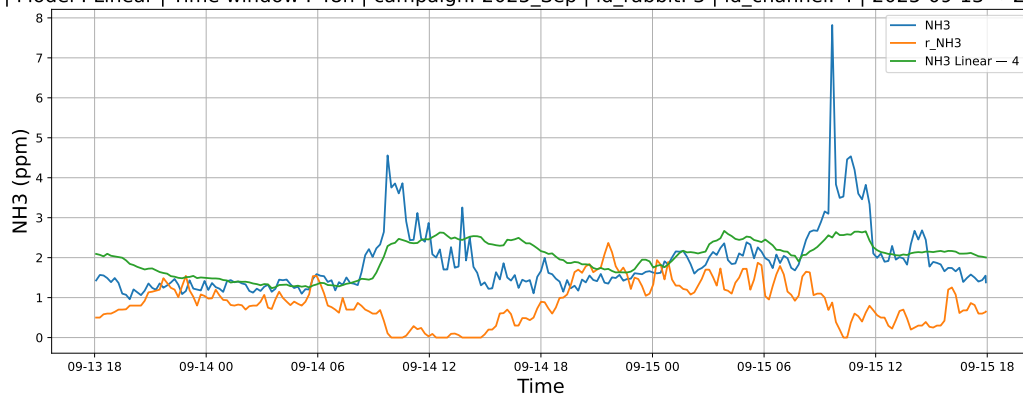


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

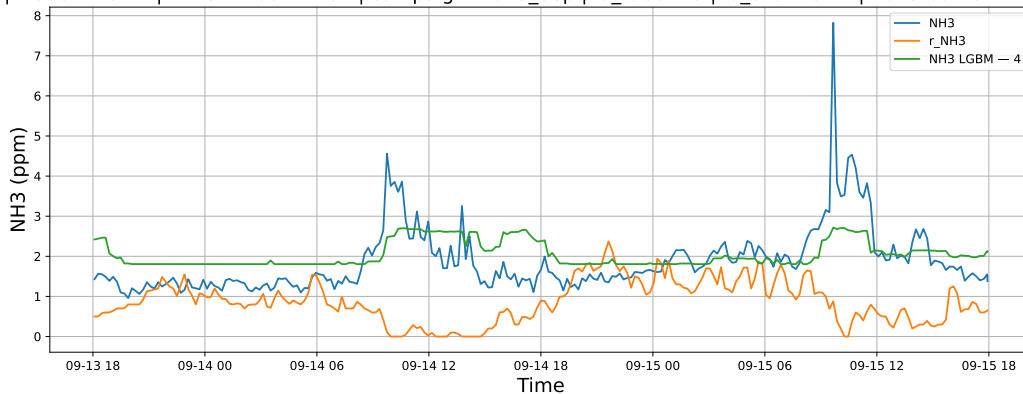


3.1.6.2 Time window : 48

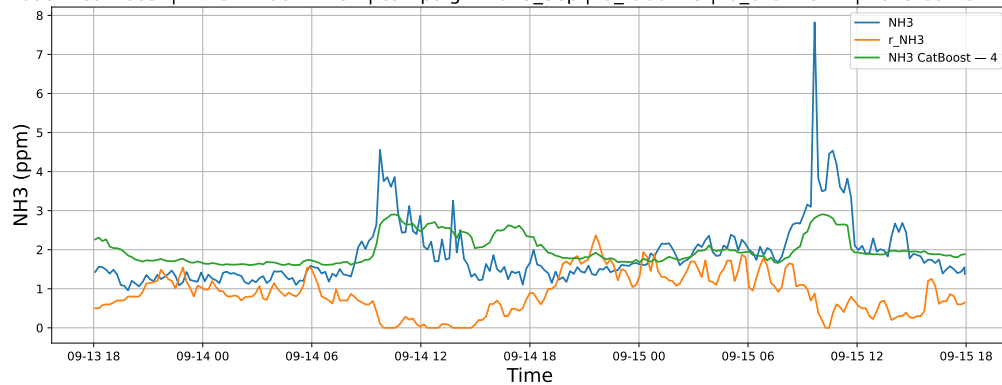
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

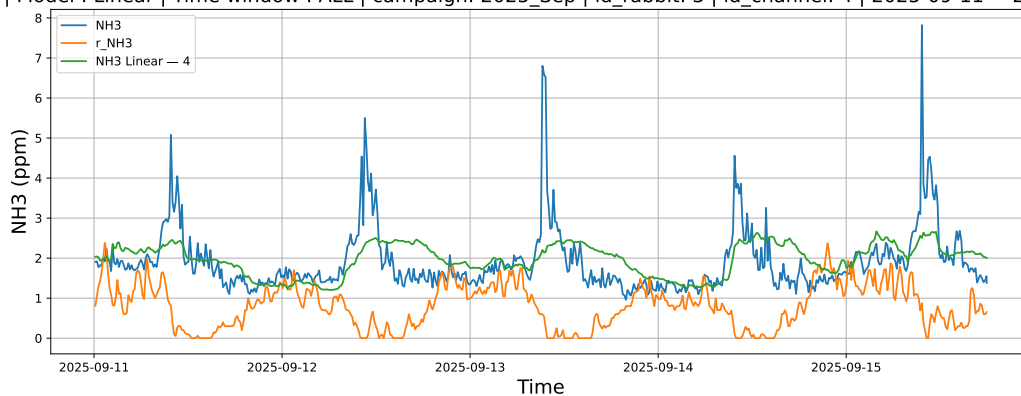


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

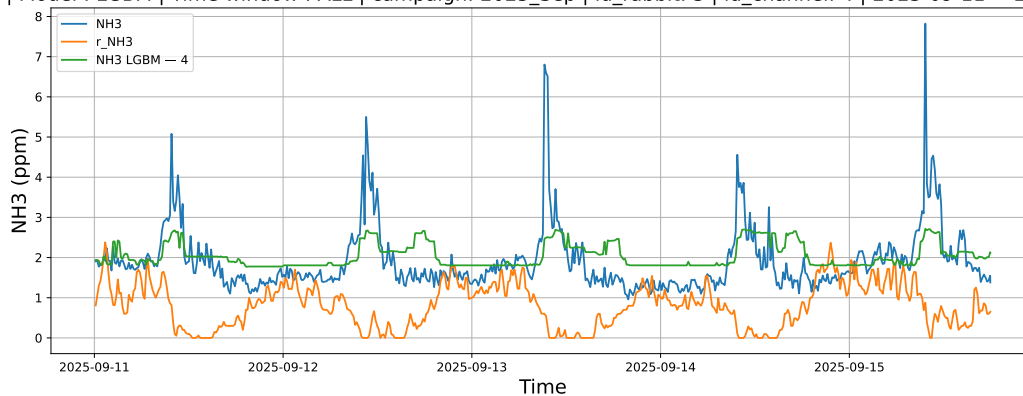


3.1.6.3 Time window : ALL

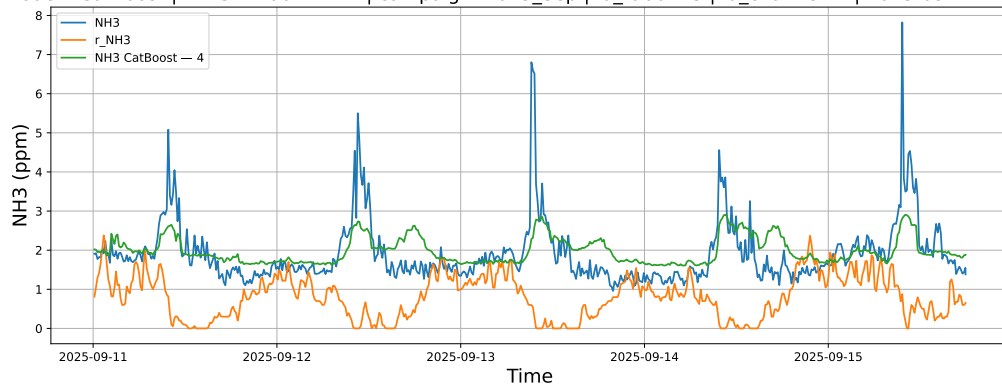
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



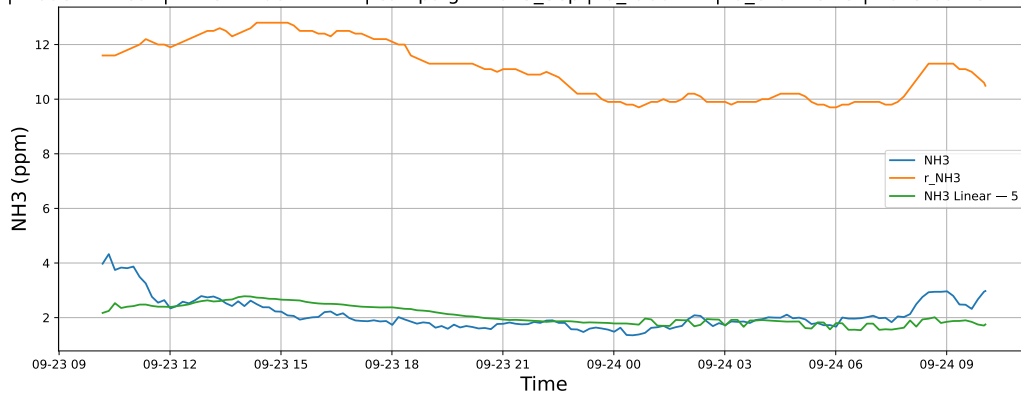
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



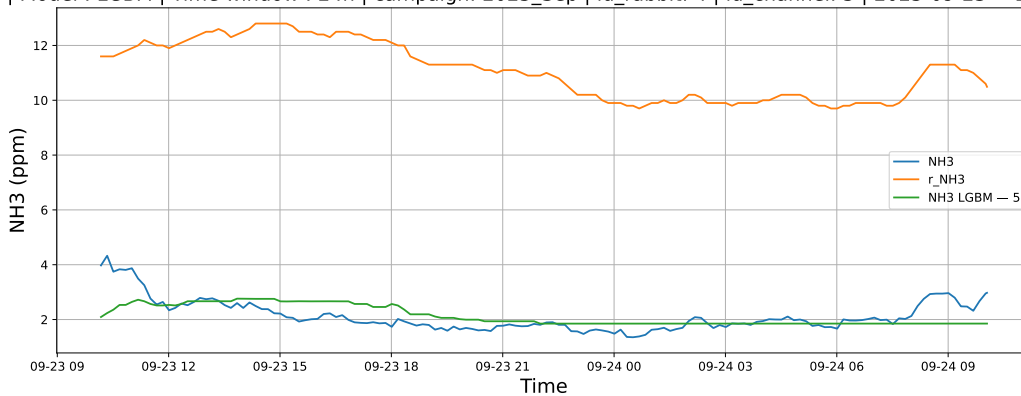
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

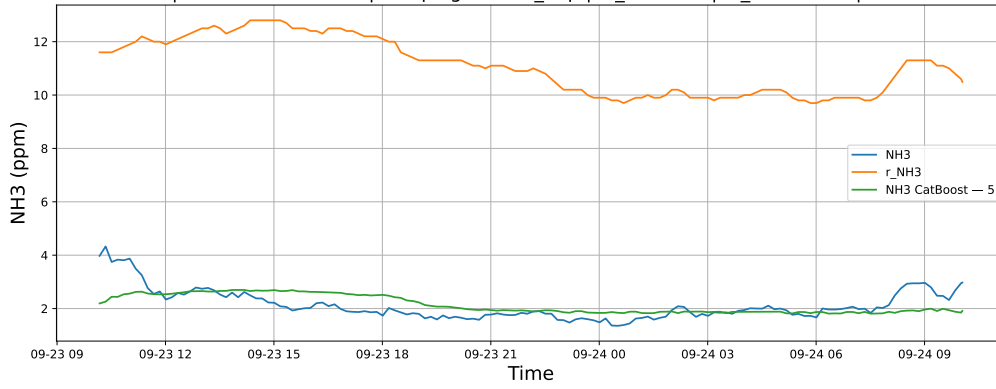
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

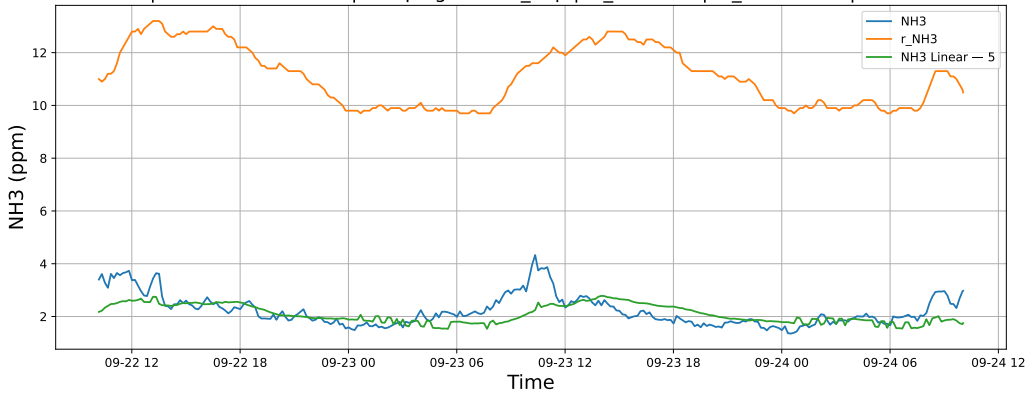


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

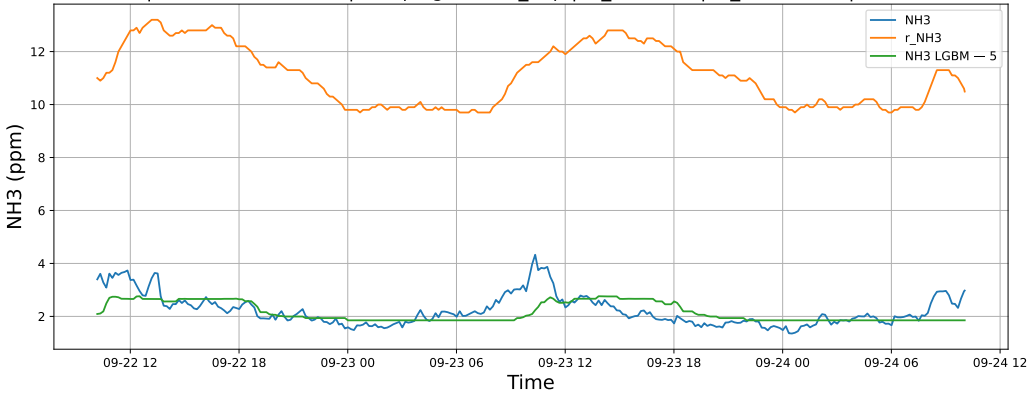


3.1.7.2 Time window : 48

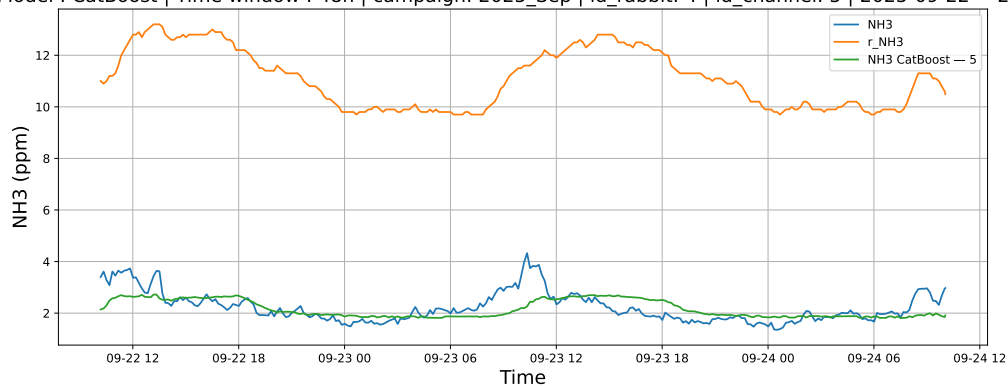
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

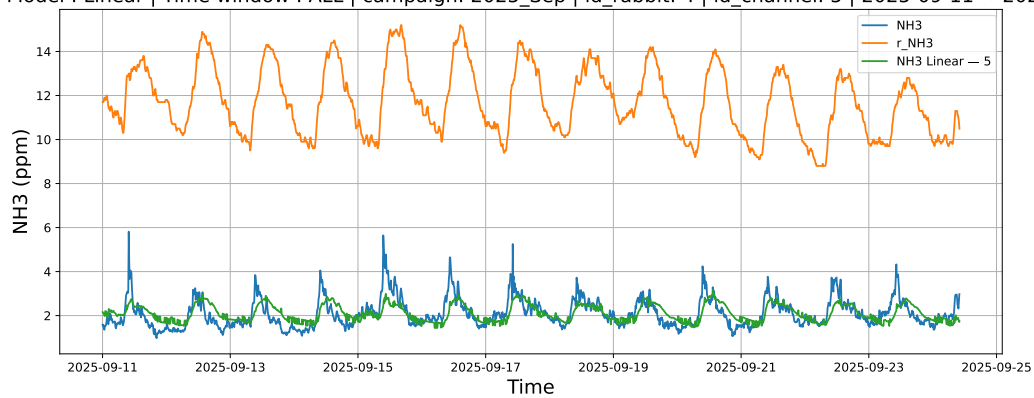


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

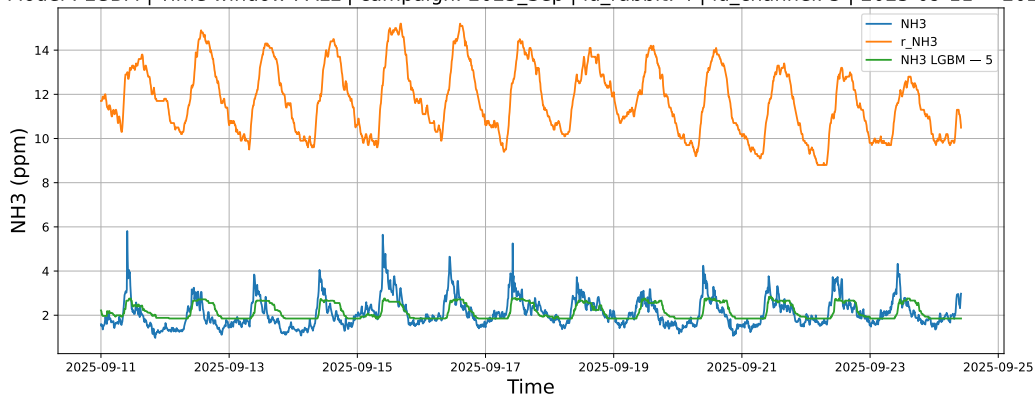


3.1.7.3 Time window : ALL

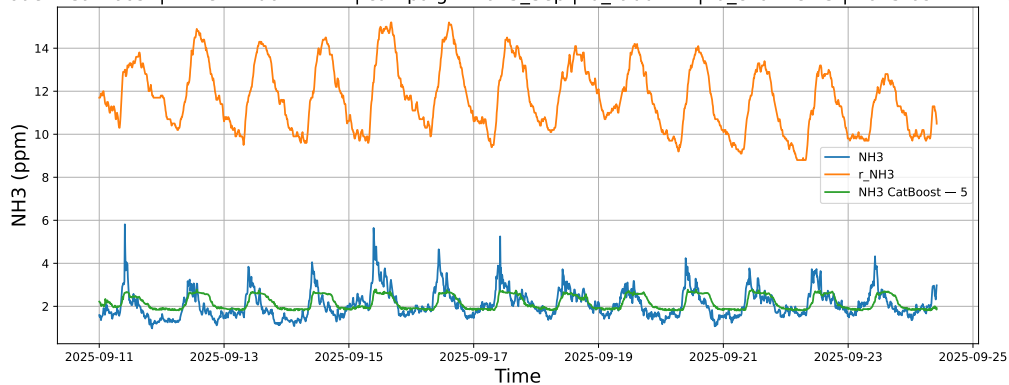
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



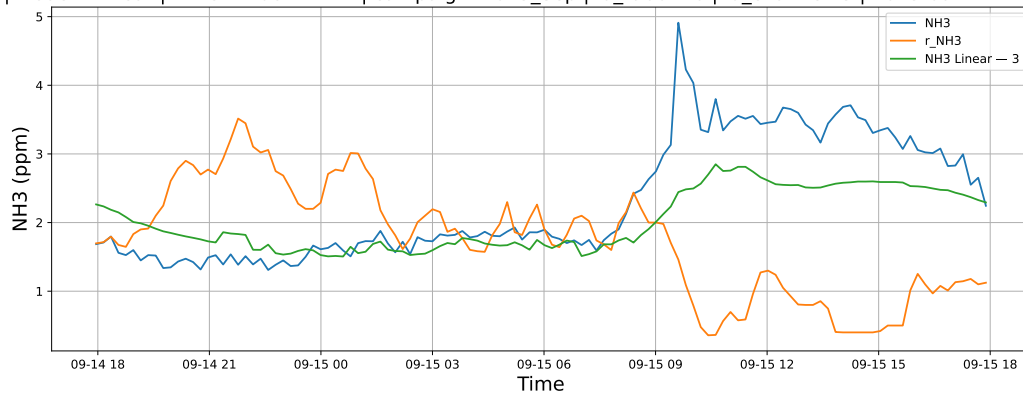
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



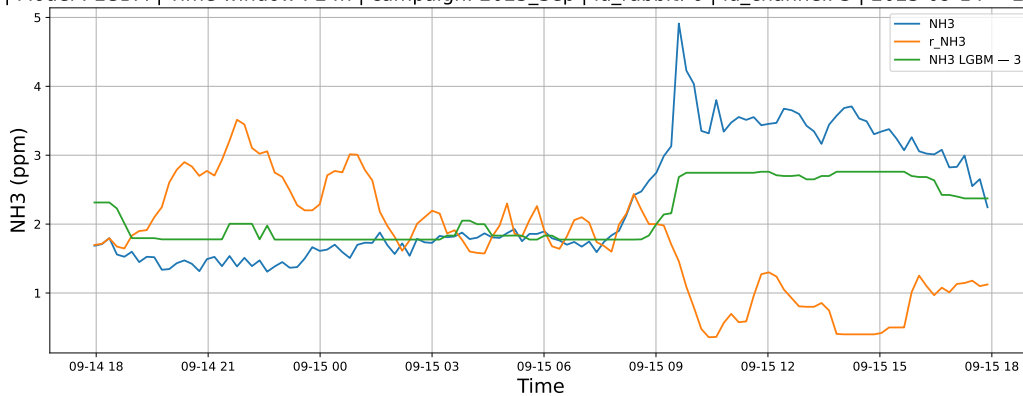
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

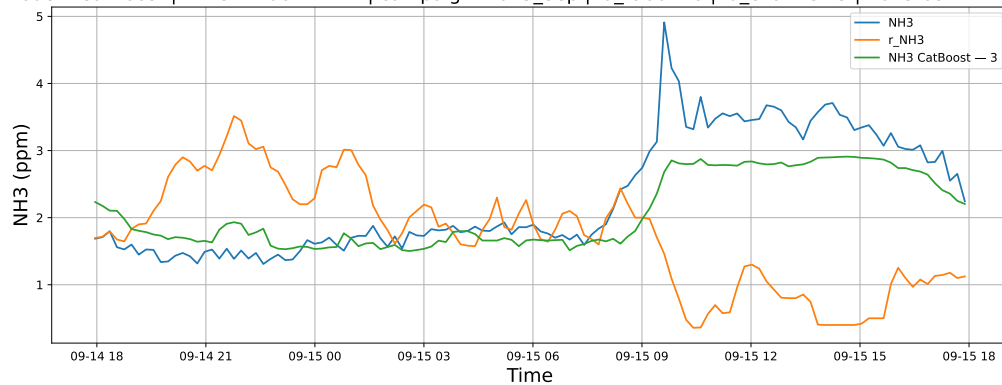
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

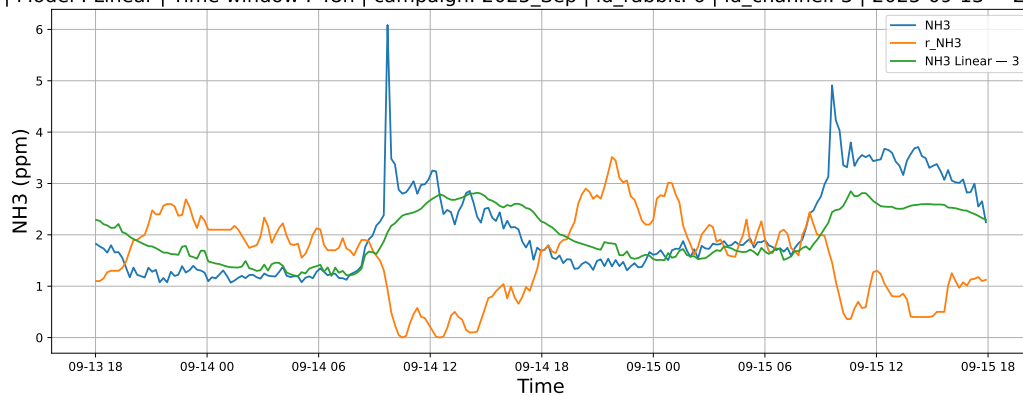


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

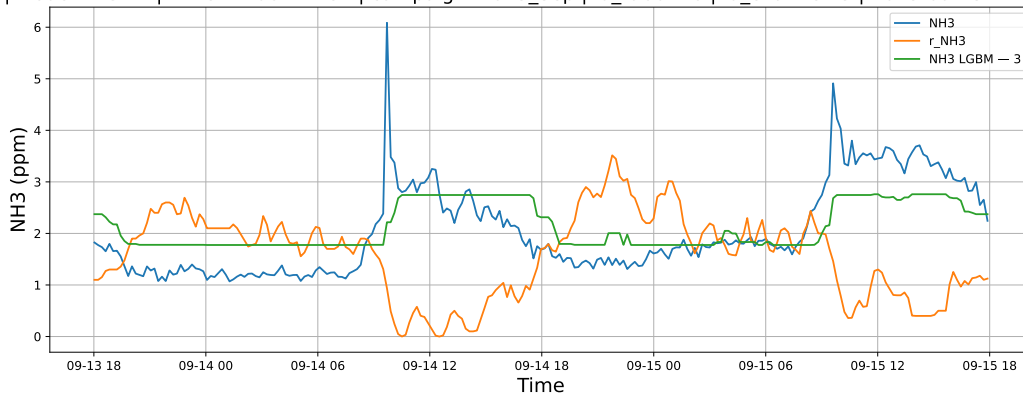


3.1.8.2 Time window : 48

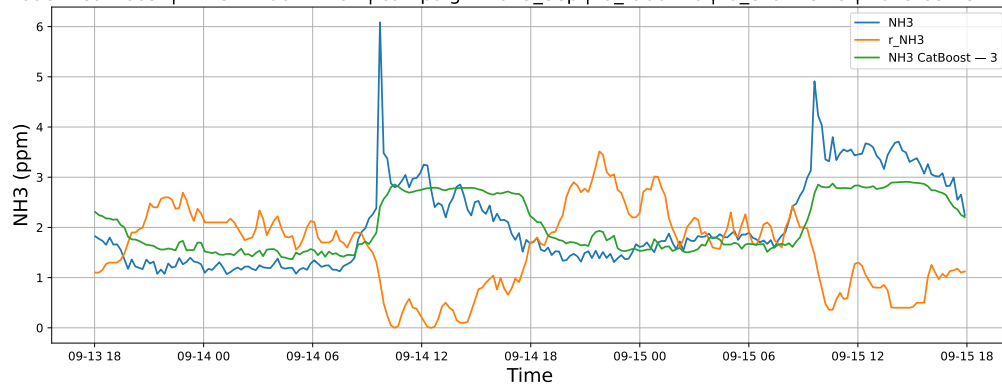
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

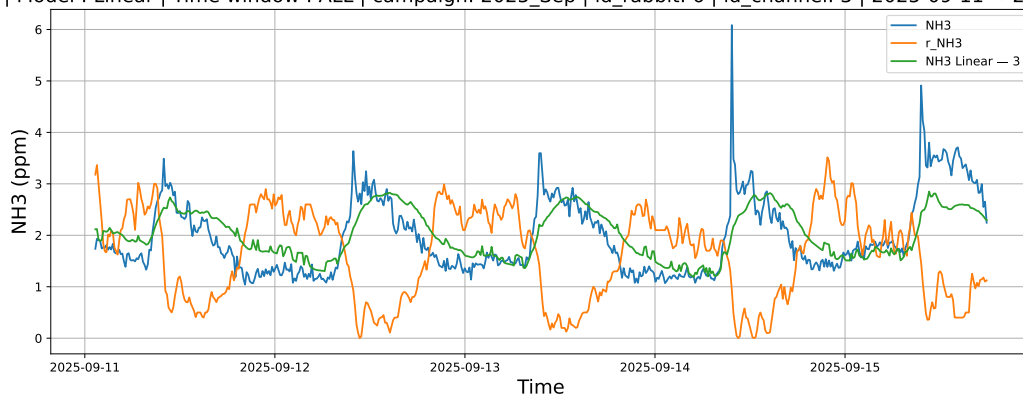


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

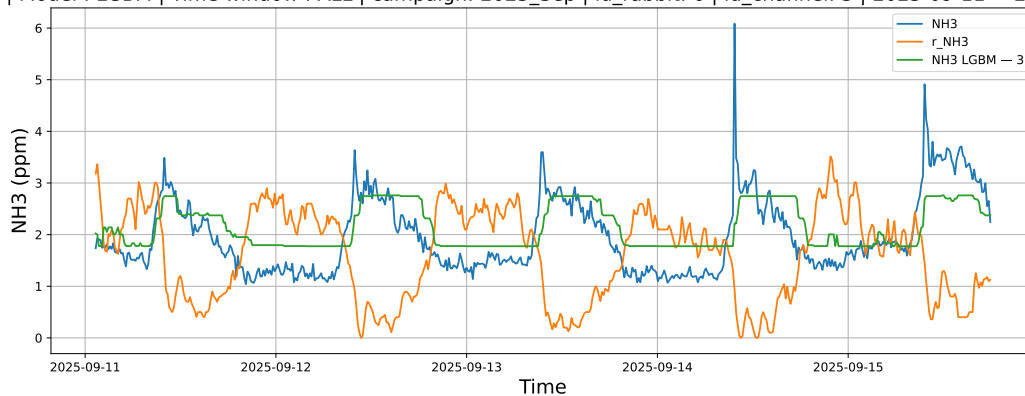


3.1.8.3 Time window : ALL

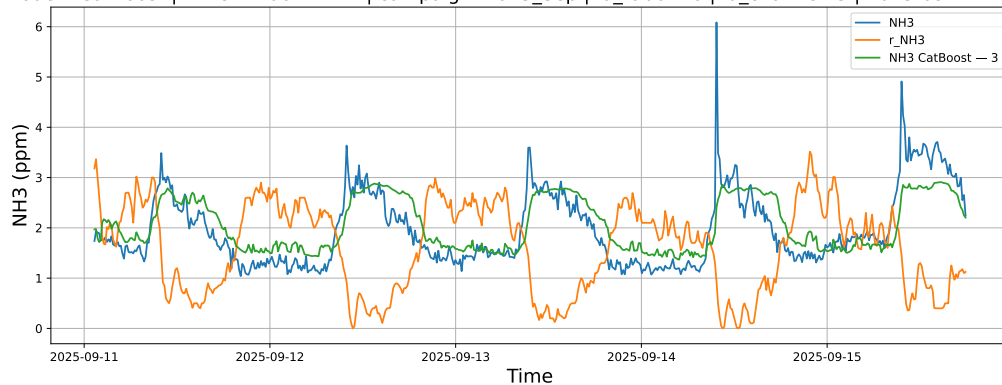
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

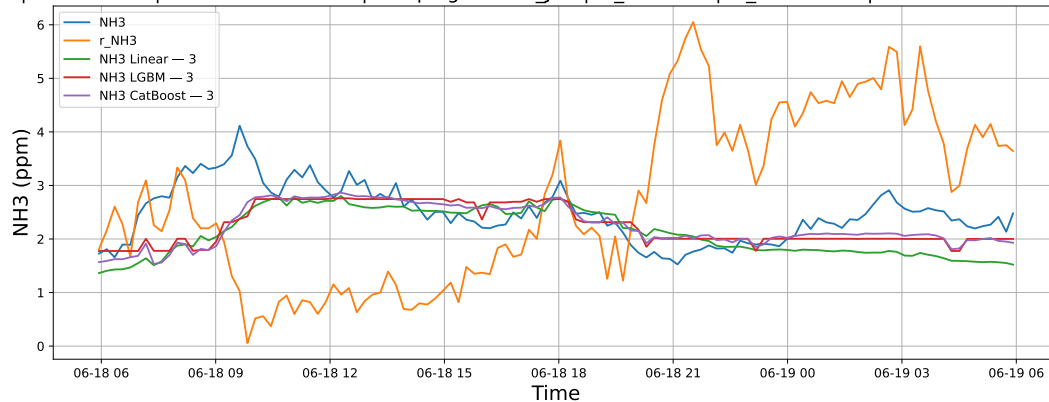


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

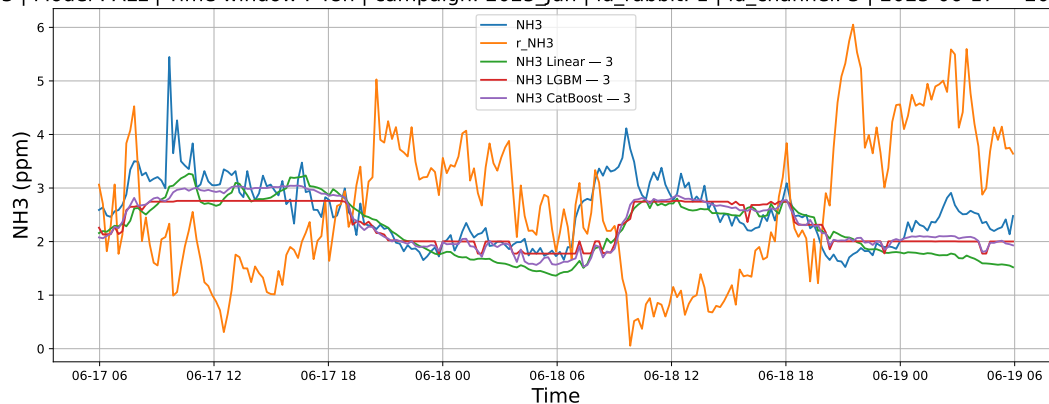
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



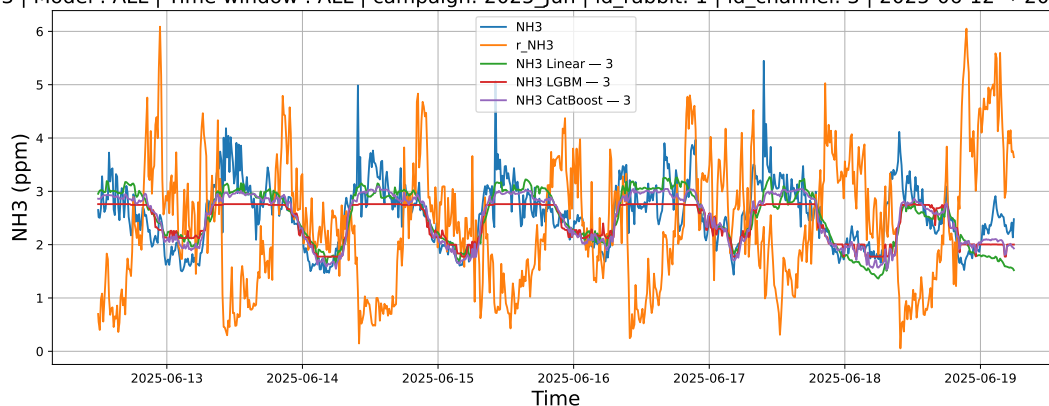
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

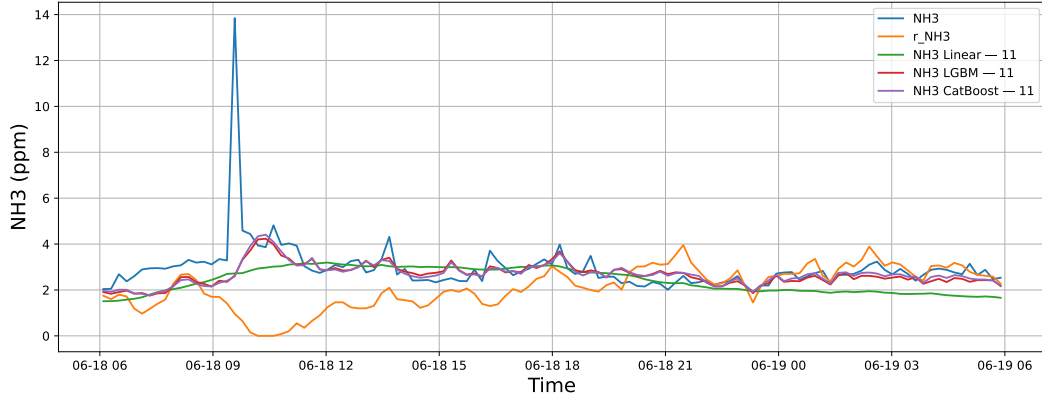
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

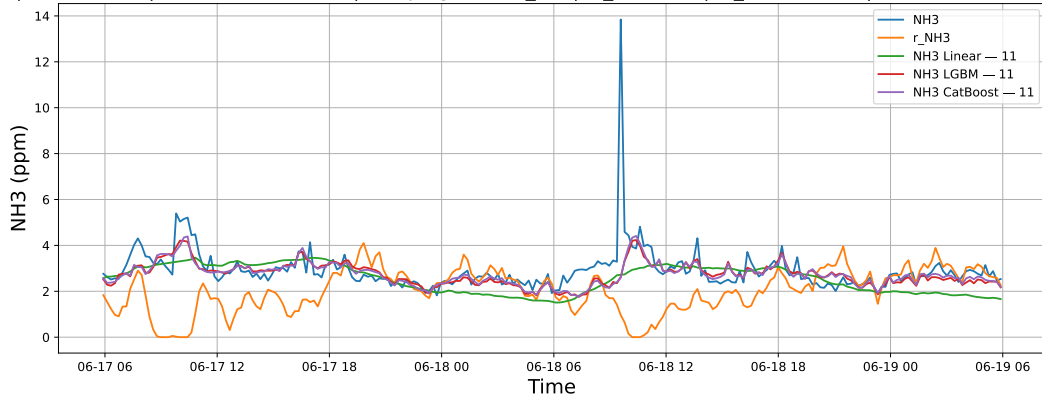
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



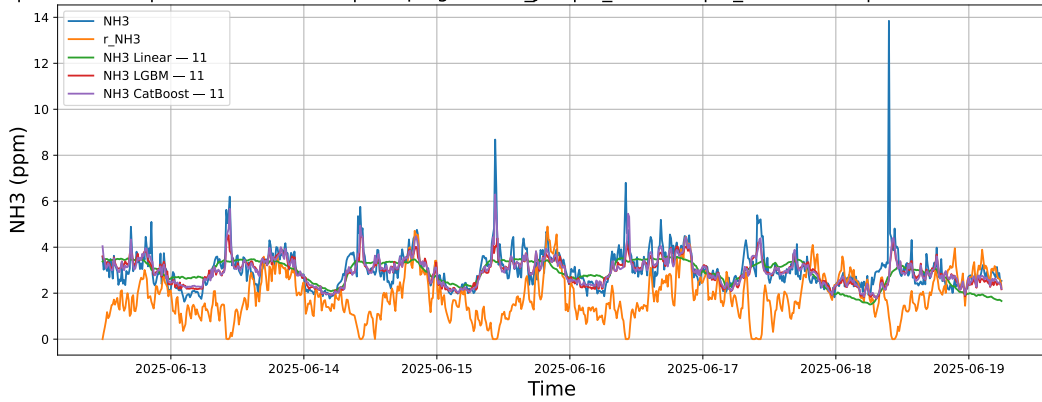
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

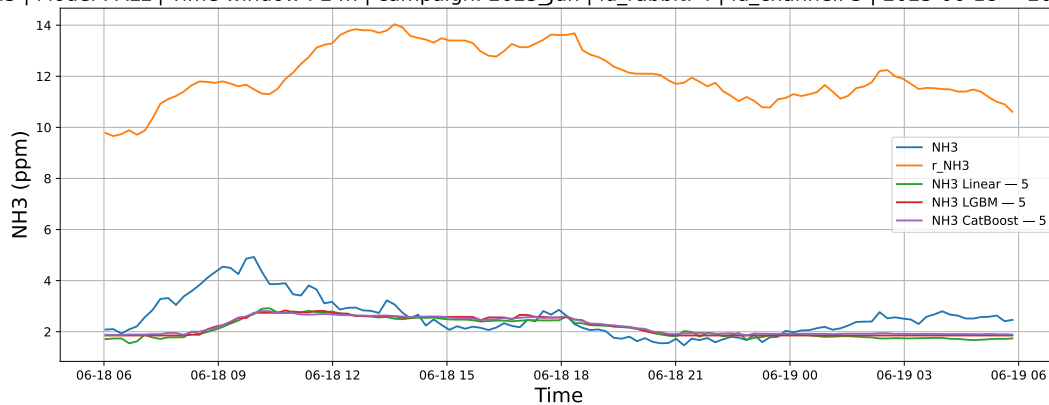
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

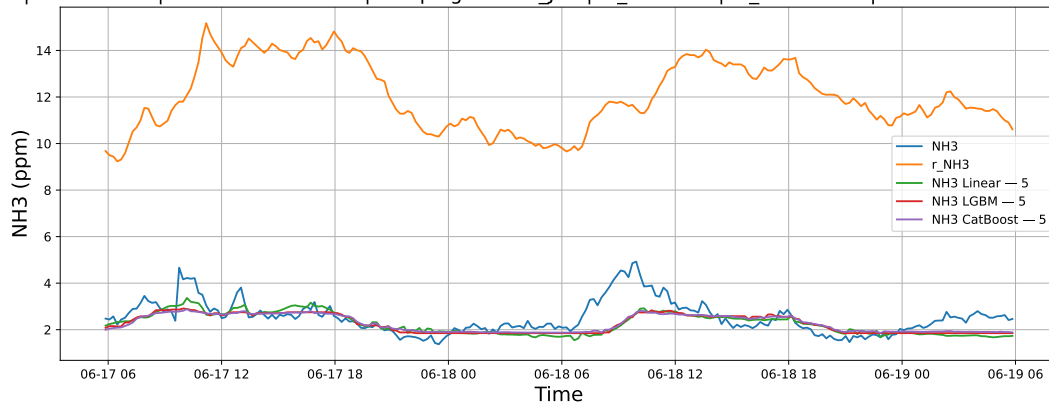
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



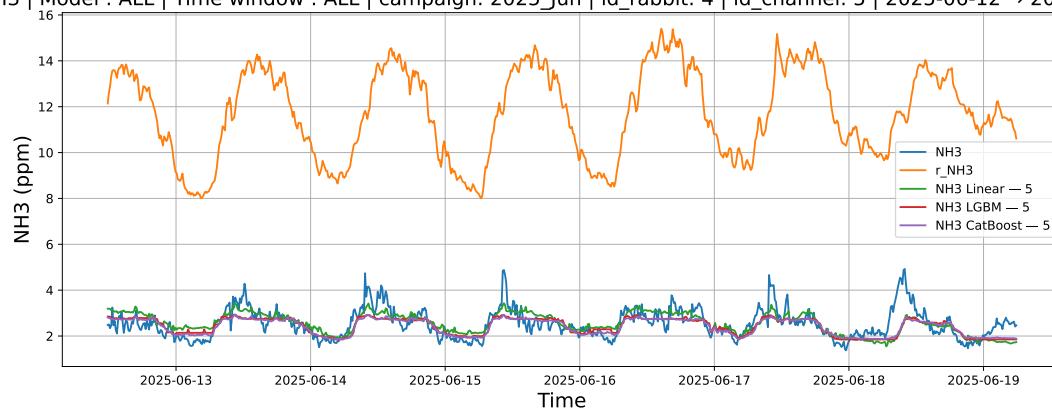
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

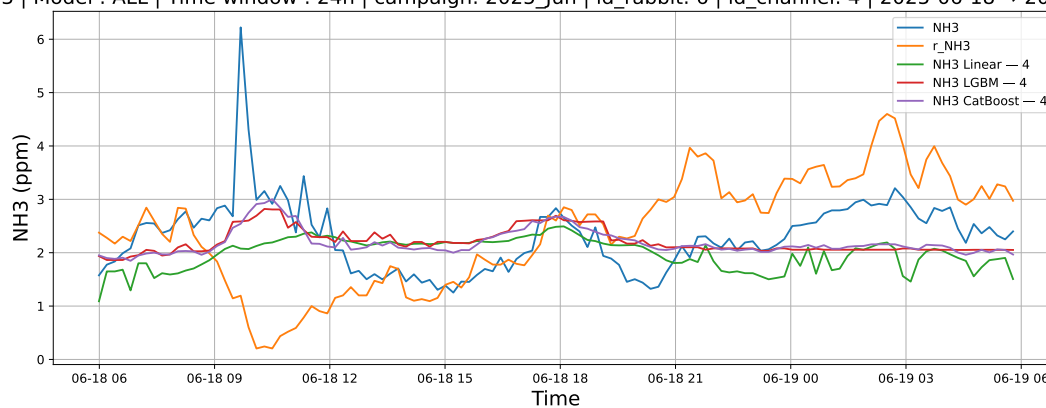
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

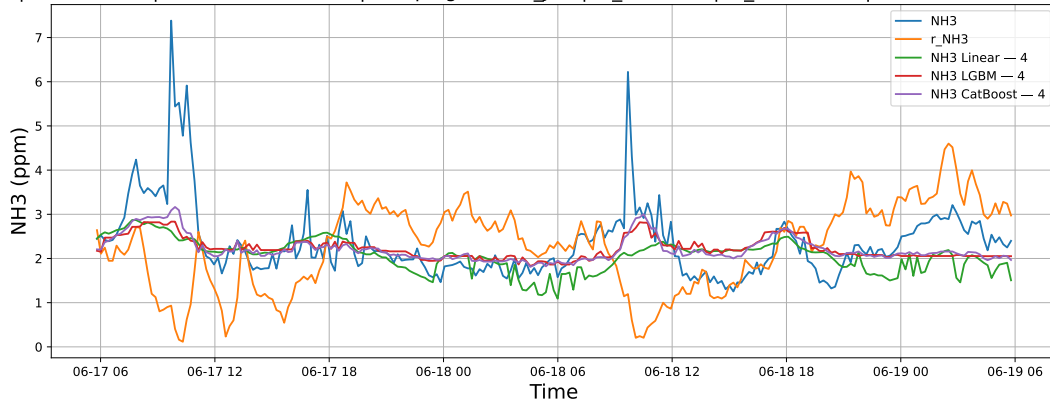
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



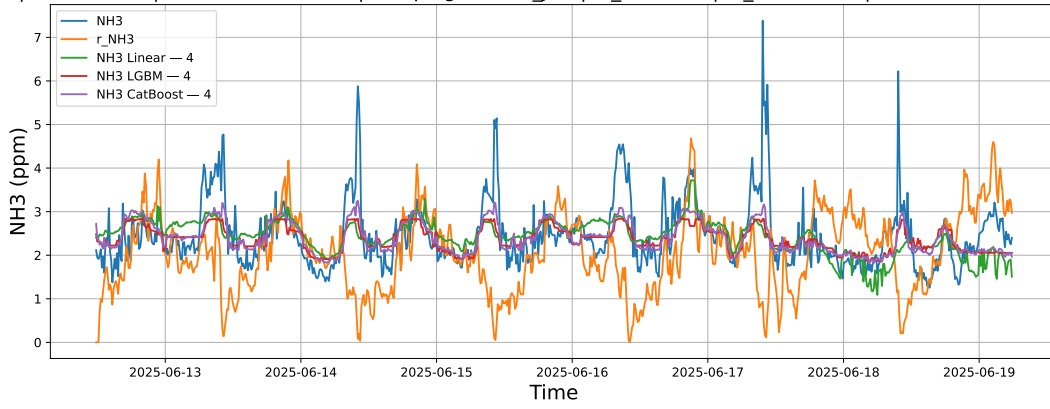
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

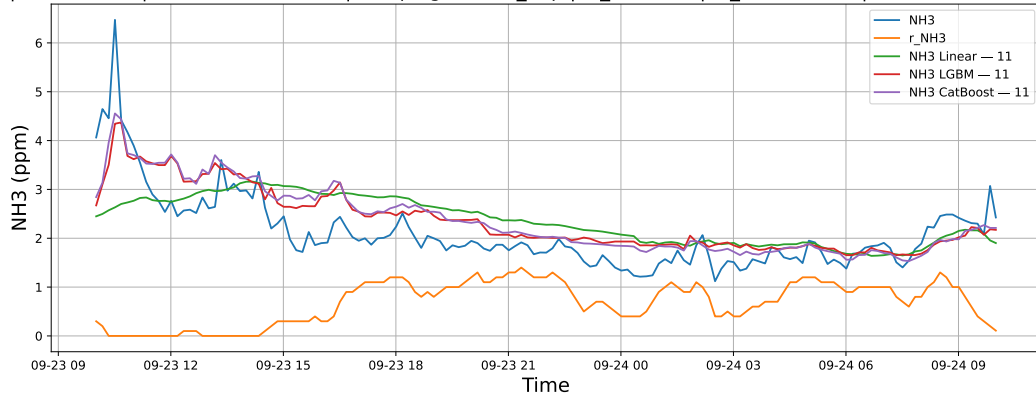
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

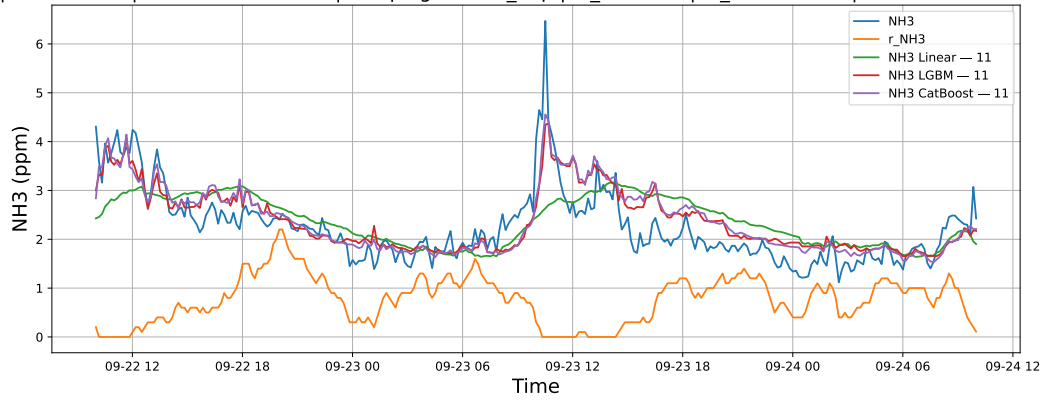
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



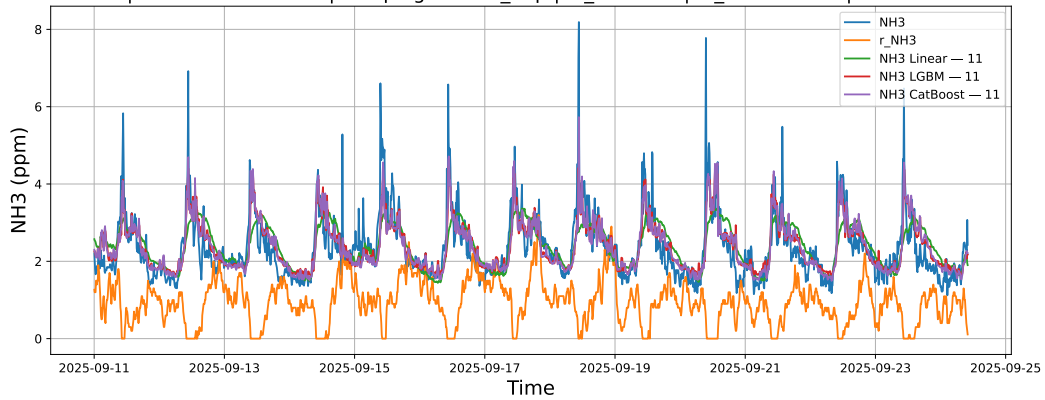
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

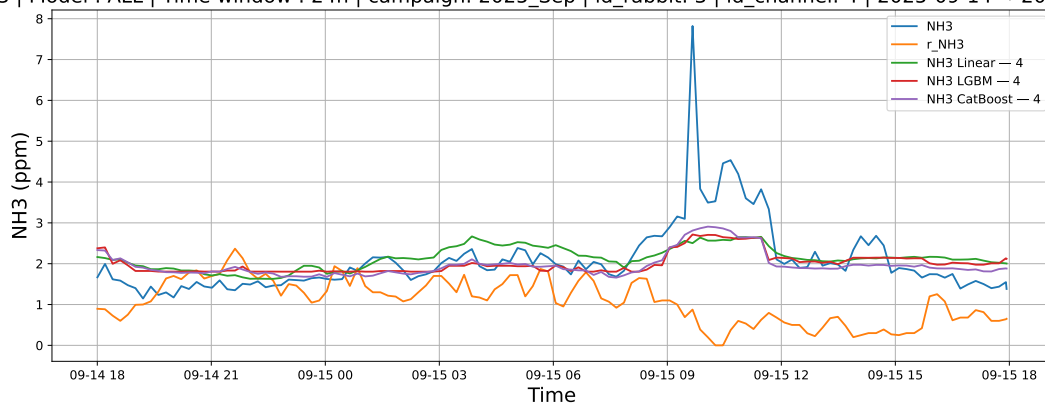
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

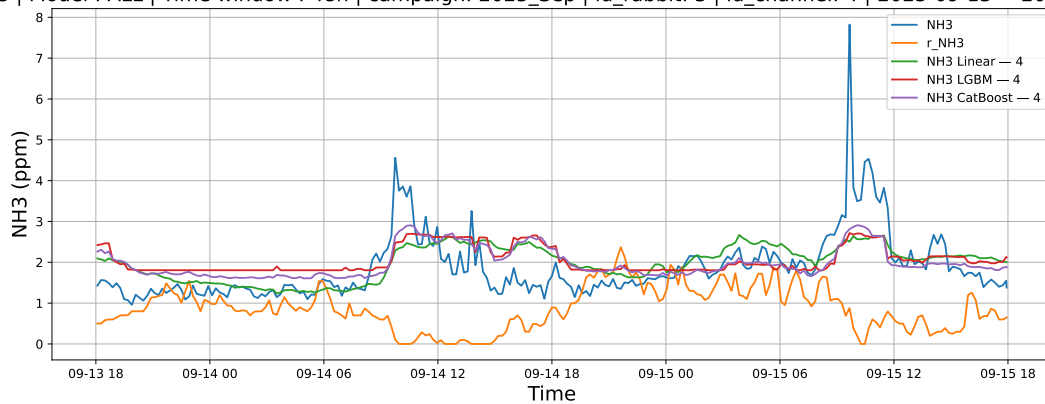
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



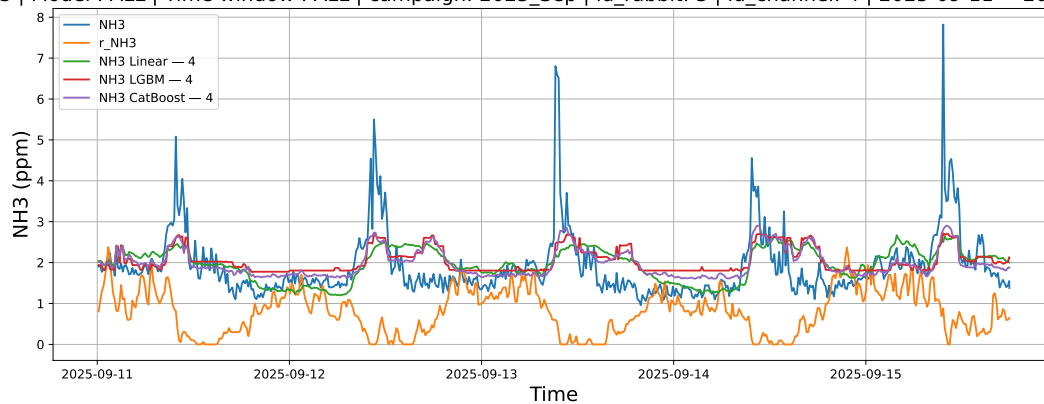
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

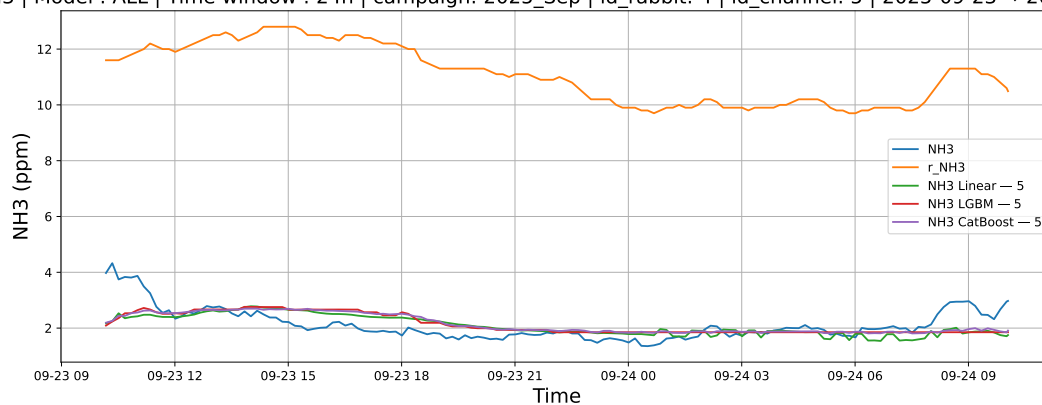
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

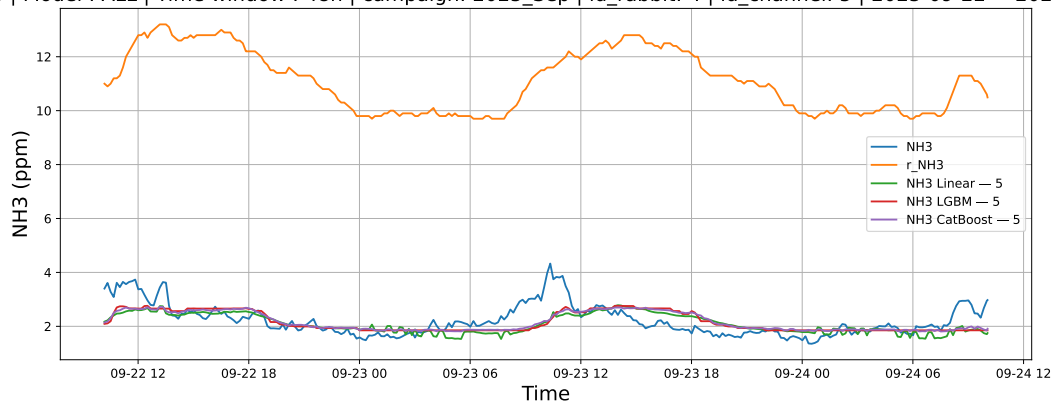
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



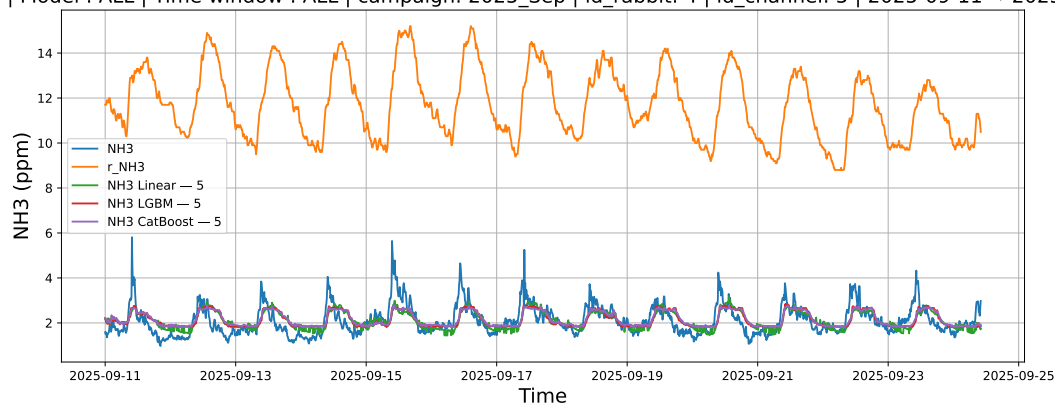
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

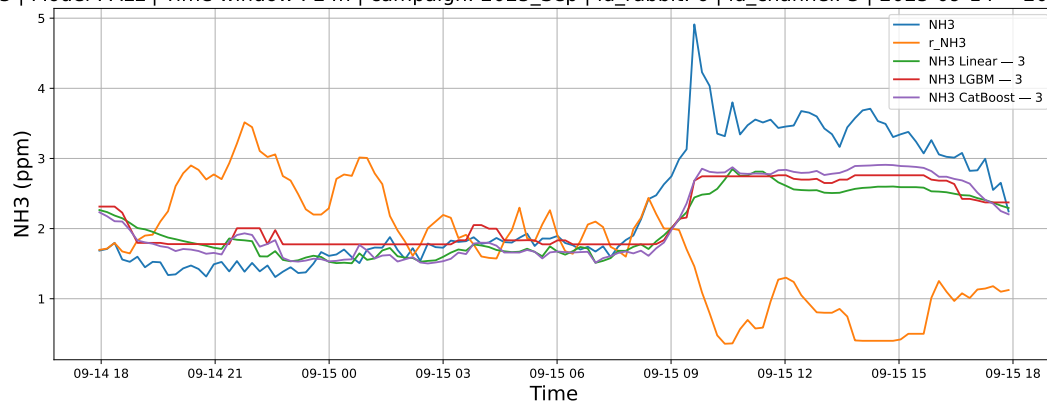
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

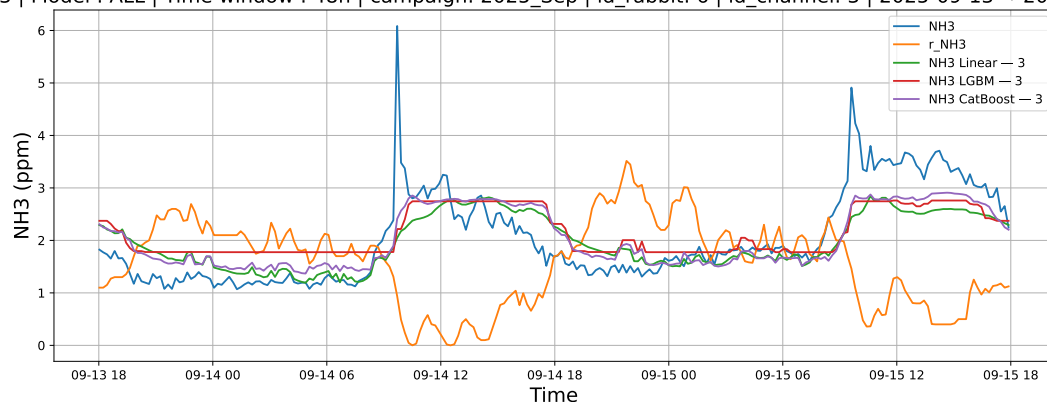
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

